A COMPLETE COURSE IN MICROSCOPIC ANATOMY

BASIC HISTOLOGY

A Learning Text

A self-contained course in microscopic anatomy, written for students who want to understand tissues, organs and the logic of their structure.

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23 chapters

100 exam questions

10 master comparison tables

1 rapid-review chapter

First edition · Text only: no histological images are reproduced

BASIC HISTOLOGY

A Learning Text

A self-contained course in microscopic anatomy for students of medicine, dentistry and the biomedical sciences

First edition · 2026

HOW THIS BOOK IS ORGANIZED

Part I · Cells and techniques: microscopy, the cytoplasm, the nucleus
Part II · The four basic tissues: epithelium, connective tissue, adipose, cartilage, bone, nerve, muscle
Part III · Transport, defense, and blood: circulation, blood, hemopoiesis, immunity
Part IV · Digestion and respiration: gut, associated organs, respiratory system, skin
Part V · Excretion and reproduction: urinary system, endocrine glands, male and female reproductive systems, eye and ear
Part VI · Rapid review, master tables, stains, vocabulary, and sources
Part VII · One hundred examination questions with explained answers

This is an independent teaching text. Its prose is original; no images, plates, captions or extended passages from another publication are reproduced. Histological facts are part of the common scientific literature, and where authorities differ the text says so rather than hiding the difference. The book is not a clinical manual and does not replace laboratory instruction or medical training. Rights in this text have not been assigned; commercial distribution requires a license and an ISBN.

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About this book

This is a complete course in histology: the microscopic structure of cells, tissues and organs, the way that structure serves function, and the way it fails in disease. It is written to be read on its own, from the first page to the last. Every explanation, table, identification task and question is complete in itself.

The book is organized in seven parts. Parts I to V teach the subject in the order in which it is best learned: the cell and the techniques that make it visible; the four basic tissues; blood, circulation and defense; digestion and respiration; and finally excretion, reproduction and the special senses. Part VI condenses the whole subject into comparison tables and a rapid-review chapter. Part VII is a hundred-question examination with explained answers.

What this book does not contain is pictures. Histology is a visual subject, and no set of printed words can replace a microscope. What this book gives you instead is a precise description of what to look for and how to tell one structure from its neighbors — the organization of a tissue, the shape and position of its cells, the staining of its matrix, the arrangement of its layers, ducts, follicles and vessels. Those descriptions are written to work with any slide collection, any virtual-microscopy set and any microscope you have access to. Read a chapter, then go and find the structures it describes. The seeing is yours; the book teaches you what to see.

A note on numbers. Values are typical adult human values unless stated otherwise. Ranges differ with method, fixation, species and observer, and even well-established figures vary a little between references. Where authorities differ, this book gives the range or says which value is being used. Percentages are approximate. No number in this book should be treated as a diagnostic threshold; clinical decisions require clinical sources.

Clinical material. The clinical notes show why histological structure matters. They are written for orientation and are not advice about diagnosis or treatment.

Scope and limits. This is an independent educational work. It is not a substitute for laboratory teaching, for a course in medicine, or for supervised clinical training. It reproduces no histological images, because the images in published atlases belong to their makers; where a structure is best learned from a picture, the text tells you exactly what the picture should show.

How this book is organized

Every chapter follows the same teaching sequence, adapted where the subject demands it:

Block	What it gives you
Learning objectives	What you should be able to do by the end of the chapter
The core idea	The one principle that makes the chapter cohere
Explanation	The structures and processes, from simple to complex, in the order they are best understood
Structure and function	Why each feature exists, and what fails when it does not
MICRO boxes	How to recognize the tissue, and how to distinguish it from its closest look-alikes
Comparison tables	Side-by-side distinctions between structures that are easily confused
CLINICAL boxes	The disease link that turns a fact into a consequence
TIP and MNEMONIC boxes	Common errors, and memory hooks that are worth the space
Chapter summary	The chapter compressed to the sentences you should be able to write
Active recall	Questions to answer from memory before checking
Examination-style question	A worked multiple-choice question with every option explained
Practical identification	Specific structures to find on a slide, and the features that identify them

The parts are cumulative. Part I establishes the cell and the methods; nothing later makes sense without it. Part II builds the four tissues. Parts III to V apply them organ by organ. Part VI is for consolidation, not for first learning. Part VII tests the whole book.

How to study

If you have six weeks. Work through Parts I to V at roughly one part per week. Read a chapter, then close it and write its comparison tables from memory, then answer its recall questions. In the sixth week, revise with Part VI and work through the hundred questions in Part VII.

If you have ten days. Two or three chapters a day. On the first pass read the explanation and the tables; leave the recall questions for the second pass. Concentrate on six comparisons: the epithelium types, the connective-tissue fibers, the cartilage types, the muscle types, the leukocytes, and the wall of the gut.

If you have 48 hours. Read Chapter 24 end to end, then the hundred questions, then the ten master tables. Do not attempt the prose. Histology rewards pattern recognition under time pressure, and patterns live in tables.

Whichever plan you use, alternate between reading and retrieving. Rereading feels productive and is not; answering a question from memory, even badly, is what builds the memory.

The Twelve Laws of Histology

These twelve rules of thumb hold across most of histology. Each has exceptions, and the chapters point them out. When you are unsure how to approach a structure, reason from them.

1. **Structure follows function, and function follows structure.** Most visible features are a solution to a physical problem — diffusion distance, mechanical stress, surface area, water loss, conduction speed.
2. **Form is set by the apical surface, the basal surface, and the junctions between cells.** Look at all three before naming a tissue.
3. **Epithelium sits on a basement membrane and is normally avascular.** Nutrients reach it by diffusion from vessels in the underlying connective tissue. The exception worth knowing is the **stria vascularis** of the cochlea, which contains capillaries within the epithelium.
4. **Connective tissue is defined by its extracellular matrix, not its cells.** The matrix tells you the mechanical job: tension, elasticity, compression, or filtration.
5. **Basophilia and eosinophilia report what a cell is doing.** Blue-purple staining (hematoxylin) reflects nucleic acid — rough ER and ribosomes in a protein-synthesizing cell. Pink staining (eosin) reflects abundant protein — cytoplasm, secretory granules, collagen. The correlation is strong but not absolute: mucin and lipid droplets stain poorly with either dye.
6. **Cells that make steroid hormone are full of smooth ER and lipid droplets** — hence foamy, pale, or vacuolated cytoplasm (adrenal cortex, Leydig cells, corpus luteum).
7. **Muscle is classified by nucleus position and striation, not by shape alone.** Peripheral multiple nuclei = skeletal; central single nucleus with striation = cardiac; central single nucleus without striation = smooth.
8. **The number of cell layers tells you the mechanical job.** One layer = exchange, absorption, secretion, or a thin barrier. Many layers = protection from abrasion.
9. **Surface area is bought with folds.** Microvilli, villi, plicae circulares, basal infoldings, and cristae all exist for the same arithmetic reason.
10. **Tubes are built in concentric layers, and each layer has one job.** In blood vessels: intima for exchange, media for tone, adventitia for anchorage. In the gut: mucosa for digestion, submucosa for supply, muscularis for movement.
11. **A capsule and a hilum identify a solid organ as lymph node, spleen, thymus, or gland.** The route of fluid flow tells you which one.
12. **Most histological facts have a clinical counterpart.** Linking a structure to a disease makes it easier to recall and easier to apply.

Exam technique

Answering a slide or image question

Never begin with a guess. Use this sequence every time:

1. **Magnification and staining.** H&E at low power versus PAS versus silver tells you what the preparation was designed to show.
2. **The boundary.** Is there a free surface? A lumen? A capsule? Blood vessels inside the tissue? This immediately separates epithelium-lined organs from solid organs.
3. **The dominant tissue.** Note the arrangement — cords, acini, tubules, follicles, layers, bundles.
4. **The one diagnostic cell.** Parietal cell, Paneth cell, podocyte, Purkinje fiber, club cell, Sertoli cell, odontoblast. Examination images often turn on a single diagnostic cell.
5. **Name it, then justify it in the same sentence.** "This is thyroid gland because the parenchyma is arranged in follicles of colloid lined by a single layer of cuboidal cells." A justified answer shows that you understand the image rather than merely recognizing it.

Answering a "describe and give the function of" essay

Use a fixed skeleton — it works for any organ:

Capsule → cortex → medulla → parenchyma arrangement → cell types → blood supply → ducts or outflow → function → clinical correlation.

The headings work for almost any organ. Write them down first, then fill them in.

Ten common errors

- "It looks like…" without saying why it looks that way.
- Naming a cell without its product or function.
- Confusing "basophilic" (blue; acidic components binding a basic dye) with "eosinophilic" (pink; basic proteins binding the acidic dye eosin).
- Saying "simple squamous" for endothelium without noting it is a type of simple squamous epithelium.
- Using "mucous" for "mucosa" — different words, different things.
- Describing the adrenal medulla and cortex as one tissue; they are embryologically unrelated.
- Calling transitional epithelium "stratified squamous"; the umbrella cells are unique.
- Forgetting that the thymus has **no** follicles and the spleen has **no** afferent lymphatics.
- Mixing up the two adrenal cortex zones that produce cortisol and aldosterone.
- Writing "bone is dead tissue" — osteocytes are metabolically active and connected by canaliculi.

HIGH-YIELD PEARL

In a viva or OSCE-style histology station, the task is usually to connect a structure to a function and a disease. Practice every fact in a three-part sentence: "Structurally X, which allows Y, and when it fails you get Z."

Chapter guide

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A note on sources

Histology is an old and well-settled science, and the facts in this book are the common property of the field. They were checked against the principal reference listed in Appendix C. Where standard accounts differ — on a range, a definition, or the weight to give a mechanism — the difference is stated in the text rather than hidden. Nothing in this book is presented as a quotation from another work, and no illustration is reproduced. Appendix C also explains how the checking was done and what it does not cover.

PART I — CELLS AND TECHNIQUES

Chapters 1–3 · microscopy, the cytoplasm, and the nucleus

1 · Histology & Its Methods of Study

In this chapter: Preparation of Tissues for Study · Light Microscopy · Electron Microscopy · Autoradiography · Cell & Tissue Culture · Enzyme Histochemistry · Visualizing Specific Molecules · Interpretation of Structures in Tissue Sections

Learning objectives

By the end of this chapter you should be able to:

1. Trace a tissue from patient to stained slide, and explain what each step preserves.
2. Predict what hematoxylin and eosin will color, and why.
3. Choose the right special technique for a target molecule (carbohydrate, lipid, protein, nucleic acid, antigen).
4. Explain how section thickness, fixation, and artifact can mislead you when you interpret a two-dimensional image.

The core idea

Histology explores the body's tissues and how they are arranged to produce functional organs. Tissues have two interacting components: **cells and extracellular matrix (ECM)**. The ECM is not packing material — it supports the cells, carries the fluid that brings nutrients and removes waste, and its molecules bind specific cell-surface receptors connected to structural components inside the cell, forming a continuum in which cells and matrix function together.

Organs form by an orderly combination of these tissues. Because cells and matrix components are small, histology depends on microscopes and molecular methods — and every image you will interpret has been fixed, dehydrated, cleared, embedded, sectioned, stained and mounted. Knowing what each step does tells you which appearances are real and which are artifacts.

Preparation of Tissues for Study

Step	What it does
Fixation	Pieces of organ are placed as soon as possible in **stabilizing or cross-linking compounds**. **Formalin** — a buffered solution of formaldehyde, made by diluting the 37–40% stock to about 4% — is widely used for light microscopy; **glutaraldehyde** for electron microscopy. Both react with the **amine groups (NH_2) of proteins**, preventing degradation by proteases; glutaraldehyde also **cross-links adjacent proteins**. Fixative must diffuse fully through the tissue, so organs are cut into small fragments, or fixed by **vascular perfusion**
Dehydration	Water extracted gradually through increasing **ethanol** solutions, ending in 100% ethanol
Clearing	Ethanol replaced by an organic solvent miscible with both alcohol and the embedding medium; the tissue becomes **translucent**
Infiltration	Tissue placed in melted **paraffin at 52–60 °C** in an oven, which evaporates the clearing solvent
Embedding	Paraffin allowed to harden in a small container at room temperature
Sectioning	Block trimmed and cut on a **microtome**. Paraffin sections are typically **3–10 µm**; EM requires sections **less than 1 µm**

A concentration note. Formalin is often described as a buffered isotonic solution of 37% formaldehyde. That 37–40% solution is the **stock** from which working fixative is made; tissue is fixed in **10% formalin**, equivalent to about 3.7–4% formaldehyde, buffered to pH 7.2–7.4 (neutral buffered formalin). Keep the two concentrations apart in practical questions.

The units you should be able to convert: 1 µm = 1/1000 mm; 1 nm = 0.001 µm = 10^{-6} mm; 1 Å = 0.1 nm = 10^{-4} µm.

Plastic resins are used for both light and electron microscopy. They are infiltrated after dehydration and harden when cross-linking polymerisers are added; plastic embedding **avoids the higher temperatures of paraffin** and therefore avoids tissue distortion.

CLINICAL CORRELATION

Biopsies are tissue samples removed during surgery or routine procedures, fixed in vials of formalin. When the result is needed before the procedure ends — typically to know whether a growth is malignant before the patient is closed — the biopsy is **frozen in liquid nitrogen**, which preserves structure and hardens the tissue, and sectioned on a **cryostat** in a subfreezing cabinet. Freezing is also used for histochemistry of **sensitive enzymes or small molecules**, because freezing, unlike fixation, does not inactivate most enzymes; and because clearing solvents dissolve lipids, **frozen sections are used whenever lipid-containing structures must be studied**.

Staining

Most cells and extracellular material are completely colorless. Dyes behave like acidic or basic compounds and form **electrostatic (salt) linkages** with ionisable radicals of tissue macromolecules.

Term	Meaning	Examples
Basophilic	Anionic (net negative) components with affinity for **basic dyes** — nucleic acids, glycosaminoglycans	Toluidine blue, alcian blue, methylene blue, **hematoxylin**
Acidophilic	Cationic components with many ionised amino groups, staining with **acidic dyes** — mitochondria, secretory granules, collagen	**Eosin**, orange G, acid fuchsin

H&E is the most common method. **Hematoxylin** stains DNA in the nucleus, RNA-rich cytoplasm, and cartilage matrix dark blue-purple; **eosin** stains other cytoplasmic structures and collagen pink and acts as the **counterstain**. **Trichrome** stains (eg, Masson) allow greater distinction among extracellular components.

The PAS reaction uses the hexose rings of polysaccharides and other carbohydrate-rich structures and stains them distinctly **purple or magenta** — the basis for demonstrating glycogen, mucin, and basement membranes. The **Feulgen reaction** is a modification of PAS that stains **DNA specifically**.

Enzyme digestion identifies what a stain is binding: pretreating a section with an enzyme that digests one substrate. Pretreatment with **ribonuclease** greatly reduces cytoplasmic basophilia with little effect on the nucleus, proving that the cytoplasmic staining depended on RNA.

Lipid-soluble dyes such as **Sudan black** reveal lipid-rich structures — provided the processing steps that remove lipids (heat and organic solvents) are avoided. **Metal impregnation** with silver salts visualizes certain ECM fibers and specific elements in nervous tissue. The staining procedures used for most light micrographs are conventional, and the ones you will meet are summarized below.

The staining palette. Learn what each one makes visible, because a report that says "Mallory trichrome" is telling you which structures were deliberately contrasted:

Stain	Nuclei	Cytoplasm	Collagen / other	Where you meet it
H&E	Blue-purple (hematoxylin)	Pink (eosin)	Collagen pink	Almost every chapter
PT (pararosaniline–toluidine blue)	Purple chromatin	Lighter violet	Collagen pale violet	Plastic sections needing fine detail
Mallory trichrome	Purple	Red-orange (also keratin, erythrocytes)	**Blue**	Connective tissue, small vessels, pituitary, kidney

Stain	Nuclei	Cytoplasm	Collagen / other	Where you meet it
Picrosirius–hematoxylin (PSH)	Purple	Pink-violet	**Collagen red**, birefringent under polarized light	Ovary, cartilage, tendon
PAS	—	Magenta where carbohydrate is abundant	Glycogen, mucin, basement membrane, glycocalyx	Gut, kidney, cartilage
Wright–Giemsa	Purple	Granules color differentially	Erythrocytes pink	Blood and marrow smears
Silver / gold	—	—	Nerve filaments and **reticulin (type III)** black-brown	Nervous tissue, lymphoid organs
Elastin stains (Weigert, aldehyde fuchsin, orcein)	—	—	Elastic fibers brown-purple	Aorta, lung, skin
Lipid stains (oil red O, Sudan black, osmium tetroxide)	—	—	Lipid droplets and myelin	Adipose tissue, steroid cells, nerve

Slide preparation, from fixation to observation, takes **12 hours to 2½ days**, depending on tissue size, embedding medium, and staining method.

Light Microscopy

The **bright-field microscope** has three lens sets:

- **Condenser** — collects and focuses a cone of light onto the slide.
- **Objective** — enlarges and projects the image; routine objectives are **×4** (large field), **×10** (medium), **×40** (detail).
- **Eyepiece (ocular)** — magnifies a further **×10**, giving total magnifications of **×40, ×100, ×400**. Total magnification = objective × ocular.

Resolving power — the smallest distance at which two structures are seen as separate — is the critical factor, and its maximum for a light microscope is about **0.2 µm**, permitting clear images at **1000–1500×**. Anything smaller or thinner than 0.2 µm (a single ribosome, a microfilament) cannot be distinguished, and two mitochondria closer than 0.2 µm appear as one. Resolution depends mainly on the **objective lens**; the ocular only enlarges and does **not** improve resolution. Magnification is of value only when accompanied by resolution.

Virtual microscopy converts a stained slide into high-resolution digital images captured in a grid at multiple magnifications, stored so a browser can navigate the original slide — now replacing glass slide collections in teaching laboratories.

Method	Principle	Use
Fluorescence	Fluorophores emit longer-wavelength light when irradiated; bright on a dark background	**DAPI** and **Hoechst** bind DNA (blue); **acridine orange** binds DNA and RNA and emits different colors, localizing them separately; fluorescein-coupled probes and labeled antibodies
Phase-contrast	Light changes speed through structures of different refractive index, producing contrast in transparent objects	**Living, unstained cultured cells**; **DIC (Nomarski optics)** gives a more three-dimensional image
Confocal	A laser point source and a **pinhole aperture** optically conjugated to the focal plane block out-of-focus light; the beam scans the specimen	Thin **optical sections**, localization with great precision, and 3-D reconstruction

Method	Principle	Use
Polarizing	A polarizing filter passes light vibrating in one direction; oriented macromolecules rotate it and appear bright against a dark background — **birefringence**	Collagen, cellulose, microtubules, actin filaments; only material with repetitive periodic structure is seen

Digital cameras extend all of these methods: images can be quantified, enhanced, and analyzed by software for features not visible through the eyepieces.

Electron Microscopy

The wavelength of an electron beam is far shorter than that of light, so the electron microscope can resolve structures far below the light-microscope limit — in principle, a thousandfold improvement.

Transmission EM (TEM) resolves to about **3 nm** in routine biological sections — the working figure, limited by section thickness and staining rather than by the instrument alone. A modern TEM can reach a point-to-point resolution of about **0.2 nm** on suitable specimens, which is why both numbers appear in teaching; use **3 nm for sectioned tissue** and **0.2 nm only when the question asks for the instrument's limit**. Either way, TEM views isolated particles magnified as much as **400,000×**. Routine tissue sections of **40–90 nm** embedded in resin are studied at up to about **120,000×**. Electrons pass through the section to produce an image in black, white, and gray: areas that transmitted electrons appear **electron lucent** (bright), while denser areas or those binding heavy metal ions appear **electron dense** (dark). The column is configured like an upside-down light microscope, with electromagnetic lenses and a tungsten filament cathode accelerating electrons at 60–120 kV.

Scanning EM (SEM) images the surface in three dimensions.

Autoradiography

A **radiolabeled precursor** is incorporated into newly synthesized macromolecules and detected by a photographic emulsion applied over the section — so the silver grains mark the **cellular sites where those macromolecules were made**. The classic application is **^{3}H-thymidine** labeling of cells in S phase.

Cell & Tissue Culture

Living cells maintained outside the body, observed with phase-contrast or DIC microscopy, allow behavior, differentiation, and drug responses to be studied directly — without fixation, which is precisely the advantage.

Enzyme Histochemistry

A substrate is converted by a specific enzyme into an **insoluble, colored product** deposited at the site of the enzyme, localizing enzyme activity within cells. **Acid phosphatase** for lysosomes is the standard example.

Visualizing Specific Molecules

Immunohistochemistry (IHC) depends on the specific reaction between an antigen and an antibody labeled with a visible marker — a fluorescent compound or **peroxidase** for light microscopy, **gold particles** for TEM.

Method	How it works	Why it is used
Direct IHC	A labeled **primary antibody** binds the antigen directly	Simple, fewer steps
Indirect IHC	An unlabeled primary antibody is detected by labeled **secondary antibodies**	**More commonly used** — the extra level of binding **amplifies the signal** and gives greater technical flexibility

In situ hybridization (ISH) uses labeled **cDNA probes** to detect specific gene sequences or mRNAs microscopically.

Interpretation of Structures in Tissue Sections

Two rules govern the interpretation of every section:

1. Many processing, preparation, and staining steps introduce **artifacts** — spaces and precipitates not present in living tissue — and these must be recognized for what they are.
2. Sections are essentially **two-dimensional planes through three-dimensional structures**, so what you see is a slice, not the whole object.

UNDER THE MICROSCOPE
Before identifying anything, decide whether it could be an artifact. A clear space around every cell usually means shrinkage; a straight edge through the tissue is a knife mark; granular debris is precipitated stain. True structure has irregular, biological boundaries.

EXAM TIP
Three numbers carry the chapter: **0.2 µm** (light microscope resolution), **3–10 µm** (paraffin section thickness), **about 3 nm** (TEM resolution of routine sections). If a question asks why the electron microscope reveals organelles that the light microscope cannot, the answer is **the shorter wavelength of the electron beam**, not the magnification.

HIGH-YIELD PEARL
A useful chain to remember from Chapter 1: **dehydration precedes clearing**, clearing makes the tissue translucent, and **freezing avoids both** — which is why frozen sections are used for rapid diagnosis, for enzymes, and for lipids.

Chapter 1 — summary

- H&E = the routine survey stain; hematoxylin is basic (nuclei, rough ER), eosin is acidic (collagen, cytoplasm).
- Resolution: light microscope about 0.2 µm; TEM about 3 nm in routine sections.
- Fixation cross-links proteins; formalin is the standard fixative, glutaraldehyde for EM.
- Paraffin sections are 3–10 µm; frozen sections are used for urgent diagnosis and for lipid.
- PAS stains carbohydrates magenta; it demonstrates glycogen, mucin, and basement membranes.
- Immunohistochemistry localizes a specific antigen with an antibody; in situ hybridization localizes a specific nucleic acid sequence.
- Artifacts — shrinkage, knife marks, precipitated stain — must be recognized before any structure is identified.
- A 2D section samples one plane, so a tube can look like a ring and a sphere like a circle.

Active recall

Answer each from memory before checking.

1. **What does formalin do to a tissue?** — Cross-links proteins, preserving structure while making the tissue firm enough to section.
2. **Which dye stains nuclei, and why?** — Hematoxylin, a basic dye that binds acidic nucleic acids (DNA and RNA).
3. **What does PAS demonstrate?** — Carbohydrates, glycogen, and glycoprotein-rich structures such as basement membranes.
4. **Light microscope versus electron microscope resolution?** — About 0.2 µm versus about 3 nm in routine sections, roughly two orders of magnitude.
5. **Why is a 5-µm section dangerous to interpret in one glance?** — It samples a single plane; a tube can look like a ring, a sphere like a circle, and a fold like a septum.

Exam-style question

A research group needs to show that a newly described protein is present in the basal layer of the epidermis. Which technique is the most direct choice?

A. Hematoxylin and eosin
B. Periodic acid–Schiff
C. Immunohistochemistry with a specific antibody
D. Sudan black

Answer: C — Immunohistochemistry localizes a specific protein by antibody binding. H&E shows general morphology, PAS shows carbohydrates, and Sudan black shows lipids.

Practical identification

Work with a real or virtual H&E section. At low power, locate the tissue edge, any free surface and any capsule; at high power, find nuclei, cytoplasm and extracellular matrix. Then look deliberately for artifacts: clear spaces around cells (shrinkage), straight knife marks, folded or torn areas, and precipitated stain. If phase-contrast, fluorescence or confocal images are available, compare what each reveals and explain why a structure may be visible in one preparation and invisible in another.

2 · The Cytoplasm

In this chapter: Cell Differentiation · The Plasma Membrane · Cytoplasmic Organelles · The Cytoskeleton · Inclusions

Learning objectives

By the end of this chapter you should be able to:

1. Describe each membrane-bound organelle and match it to the cell type that needs it most.
2. Explain how proteins are sorted from ribosome to final destination.
3. Distinguish the three cytoskeletal systems by subunit, diameter, and motor protein.
4. Predict the ultrastructure of a cell from its function (steroid synthesis, protein export, phagocytosis, movement).

The core idea

A cell is a compartmentalization machine. Each organelle is a membrane-bound solution to a specific logistics problem: making protein, making lipid, generating ATP, degrading waste, or moving cargo. Learn the job, and the morphology follows automatically.

Cell Differentiation

The chapter opens with differentiation, because it explains why one genome produces hundreds of cell shapes. Cells become specialized by **differential gene expression** — the same DNA, different genes switched on. Differentiation is accompanied by changes in the amount and type of organelles, which is exactly what you recognize down a microscope: an erythroblast loses its nucleus, a plasma cell fills with rough ER, a muscle cell fills with contractile filaments.

Stem cells retain the capacity to divide and to produce differentiated progeny. They divide **asymmetrically**: one daughter renews the stem-cell pool, the other commits to differentiation. This mechanism explains tissue renewal in the gut, epidermis, and bone marrow (Chapter 3).

The Plasma Membrane

A **7.5–10 nm phospholipid bilayer** with embedded proteins — the **fluid mosaic model**. Amphipathic phospholipids form the bilayer spontaneously; cholesterol modulates fluidity; glycolipids and glycoproteins project outward as the **glycocalyx**, which mediates cell recognition, adhesion, and immune identity.

Protein class	Position	Function
Integral (transmembrane)	Span the bilayer	Channels, pumps, receptors, adhesion
Peripheral	Attached to one face	Signaling, cytoskeletal anchoring, enzymes

Lipid rafts are cholesterol- and sphingolipid-enriched microdomains that concentrate signaling molecules. **Caveolae** are small flask-shaped invaginations involved in uptake and signaling.

Transport across the membrane

- **Simple diffusion** — small, non-polar molecules down a gradient.
- **Osmosis** — water movement through the bilayer and aquaporins.
- **Facilitated diffusion** — carrier or channel, down the gradient, no energy.
- **Active transport** — against the gradient, ATP-dependent. The **Na^+/K^+-ATPase** exchanges 3 Na^+ out for 2 K^+ in and consumes a large fraction of the cell's resting energy budget.
- **Endocytosis** — **phagocytosis** (large particles, actin-driven), **pinocytosis** (fluid), **receptor-mediated endocytosis** (clathrin-coated pits; the classic example is LDL uptake, and its

failure causes familial hypercholesterolemia).

- **Exocytosis** — vesicle fusion with the plasma membrane; requires SNARE proteins and calcium.
- **Transcytosis** — endocytosis on one face plus exocytosis on the other; used by endothelial cells and antibodies crossing epithelia.

Cytoplasmic Organelles

Mitochondria. Double membrane; the inner membrane folds into **cristae** carrying the electron transport chain and ATP synthase. The matrix contains circular mitochondrial DNA, mitochondrial ribosomes, and calcium granules. Mitochondria are **maternally inherited** and semi-autonomous. Cells with huge energy demand (cardiac muscle, brown fat, proximal tubule, sperm midpiece) are packed with them. **Cells that synthesize steroid hormone have mitochondria with tubular rather than lamellar cristae**, a reliable ultrastructural marker.

Ribosomes. 15–20 nm particles of rRNA plus protein, made of a large and small subunit. Free ribosomes make cytosolic protein for internal use; ribosomes bound to the rough ER make protein for secretion, membranes, or lysosomes. **Polyribosomes (polysomes)** are strings of ribosomes on one mRNA, and a **polyribosome** is the functional unit of translation.

Endoplasmic reticulum (ER). An anastomosing network of intercommunicating channels and cisternae, continuous with the outer nuclear membrane. **Rough ER (RER)** — flattened cisternae studded with ribosomes. Sites of synthesis of secretory proteins, integral membrane proteins, and lysosomal enzymes. It is the structure responsible for **basophilia** — and in neurons, stacked RER plus polysomes form the **Nissl bodies**.

Smooth ER (SER) — tubular and ribosome-free. Three jobs: **lipid and steroid synthesis**, **detoxification** (hepatic cytochrome P450), and **calcium storage** (sarcoplasmic reticulum of muscle). Hepatocytes, adrenal cortical cells, Leydig cells, and the corpus luteum are rich in SER — which is why steroid-secreting cells look pale and foamy rather than blue.

Golgi apparatus. Vesicles leave the ER in coats marked by **COPII** and fuse with the cis (forming) face; retrograde transport back to the ER is mediated by **COPI**-coated vesicles. It is a stack of 4–8 flattened cisternae with a **cis (forming) face** receiving vesicles from the ER and a **trans (maturing) face** releasing vesicles. Functions: modification (glycosylation, sulfation, phosphorylation), sorting, and packaging. The **trans-Golgi network** is the dispatch center routing proteins to secretory granules, lysosomes, or the membrane. Cells specialized for secretion (plasma cells, pancreatic acinar cells, goblet cells) have an obvious, often supra-nuclear Golgi.

Lysosomes. 0.05–0.5 µm vesicles of acid hydrolases at pH ≈ 5 maintained by a proton pump. **Primary lysosomes** are newly formed; **secondary lysosomes** are the product of fusion with a phagosome or autophagosome; **residual bodies** remain after digestion and accumulate as **lipofuscin**, the "wear-and-tear" pigment of long-lived cells such as cardiac myocytes and neurons.

Peroxisomes. Small, membrane-bound oxidative organelles containing **catalase** and oxidases. They perform β-oxidation of very long-chain fatty acids, detoxify ethanol, and destroy H_2O_2.

Proteasomes. Cytosolic barrel-shaped complexes that degrade ubiquitin-tagged proteins.

The Cytoskeleton

Element	Diameter	Protein	Main roles
Microfilament	6–8 nm	**Actin**	Microvilli core, cell cortex, stress fibers, cleavage furrow, muscle contraction
Intermediate filament	10 nm	**Keratin, vimentin, desmin, GFAP, neurofilament, lamin**	Mechanical strength; tissue-specific identity
Microtubule	25 nm	**Tubulin (α/β dimer)**	Spindle, cilia and flagella, intracellular transport

Intermediate filaments are the histologist's best friend, because each cell type keeps its own: **keratin** in epithelium, **vimentin** in mesenchyme, **desmin** in muscle, **GFAP** in astrocytes, **neurofilament** in neurons, **lamins** in the nucleus. This is why immunohistochemistry can identify the origin of a tumor that looks undifferentiated.

Microtubules are polar. Kinesin walks cargo toward the plus end (anterograde, away from the cell body); **dynein** walks toward the minus end (retrograde, back to the cell body). Rabies and herpes viruses exploit dynein to travel to the neuronal soma.

Cilia and flagella have a **9 + 2** arrangement with dynein arms; the **basal body** has **9 + 0**. Defective dynein causes **primary ciliary dyskinesia**, of which **Kartagener syndrome** (situs inversus, bronchiectasis, sinusitis, male infertility) is the classic form.

Inclusions

Non-membranous accumulations of metabolites or pigments: **glycogen** (PAS-positive, liver and muscle), **lipid droplets** (Sudan-positive), **melanin**, **lipofuscin**, **hemosiderin**, and crystals.

CLINICAL CORRELATION

Lysosomal storage diseases are the direct consequence of a missing acid hydrolase. In **Tay–Sachs disease** hexosaminidase A is absent, so GM2 ganglioside accumulates in neurons, producing developmental regression, a cherry-red macular spot, and death in early childhood. In **I-cell disease** the enzyme that tags hydrolases with mannose-6-phosphate is defective, so the enzymes are secreted instead of delivered to lysosomes.

UNDER THE MICROSCOPE

Learn to read a cell's job from its EM. Abundant RER and a large Golgi = protein secretion. Abundant SER and lipid droplets, mitochondria with tubular cristae = steroid secretion. Many mitochondria and few other organelles = energy transduction. Numerous lysosomes and residual bodies = phagocytosis or long life. Bundles of filaments = mechanical work.

MEMORY HOOK

"Keratins Keep Epithelium; Vimentin is Versatile mesenchyme; Desmin Does muscle; GFAP Glues astrocytes; Neurofilaments are Neuronal."

EXAM TIP

Keep two distinctions straight. The Golgi **modifies and sorts** proteins; the RER makes them. The mitochondrion generates ATP; the lysosome degrades material.

Chapter 2 — summary

- Rough ER makes membrane and secretory proteins; smooth ER makes lipids and steroids and detoxifies.
- The Golgi is polarized: cis face receives, medial cisternae modify, trans face sorts and packages.
- Mitochondria have outer and inner membranes; cristae shape reflects the cell's energy demand.
- The peroxisome contains catalase and performs β-oxidation of very long-chain fatty acids.
- The lysosome is the acid hydrolase compartment; its failure produces storage diseases.
- Microtubules are 25 nm, actin filaments 7 nm, intermediate filaments 10 nm.
- Kinesin walks toward the microtubule plus end (anterograde); dynein toward the minus end (retrograde).
- Cilia are 9 + 2 and motile; basal bodies are 9 + 0; microvilli have an actin core.
- Apoptosis is programmed, ATP-dependent, and non-inflammatory; necrosis is classically passive and inflammatory, though some regulated lytic death programs blur the distinction.

Active recall

1. **Rough versus smooth ER — what distinguishes them functionally?** — Rough ER carries ribosomes and makes membrane and secretory proteins; smooth ER makes lipids and steroids and detoxifies.
2. **Why is the Golgi described as polarized?** — Cis face receives, medial cisternae modify, trans face packages and sorts.
3. **Which organelle contains catalase, and what does it do?** — The peroxisome; catalase breaks down hydrogen peroxide, and peroxisomes also perform β-oxidation of very long-chain fatty acids.
4. **Which cytoskeletal element forms the mitotic spindle, and which motor walks toward its plus end?** — Microtubules; kinesin moves toward the plus end (anterograde), dynein toward the minus end (retrograde).
5. **What ultrastructural feature identifies a cell that secretes steroid hormone?** — Abundant smooth ER and lipid droplets, with mitochondria that have tubular cristae.

Exam-style question

A cell in the adrenal cortex is examined by electron microscopy. Which combination of features is most consistent with its function?

A. Abundant rough ER and apical zymogen granules
B. Abundant smooth ER, lipid droplets, and tubular mitochondrial cristae
C. Numerous lysosomes and residual bodies
D. Dense arrays of actin filaments and desmosomes

Answer: B — Steroid synthesis requires cholesterol (stored as lipid droplets), smooth ER for the enzymatic steps, and mitochondria with tubular cristae.

Practical identification

On an electron micrograph, identify the plasma membrane, rough ER (parallel cisternae bearing ribosomes), smooth ER (tubules without ribosomes), Golgi (stacked flattened cisternae with vesicles), mitochondria (double membrane and cristae), lysosomes (dense bodies) and inclusions (glycogen rosettes, lipid droplets, pigment). On a light micrograph, use basophilia to locate rough ER and eosinophilia to locate mitochondria and secretory granules, and justify each assignment from the chemistry of the dyes.

3 · The Nucleus

In this chapter: Components of the Nucleus · The Cell Cycle · Mitosis · Stem Cells & Tissue Renewal · Meiosis · Apoptosis

Learning objectives

By the end of this chapter you should be able to:

1. Describe the nuclear envelope, lamina, and pore complex as a selective transport system.
2. Explain how DNA is packaged into nucleosomes, chromatin, and chromosomes.
3. Sequence the cell cycle and name the checkpoints that guard it.
4. Contrast mitosis, meiosis, and apoptosis, and explain what goes wrong when each fails.

The core idea

The nucleus stores the genome, transcribes it, and duplicates it before division. Its morphology is a readout of transcriptional activity: **dark, condensed chromatin means silent genes; pale, dispersed chromatin means active genes.**

Components of the Nucleus

Nuclear envelope. Two membranes separated by a **perinuclear cisterna**, continuous with the rough ER. Beneath the inner membrane lies the **nuclear lamina**, a meshwork of **lamins A, B, and C** that supports the envelope and organizes chromatin. **Mutations in lamin A cause Hutchinson–Gilford progeria.**

Centrosomes. Each cell has a pair of **centrioles** — nine microtubular triplets — surrounded by pericentriolar material: the **centrosome**, the microtubule-organizing center that builds the mitotic spindle, and the **basal body** of a cilium.

Nuclear pores. 3,000–4,000 per nucleus. Each is a **nuclear pore complex** about 100 nm wide, built from roughly 34 nucleoporins, and can pass up to about 1,000 macromolecules per second. Small molecules diffuse; proteins and RNA require **importins and exportins** and a Ran-GTP gradient.

Chromatin. Each human cell holds about **2 meters of DNA — 3.2 billion base pairs** — which must be packaged into a nucleus a few micrometers across. The DNA wraps around a core of **eight histones** (two copies each of H2A, H2B, H3, H4) with **about 150 bp of DNA per nucleosome** (about 1.65 turns around the octamer), plus a larger **H1** histone on the outside: that unit is the **nucleosome**, which in EM looks like **10-nm beads on a string**. Further folding into 30-nm fibers and beyond involves nonhistone scaffold proteins and is less well understood. **Heterochromatin** is condensed and transcriptionally inactive; **euchromatin** is dispersed and active. **Constitutive** heterochromatin is permanently silent; **facultative** heterochromatin can be reactivated. The **Barr body** is a facultatively heterochromatic, inactivated X chromosome — one per somatic cell in a female.

Nucleolus. Not membrane-bound; a factory for ribosomal subunits, with a **pars fibrosa** (rRNA transcription) and a **pars granulosa** (ribosome assembly), organized around **nucleolar organizer regions** on the acrocentric chromosomes **13, 14, 15, 21, and 22**. Cells that make large amounts of protein have prominent nucleoli, and malignant cells often do too.

The Cell Cycle

Phase	What happens	Duration (typical cycling cell)
G1	Cell grows, checks nutrients and DNA integrity; restriction point	~10 h
S	DNA replication; histones synthesized	~8 h
G2	Repair check, prepare for mitosis	~4 h
M	Mitosis and cytokinesis	~1 h

The durations above are typical of a **rapidly dividing cultured cell**; most adult tissues cycle far more slowly or not at all, so do not transplant these numbers to the body. Progression is driven by **cyclin–CDK complexes**. **p53** halts the cycle at G1 to allow repair and triggers apoptosis if repair fails — hence "guardian of the genome" — and it is mutated in about half of human cancers. **Rb** is the G1/S brake lost in retinoblastoma. **G0** is the quiescent state of cells that have left the cycle.

Mitosis

- **Prophase** — chromosomes condense, centrioles move to opposite poles, the nuclear envelope begins to break down.
- **Metaphase** — chromosomes align at the equatorial plate; the spindle attaches to kinetochores.
- **Anaphase** — sister chromatids separate and are pulled poleward.
- **Telophase** — nuclear envelopes re-form, chromosomes decondense.
- **Cytokinesis** — an **actin** contractile ring pinches the cell in two.

Stem Cells & Tissue Renewal

- **Totipotent** — can form the whole organism (zygote, early blastomeres).
- **Pluripotent** — can form all three germ layers (inner cell mass; embryonic stem cells; induced pluripotent stem cells reprogrammed with the Yamanaka factors Oct4, Sox2, Klf4, c-Myc).
- **Multipotent** — form a restricted set of cell types (hematopoietic stem cell).
- **Unipotent** — one cell type.

Tissues are **labile** (continuous turnover — epidermis, gut, marrow), **stable** (low turnover, can be triggered — liver, kidney, fibroblasts), or **permanent** (most neurons, cardiac myocytes, lens fibers), with little or no regenerative capacity under normal conditions.

Meiosis

Two divisions produce **four haploid cells** from one diploid cell, with **crossing over** between homologous chromosomes during **pachytene of prophase I**, held together by the **synaptonemal complex**. Meiosis generates genetic diversity and halves the chromosome number; **non-disjunction** produces aneuploid gametes, the mechanism behind trisomy 21.

Apoptosis

Programmed cell death: **caspases** are activated, **cytochrome c** escapes mitochondria, the **Bcl-2 family** decides survival versus death, chromatin condenses, the cell shrinks and buds into **apoptotic bodies** that are phagocytosed **without inflammation**.

Feature	Apoptosis	Necrosis
Trigger	Physiological or programmed	Injury, ischemia, toxin
Energy	ATP-dependent	Passive
Cell size	Shrinks	Swells
Nucleus	Condenses, fragments	Pyknosis, karyorrhexis, karyolysis

Feature	Apoptosis	Necrosis
Membrane	Intact, blebbing	Ruptures
Inflammation	**Absent**	**Present**
Spread	Single cells	Contiguous sheets

The contrast is a teaching simplification. **Necroptosis** and **pyroptosis** are regulated lytic deaths, and apoptotic cells that are not cleared promptly can provoke inflammation, so the presence or absence of inflammation is a strong rule of thumb rather than an absolute distinction.

CLINICAL CORRELATION

Lamin A mutation causes progeria with premature aging. Changes in nuclear shape and chromatin texture are also one of the oldest cancer-screening tools: the irregular, hyperchromatic nucleus of a dysplastic cervical cell is what a Pap smear is looking for.

UNDER THE MICROSCOPE

An active cell has a pale nucleus with a prominent, reticulated nucleolus and lots of euchromatin. A quiescent cell — a small lymphocyte, a spermatozoon, a superficial keratinocyte — has dense heterochromatin and little visible nucleolus. **Nuclear texture is a functional readout.**

EXAM TIP

One Barr body in a normal female somatic cell, **zero** in a male, one in Klinefelter syndrome (47, XXY), none in Turner syndrome (45, X). Count Barr bodies and you have counted X chromosomes minus one.

Chapter 3 — summary

- Nuclear lamina = lamins; lamin A mutation = progeria.
- Nuclear envelope = two membranes + perinuclear cisterna, continuous with rough ER.
- Nuclear pores = 3,000–4,000 per nucleus; importins and exportins, Ran-GTP gradient.
- Nucleosome = eight histones + about 150 bp of DNA; H1 sits outside; 10-nm beads on a string.
- Heterochromatin = condensed and silent; euchromatin = dispersed and active.
- Barr body = facultative heterochromatin of one inactivated X chromosome.
- Nucleolus = rRNA synthesis and ribosome assembly; NORs on 13, 14, 15, 21, 22.
- Cell cycle = G1 → S → G2 → M; DNA replicates in S.
- Cyclin–CDK complexes drive the cycle; p53 arrests at G1; Rb is the G1/S brake.
- p53 = G1 checkpoint and apoptosis; mutated in about half of human cancers.
- Mitosis order = prophase, metaphase, anaphase, telophase; actin ring divides the cell.
- Meiosis = four haploid cells, crossing over in pachytene, held by the synaptonemal complex.
- Non-disjunction = aneuploid gametes, the mechanism of trisomy 21.
- Apoptosis = caspases, cytochrome c, Bcl-2 family, apoptotic bodies, no inflammation.
- Necrosis = swelling, membrane rupture, inflammation.
- Labile tissue turns over constantly; stable tissue can be triggered; permanent tissue has little or no regenerative capacity.

Active recall

1. **What is a nucleosome?** — About 150 bp of DNA wrapped around an octamer of histones (two each of H2A, H2B, H3, H4), with H1 on the outside.
2. **Heterochromatin versus euchromatin?** — Heterochromatin is condensed and transcriptionally silent; euchromatin is dispersed and active.
3. **What is a Barr body?** — The facultatively heterochromatic, inactivated X chromosome in a female somatic cell.

4. **Which checkpoint does p53 guard, and what happens if repair fails?** — The G1 checkpoint; p53 triggers apoptosis if the DNA cannot be repaired.
5. **Name two ways apoptosis differs from necrosis.** — Apoptosis is ATP-dependent, shrinks the cell, keeps the membrane intact, and causes no inflammation; necrosis is classically passive — it swells and ruptures the cell and inflames the tissue — although regulated lytic deaths such as necroptosis blur the boundary.

Exam-style question

A cell is blocked in metaphase with colchicine. Its chromosomes number 46. Which statement is correct?

A. It is a haploid gamete
B. It is a diploid somatic or pre-meiotic germ cell
C. It has completed meiosis II
D. It is an anucleate cell

Answer: B — Colchicine arrests cells in metaphase, when sister chromatids are still attached; a count of 46 means diploid. Haploid cells would show 23.

Practical identification

Find a cell with a large, pale nucleus and a prominent nucleolus, such as a neuron or a plasma cell, and contrast it with a small lymphocyte, whose dense nucleus almost fills the cell. Identify mitotic figures by their condensed chromosomes and absent nuclear envelope, and distinguish them from apoptotic bodies, which are fragmented, membrane-bound and lack a spindle. In a buccal smear from a female, look for the Barr body at the nuclear margin.

PART II — THE FOUR BASIC TISSUES

Chapters 4–10 · the four basic tissues and their specializations

4 · Epithelial Tissue

In this chapter: Characteristic Features of Epithelial Cells · Specializations of the Apical Cell Surface · Types of Epithelia · Transport Across Epithelia · Renewal of Epithelial Cells

Learning objectives

By the end of this chapter you should be able to:

1. Classify any epithelium by the number of layers and the shape of its surface cells.
2. Match each junctional complex to its molecular components and its function.
3. Explain how microvilli, stereocilia, and cilia differ in structure and purpose.
4. Predict where a given epithelium will be found from the mechanical or transport job it must do.

The core idea

Epithelial tissues are composed of closely aggregated polyhedral cells adhering strongly to one another and to a thin layer of ECM, forming cellular sheets that line the cavities of organs and cover the body surface. Epithelia (Gr. epi, upon + thele, nipple) line all external and internal surfaces of the body, and **all substances that enter or leave an organ must cross this type of tissue**. Everything else — the number of layers, the shape of the cells, the specializations on the apex — is a solution to one of four problems: **protection, absorption, secretion, or transport**.

The four basic tissues (the framework for the whole subject):

Tissue	Cells	Extracellular matrix	Main functions
Epithelial	Aggregated polyhedral cells	**Small amount**	Lining of surface or body cavities; glandular secretion
Connective	Several types of fixed and wandering cells	**Abundant amount**	Support and protection of tissues/organs
Muscle	Elongated contractile cells	Moderate amount	Strong contraction; body movements
Nervous	Elongated cells with extremely fine processes	**Very small amount**	Transmission of nerve impulses

Most organs are divided into the **parenchyma** — the cells responsible for the organ's specialized functions — and the **stroma**, which plays a supporting role. **Except in the brain and spinal cord, the stroma normally consists of connective tissue**, which is why the CNS has no connective-tissue framework of its own.

Characteristic Features of Epithelial Cells

The principal functions of epithelial tissues are **covering, lining, and protecting surfaces** (eg, epidermis), **absorption** (eg, the intestinal lining), and **secretion** (eg, parenchymal cells of glands). Specific cells of certain epithelia may be contractile (**myoepithelial cells**) or specialized sensory cells, such as those of taste buds or the olfactory epithelium.

Cell shape and nuclear shape correspond: **columnar** cells have elongated nuclei, **squamous** cells flattened nuclei, and **cuboidal or pyramidal** cells spherical nuclei. This is the fastest identification rule in the whole subject.

The defining features:

1. Cells are closely apposed with minimal intercellular space.
2. There is a **free (apical) surface** facing air, a lumen, or a cavity.
3. Cells rest on a **basement membrane**.

4. The tissue is **avascular** — nourishment arrives by diffusion from vessels in the underlying connective tissue.
5. Cells are **polarized**: apical, lateral, and basal domains differ in structure and function.

The basement membrane, visible with PAS or silver in the light microscope and as two layers in EM:

Layer	Composition	Made by
Lamina lucida	Laminin, entactin, perlecan	Epithelial cells
Lamina densa	**Type IV collagen**, heparan sulfate proteoglycan	Epithelial cells
Lamina reticularis	**Type III (reticular) collagen**	Fibroblasts of underlying connective tissue

Cell junctions

Junction	Transmembrane protein	Cytoskeletal link	Function
Tight junction (zonula occludens)	Claudins, occludins	Actin	Seals the intercellular space; creates the **blood–brain barrier** and apical/basal polarity
Adherent junction (zonula adherens)	**E-cadherin** (Ca^{2+}-dependent)	Actin	Mechanical adhesion; the belt that holds sheets together
Desmosome (macula adherens)	**Desmoglein, desmocollin**	**Keratin**	Spot-welds resisting shear; autoantibodies cause **pemphigus vulgaris**
Gap junction (nexus)	**Connexins**	None	Direct cytoplasmic communication
Hemidesmosome	**Integrins**	Keratin → laminin	Anchors basal cells; antibodies cause **bullous pemphigoid**

The pattern is worth memorizing: **tight seals, adherens belts to actin, desmosomes to keratin, gap junctions for talk, hemidesmosomes for feet.**

Specializations of the Apical Cell Surface

Microvilli. Finger-like extensions with a core of **actin** filaments cross-linked by actin-bundling proteins (villin, fimbrin) and anchored in a **terminal web**. They multiply surface area for absorption. In the proximal tubule and small intestine they form a **brush border** visible as a fuzzy eosinophilic rim. In the epididymis and inner ear, unusually long, stiff microvilli are called **stereocilia** — a misnomer, because they contain no microtubules.

Cilia. Motile, **9 + 2** microtubule axoneme with dynein arms, anchored to a **basal body (9 + 0)**. They move mucus, cerebrospinal fluid, and the ovum. A **primary cilium** is a single non-motile 9 + 0 cilium acting as a sensory antenna — its failure underlies polycystic kidney disease.

Basal infoldings. Deep folds of the basal membrane packed with mitochondria — the "striated" appearance in the ducts of salivary glands and the distal tubule of the kidney. They are the morphological signature of **active ion transport**.

Types of Epithelia

Type	Structure	Location	Function
Simple squamous	One flat layer, central nucleus	Alveoli, endothelium, mesothelium, Bowman capsule	Diffusion, filtration, lubrication
Simple cuboidal	One layer, nuclei round and central	Kidney tubules, small ducts, thyroid follicles, surface of ovary	Secretion, absorption, conduit
Simple columnar	One layer, tall cells	Stomach, intestine, gallbladder, uterine tube	Absorption, secretion, protection
Pseudostratified ciliated columnar	Single layer with nuclei at different heights; cilia and goblet cells	Trachea, bronchi, nasal cavity	Mucus clearance ("respiratory epithelium")
Stratified squamous, keratinized	Many layers, outermost anucleate keratin	Epidermis	Protection against abrasion and dehydration
Stratified squamous, non-keratinized	Many layers, outermost live nucleated cells	Oral cavity, esophagus, vagina, anal canal, cornea	Protection in a wet environment
Stratified cuboidal / columnar	Two or more layers	Ducts of sweat glands, conjunctiva	Conduit with protection
Transitional (urothelium)	Stratified, with dome-shaped **umbrella cells** whose apical membrane is stiffened by **uroplakin** plaques	Renal pelvis, ureter, bladder, proximal urethra	Stretchable barrier to urine
Epithelioid	Cells in clusters with no free surface	Adrenal cortex, islets of Langerhans, liver cords	Endocrine secretion

Two naming traps to avoid. **Endothelium and mesothelium are both simple squamous epithelium** — one lines blood vessels, the other body cavities; they are not separate tissue types. And **pseudostratified** epithelium is truly a **single layer** in which every cell touches the basement membrane, even though the nuclei sit at different levels.

Transport Across Epithelia

An epithelium that moves material from one side to the other is built like a machine: **entry at the apical face, exit at the basal face**, with the tight junction preventing back-leak and mitochondrial-rich basal infoldings powering the pumps. In the intestinal enterocyte, glucose enters apically via **Na^+-glucose cotransport (SGLT1)** and exits basolaterally via **GLUT2**; the Na^+ gradient is maintained by basolateral Na^+/K^+-ATPase. This is **transcellular transport**. **Paracellular** transport passes between cells and is limited by tight junctions.

Renewal of Epithelial Cells

Epithelia wear out. They are replaced from **stem cells** whose location varies with the tissue: the **basal layer** of stratified squamous epithelium, the **crypt base** of the intestine, the **isthmus** of the gastric gland, and the basal layer of the epidermis. Renewal rate is a diagnostic index: the intestinal epithelium turns over in days, the epidermis in weeks. Because the stem cells rest on the basement membrane, an epithelium can regenerate completely after superficial injury — but not after the basement membrane itself is destroyed.

Glands

- **Exocrine** — connected to a surface by a duct. **Endocrine** — ductless, secreting into the blood.
- **Unicellular** — the goblet cell (mucin; empty cup-shaped apex after secretion).
- **Simple** — one unbranched duct (simple tubular = intestinal crypt; simple coiled tubular = eccrine sweat gland; simple branched acinar = sebaceous gland).

- **Compound** — branched duct (salivary glands, pancreas, mammary gland).
- **Mechanism: merocrine** (exocytosis — pancreas, sweat glands), **apocrine** (membrane and cytoplasm bud off — the lipid component of mammary secretion), **holocrine** (whole cell disintegrates — sebaceous gland).

Myoepithelial cells sit between secretory cells and the basement membrane and contract to expel product — visible in mammary and salivary glands.

CLINICAL CORRELATION

Epithelial junctions are the target of autoimmune blistering. In **pemphigus vulgaris**, antibodies against **desmoglein** break desmosomes, so the epidermis detaches within the epithelium and thin-walled blisters form. In **bullous pemphigoid**, antibodies against **hemidesmosomal** antigens split the epidermis from the basement membrane and the blister is thick-walled and tense. The level of the split is dictated by which junction fails.

UNDER THE MICROSCOPE

Count layers, not nuclei. If every cell rests on the basement membrane and only the nuclei sit at different heights, the epithelium is **pseudostratified** — tall, ciliated, respiratory. If cells are stacked on one another, it is **stratified**. Then the top layer: **flat and anucleate** = keratinized; **flat with nuclei** = non-keratinized; **dome-shaped and large** = transitional; **tall with cilia** = respiratory.

MEMORY HOOK

"Simple for exchange, stratified for stress, transitional for stretch." For cilia, remember **9 + 2 moves, 9 + 0 anchors.**

EXAM TIP

Where is ciliated simple columnar epithelium found? The **uterine tube** — cilia move the ovum — and parts of the respiratory tract, where the epithelium is pseudostratified. Students often wrongly assign all ciliated epithelium to the lung.

Chapter 4 — summary

- Epithelium sits on a basal lamina, is avascular, and is normally adjacent to connective tissue.
- Simple = one layer (exchange, absorption, secretion); stratified = protection; pseudostratified = all cells touch the basement membrane.
- Squamous = flat; cuboidal = height ≈ width; columnar = height > width.
- Keratinized stratified squamous = dry, abrasive surfaces; nonkeratinized = wet surfaces (mouth, esophagus, vagina).
- Urothelium (transitional) = dome-shaped umbrella cells; stretches.
- Tight junction (zonula occludens) = seal and fence; it creates apical–basal polarity.
- Zonula adherens anchors actin; desmosomes anchor intermediate filaments; gap junctions couple cells.
- Hemidesmosomes attach the basal cell to the basal lamina.
- Basal lamina = lamina lucida + lamina densa (type IV collagen, laminin); the reticular lamina (type III) belongs to connective tissue.
- Microvilli = actin core, increase surface; stereocilia = long, stiff microvilli; cilia = 9 + 2 microtubules, motile.
- Basal bodies anchor cilia and have a 9 + 0 arrangement.
- Apical specializations serve absorption or movement; basal specializations anchor and receive signals.
- Epithelia renew from stem cells: labile (gut, skin) versus stable (liver, kidney).

Active recall

1. **How do you classify an epithelium in one pass?** — First count the layers (simple, stratified, pseudostratified), then name the surface cell shape (squamous, cuboidal, columnar).
2. **What does a tight junction do besides sealing?** — It fences membrane proteins, creating and maintaining apical–basal polarity.
3. **Which junction attaches a cell to the basal lamina?** — The hemidesmosome.
4. **Microvillus, stereocilium, cilium — core structure?** — Microvillus: actin core. Stereocilium: long, stiff actin bundle. Cilium: 9 + 2 microtubules anchored by a 9 + 0 basal body.
5. **Why is urothelium suited to the bladder?** — Umbrella cells with uroplakin plaques and tight junctions form an osmotic barrier, and the epithelium thins as the bladder fills.

Exam-style question

An epithelium lines a dry, abrasion-prone surface. Which description fits it?

A. Simple columnar with goblet cells
B. Stratified squamous keratinized
C. Simple squamous with fenestrations
D. Pseudostratified ciliated columnar

Answer: B — Keratinized stratified squamous epithelium is the body's answer to mechanical wear plus water loss (skin, hard palate, gingiva).

Practical identification

Classify any epithelium with two questions: how many layers, and what shape is the surface cell? Then find the basal lamina, the underlying connective tissue and the apical specialization — microvilli, cilia, stereocilia or keratin. Distinguish pseudostratified from stratified by checking whether every cell reaches the basement membrane. In a gland, decide whether the secretory unit is serous, mucous or mixed, and whether the duct is intercalated, striated or excretory.

5 · Connective Tissue

In this chapter: Cells of Connective Tissue · Fibers · Ground Substance · Types of Connective Tissue

Learning objectives

By the end of this chapter you should be able to:

1. Name the cells, fibers, and ground substance of connective tissue and give each one's job.
2. Match collagen types I–IV to their locations and to the diseases caused by their failure.
3. Distinguish loose from dense connective tissue, and regular from irregular, by collagen arrangement.
4. Explain how connective tissue repairs and why scurvy and Ehlers–Danlos syndrome look the way they do.

The core idea

Connective tissue is the tissue whose defining feature is **abundance of extracellular matrix**. The cells are widely separated because the matrix — not the cells — does the mechanical work. Name the matrix and you have named the tissue's job: collagen for tension, elastin for recoil, ground substance for compression and diffusion.

Cells of Connective Tissue

Cell	Origin	Function	Identifying feature
Fibroblast / fibrocyte	Mesenchyme	Synthesizes collagen, elastin, GAGs; **fibrocyte** is the quiescent form	Elongated, pale, abundant RER; **myofibroblast** also has actin for wound contraction
Macrophage	Monocyte	Phagocytosis, antigen presentation, cytokines	Irregular, abundant lysosomes; **Kupffer cell** in liver, **microglia** in CNS, **osteoclast** in bone (a fused macrophage)
Mast cell	Bone marrow	Releases **histamine** and **heparin**; IgE-mediated allergy	Round, many dark granules; degranulates on IgE cross-linking
Plasma cell	B lymphocyte	Secretes antibody	**Clock-face nucleus**, eccentric, perinuclear clear zone (Golgi)
Adipocyte	Mesenchyme	Store fat	Signet-ring when unilocular
Pericyte	Mesenchyme	Contractile support of capillaries	On the abluminal surface of capillaries

Fibers

Fiber	Protein	Properties	Where
Collagen	Tropocollagen (types I–XXVIII)	High tensile strength, inelastic	Everywhere; the most abundant protein in the body
Reticular	**Type III collagen**	Fine, branched; silver-staining	Lymphoid organs, bone marrow, liver, endocrine glands
Elastic	**Elastin** core + **fibrillin** microfibrils	Stretch and recoil	Aorta, skin, lung, elastic cartilage, ligaments

Type	Form	Main location	Disease if defective
I	Thick fibrils	Bone, dermis, tendon, cornea	**Osteogenesis imperfecta**
II	Fibrils	Hyaline and elastic cartilage, vitreous	Chondrodysplasias

Type	Form	Main location	Disease if defective
III	Reticular fibers	Lymph node, spleen, bone marrow, skin	**Ehlers–Danlos vascular type**
IV	Network	**Basement membrane**	Alport syndrome
V	Fibrils with type I	Cornea, bone, placenta	Ehlers–Danlos

Collagen synthesis in five steps — a reliable structured-question answer:

1. **Translation** on the RER as preprocollagen.
2. **Hydroxylation** of proline and lysine — **requires vitamin C**.
3. **Glycosylation** in the ER and Golgi.
4. **Procollagen** is secreted and the propeptides cleaved → **tropocollagen**.
5. **Fibril assembly** with cross-links formed by **lysyl oxidase**, which requires **copper**.

This pathway explains two diseases. Without vitamin C, hydroxylation fails and the result is **scurvy** — bleeding gums, poor wound healing, perifollicular hemorrhage. Without copper, cross-linking fails, as in **Menkes disease**. In **Ehlers-Danlos syndrome** the defect is in collagen processing or type; in **Marfan syndrome** the defect is in **fibrillin**, so the aorta dilates and the lens dislocates.

Ground Substance

An amorphous, hydrated gel of **glycosaminoglycans (GAGs)**, **proteoglycans**, and **glycoproteins**, which resists compression, lubricates, and controls diffusion.

- **GAGs** (also called mucopolysaccharides) are long polymers of repeating disaccharide units, usually with many negative charges that attract water: **hyaluronic acid** (the only **unsulfated** one, not attached to protein, and the backbone of the gel) and the **sulfated GAGs** — chondroitin sulfate, keratan sulfate, dermatan sulfate, and heparan sulfate.
- **Proteoglycans** are a core protein with GAG side chains — the classic is **aggrecan**, a very large molecule with a **250-kDa core protein** heavily bound with chondroitin and keratan sulfate and joined to hyaluronan by a **link protein**. Its water-binding is why cartilage resists compression.
- **Perlecan** is the key proteoglycan of **all basal laminae** — the one name that links ground substance to the basement membrane.
- **Glycoproteins** — **fibronectin** (binds cells to matrix via integrins, important in wound healing) and **laminin** (basement membrane).

Types of Connective Tissue

- **Loose (areolar)** — abundant ground substance, scattered fibers, many cells. Fills spaces, surrounds vessels, underlies epithelia.
- **Dense irregular** — thick collagen bundles in many directions; dermis, organ capsules. Resists stress from all directions.
- **Dense regular** — parallel collagen bundles; **tendon** and ligament. Resists stress in one direction.
- **Elastic** — many elastic fibers; elastic ligament, vocal cord, large arteries.
- **Reticular** — type III collagen network; lymphoid organs and marrow.
- **Mucous (Wharton's jelly)** — gelatinous, in the umbilical cord.
- **Embryonic** — mesenchyme.

Besides fibroblasts, the resident population includes **adipocytes**, **macrophages**, mast cells, and the **leukocytes** that migrate in (especially **lymphocytes** and plasma cells in chronic inflammation) — which is why older descriptions divided them into fixed and wandering cells.

Repair. Inflammation → **granulation tissue** (proliferating capillaries, fibroblasts, macrophages) → collagen deposition → **scar**. **Primary intention** occurs in clean apposed wounds; **secondary intention** in larger, open wounds with more granulation and contraction.

CLINICAL CORRELATION

Scurvy is the purest histology-to-clinic link in the chapter: inadequate hydroxylation produces defective collagen, which produces fragile capillaries (bleeding gums, petechiae) and failed wound healing. **Scleroderma** is the opposite failure — excess collagen in skin and internal organs. **Keloid** is an overgrowth of dense collagen beyond the original wound margin, common in darker skin.

UNDER THE MICROSCOPE

Collagen is pink, wavy, and in bundles; elastin is sparse, often darker, and best shown by orcein. A plasma cell has an **eccentric clock-face nucleus with a pale perinuclear halo**; a mast cell's granules stain metachromatically; the eosinophil's bilobed nucleus is unmistakable.

MEMORY HOOK

"1 bone, 2 cartilage, 3 reticulin, 4 basement membrane."

EXAM TIP

The distinction that matters most here is **dense regular versus dense irregular**, and the answer is one word: **direction**. Regular = parallel = tendon = one axis of force. Irregular = random = dermis = many axes.

Chapter 5 — summary

- Connective tissue = cells + fibers + ground substance; it is vascular except in cartilage.
- Type I collagen = bone, tendon, dermis, scar. Type II = hyaline and elastic cartilage. Type III = reticulin. Type IV = basal lamina.
- Collagen synthesis needs vitamin C for proline and lysine hydroxylation; deficiency = scurvy.
- Elastic fibers = elastin core + fibrillin microfibrils; orcein stains them.
- Ground substance = GAGs and proteoglycans; **hyaluronan is the only unsulfated GAG**.
- Fibroblast makes collagen; myofibroblast contracts (wound closure).
- Plasma cell = eccentric clock-face nucleus, perinuclear Golgi halo, antibody factory.
- Mast cell = heparin, histamine, proteases; metachromatic granules; IgE receptor.
- Macrophage = monocyte-derived phagocyte; foreign-body and Langhans giant cells are fused macrophages.
- Adipocytes, mast cells, and leukocytes are resident connective tissue cells.
- Loose (areolar) tissue = cells and vessels; dense irregular = dermis; dense regular = tendon.
- Reticular tissue = type III collagen framework of lymphoid organs and marrow.
- Edema = excess fluid in ground substance; it follows increased pressure, leaky vessels, or low albumin.

Active recall

1. **Which collagen type forms the basal lamina?** — Type IV (with laminin), over a reticular lamina of type III.
2. **Which collagen type is defective in osteogenesis imperfecta?** — Type I.
3. **What is the only unsulfated GAG?** — Hyaluronan.
4. **How do you tell dense regular from dense irregular connective tissue?** — Direction: regular has parallel bundles (tendon), irregular has randomly oriented bundles (dermis).
5. **What does a plasma cell look like, and what does it make?** — Eccentric clock-face nucleus with a pale perinuclear Golgi halo; it secretes antibody.

Exam-style question

A patient has bleeding gums, poor wound healing, and perifollicular hemorrhages. Which step in collagen synthesis is defective?

A. Transcription of collagen genes
B. Hydroxylation of proline and lysine
C. Cleavage of propeptides
D. Cross-linking by lysyl oxidase

Answer: B — Vitamin C is the cofactor for prolyl and lysyl hydroxylase; without hydroxylation the triple helix is unstable — scurvy.

Practical identification

Identify the three fiber types by their appearance and staining: collagen bundles (pink with eosin, blue with trichrome), elastic fibers (wavy, refractile, black with orcein) and reticular fibers (fine and argyrophilic). Name the resident cells — fibroblast, macrophage, mast cell with its heparin granules, adipocyte, and plasma cell with its clock-face nucleus — and the ground substance that fills the spaces. Distinguish loose from dense irregular and dense regular tissue by fiber orientation and cell density.

6 · Adipose Tissue

In this chapter: White Adipose Tissue · Brown Adipose Tissue

Learning objectives

By the end of this chapter you should be able to:

1. Distinguish white and brown adipocytes by light and electron microscopy.
2. Explain how white adipose tissue works as an endocrine organ.
3. Explain the mechanism of non-shivering thermogenesis and where brown fat is found.
4. Relate adipose storage limits to obesity, lipotoxicity, and lipodystrophy.

The core idea

Adipose tissue is an endocrine and metabolic organ disguised as a filling tissue. White fat stores energy; brown fat burns it for heat. Both are built from the same cell with a completely different internal architecture.

White Adipose Tissue

Unilocular. Each white adipocyte is a very large cell, **50–150 µm** in diameter, containing one enormous lipid droplet that displaces the cytoplasm to a thin rim and flattens the nucleus against the plasmalemma — the **signet-ring** appearance. Adipose tissue is normally **15–20% of body weight in men** and somewhat more in women. The lipid dissolves during routine processing, so on an H&E slide the cell appears as an empty circle with a thin border.

Functions: energy storage, insulation, cushioning of organs (retro-orbital, perirenal, mesenteric), and **endocrine signaling**.

Where the stored fat comes from — three sources: dietary fat carried as **chylomicrons**; triglycerides made in the **liver** and carried as **VLDLs**; and fatty acids and glycerol **synthesized by the adipocytes themselves**. Mobilization is by **hormone-sensitive lipase**, activated by **norepinephrine** from the adrenal gland and sympathetic nerves, and by growth hormone; **insulin inhibits** that lipase and stimulates lipid synthesis. The cells are supported by **reticular fibers**, with connective tissue septa dividing the tissue into lobules.

Adipose-derived factors — the ones worth knowing (the last is an enzyme, not an adipokine):

- **Leptin** — from adipocytes, acts on the hypothalamus to reduce appetite.
- **Adiponectin** — increases insulin sensitivity; low in obesity.
- **Resistin, TNF-α, IL-6** — pro-inflammatory, part of why obesity is a chronic inflammatory state.
- **Aromatase** — an enzyme, not a cytokine: it converts androgens to estrogens, making adipose tissue a peripheral estrogen source (the reason obesity raises estrogen levels).

Brown Adipose Tissue

Multilocular. Many small lipid droplets, a **central round nucleus**, and — the diagnostic feature — **numerous mitochondria with abundant cristae** containing **uncoupling protein 1 (UCP1, thermogenin)**, which short-circuits the proton gradient so energy is released as heat instead of ATP.

Found in the fetus and newborn (interscapular, axillary, perirenal, along the great vessels), much reduced in adults. Sympathetic innervation drives it. **"Beige" adipocytes** appear within white fat after cold exposure.

Feature	White	Brown
Lipid droplets	One (unilocular)	Many (multilocular)
Nucleus	Peripheral	Central

Feature	White	Brown
Mitochondria	Few	**Many, cristae-rich**
UCP1	Absent	**Present**
Main role	Energy storage, endocrine	**Non-shivering thermogenesis**
Sympathetic innervation	Sparse	Rich

CLINICAL CORRELATION

In **lipodystrophy** fat is absent or redistributed, and the metabolic consequences are paradoxical: severe insulin resistance with hypertriglyceridemia, because the remaining fat cannot store lipid safely. In **obesity** the same storage limit is approached, so lipid spills into liver (steatosis), muscle, and pancreas — **lipotoxicity**. Brown fat is a therapeutic target: activating it increases energy expenditure, and PET imaging shows active depots in adults.

UNDER THE MICROSCOPE

One big hole per cell with a flattened nucleus at the rim = white. Many small holes with a round central nucleus and granular eosinophilic cytoplasm = brown.

EXAM TIP

Do not confuse adipose tissue with **steatosis**. Adipocytes are a normal tissue with a capsule and vessels; steatosis is lipid accumulating inside hepatocytes or myocytes as vacuoles.

Chapter 6 — summary

- White adipocyte: **one large lipid droplet**, nucleus flattened at the rim, few mitochondria.
- Brown adipocyte: **many small droplets**, round central nucleus, many cristae-rich mitochondria.
- White fat = energy storage plus an **endocrine organ**: leptin, adiponectin, and other adipokines.
- Brown fat = **non-shivering thermogenesis** through **UCP1**, which uncouples the proton gradient.
- Brown fat is richest in the newborn (neck, back, mediastinum) and persists in adult neck and supraclavicular depots.
- Beige/brite adipocytes are an inducible, brown-like population within white fat.
- Adipose tissue is vascular and innervated; white fat's sympathetic supply is sparse, brown fat's is rich.
- Obesity is expansion by hypertrophy and hyperplasia; the storage limit produces lipotoxicity.
- Lipodystrophy = absent or redistributed fat with severe insulin resistance.
- Leptin signals energy stores to the hypothalamus; adiponectin increases insulin sensitivity.

Active recall

1. **One histological difference between white and brown fat?** — White: one large droplet, peripheral flattened nucleus. Brown: many small droplets, central round nucleus, eosinophilic granular cytoplasm.
2. **What is UCP1 and where does it act?** — Uncoupling protein 1 in the inner mitochondrial membrane; it lets protons leak back without ATP synthesis, releasing heat.
3. **Name two adipokines.** — Leptin (signals stored energy) and adiponectin (increases insulin sensitivity).
4. **When is brown fat most abundant?** — In the newborn (neck, back, mediastinum); it persists in adult neck and supraclavicular depots.
5. **What is lipotoxicity?** — When adipose storage capacity is exceeded, lipid accumulates in liver, muscle, and pancreas, impairing their function.

Exam-style question

A newborn's PET scan shows active supraclavicular fat that burns energy without shivering. Which feature explains this?

A. Unilocular droplets and few mitochondria
B. Multilocular droplets, abundant mitochondria, and UCP1
C. Dense collagen bundles and few cells
D. Abundant rough ER and zymogen granules

Answer: B — Brown adipocytes are built for heat production: many small droplets for rapid lipolysis, cristae-rich mitochondria, and UCP1.

Practical identification

Distinguish white from brown adipose tissue in a single section. White adipocytes are large, unilocular and appear empty after lipid extraction; brown adipocytes are smaller, multilocular, with central nuclei and eosinophilic cytoplasm packed with mitochondria. Identify the capillary network around brown adipocytes and explain why the two tissues look so different after routine processing, and what a frozen section would add.

7 · Cartilage

In this chapter: Hyaline Cartilage · Elastic Cartilage · Fibrocartilage · Cartilage Formation, Growth, & Repair

Learning objectives

By the end of this chapter you should be able to:

1. Distinguish hyaline, elastic, and fibrocartilage by matrix, cells, and perichondrium.
2. Explain why cartilage heals so poorly and what tissue replaces it.
3. Describe appositional and interstitial growth and when each is possible.
4. Identify cartilage types in a micrograph and relate each to its mechanical role.

The core idea

Cartilage is a **tough, resilient** connective tissue in which the extracellular matrix — collagen, proteoglycans, and water — is the bulk of the tissue; **chondrocytes** occupy only a small fraction of its volume, sitting in lacunae. **Aggrecan** binds a large amount of water, giving the matrix its compressive resilience. Cartilage supports soft tissues, notably in the respiratory tract, and provides **cushioned, low-friction surfaces in joints**. It is **avascular and aneural** — no blood vessels, lymphatics, or nerves — and is usually surrounded by a vascularized **perichondrium** of dense connective tissue whose fibroblasts form its outer fibrous layer. The absence of vessels explains both its durability under load and its poor healing.

The shared architecture

- Cells: **chondroblasts** (immature, at the surface) and **chondrocytes** (mature, in **lacunae**).
- Matrix: **territorial matrix** immediately around the lacuna (rich in proteoglycan, basophilic) and **interterritorial matrix** between cells (more collagen, eosinophilic).
- **Isogenous groups** — clusters of 2–8 chondrocytes from one parent cell, indicating recent **interstitial growth**.
- **Perichondrium** — fibrous outer layer plus chondrogenic inner layer; present on most cartilage, **absent on fibrocartilage and on articular surfaces**.

Hyaline Cartilage

The commonest type and the template for most of the fetal skeleton. Glassy, homogeneous matrix; **collagen makes up 40% of hyaline cartilage's dry weight**, all of it embedded in a hydrated gel of proteoglycans and structural glycoproteins, with more proteoglycan and less collagen immediately around the lacunae (the **territorial matrix**). Perichondrium is present except on **articular surfaces** and the epiphyses of growing long bones, where the cartilage itself forms the smooth, low-friction bearing surface of joints. Found in the **trachea, bronchi, nose, ribs, epiphyseal plate, and articular surfaces**.

Elastic Cartilage

Type II collagen plus a network of **elastic fibers**, giving flexibility with recoil. Found in the **epiglottis, pinna, external auditory canal, and Eustachian tube** — that is, everywhere something must bend and spring back.

Fibrocartilage

Type I and type II collagen in coarse bundles, with chondrocytes arranged in **parallel rows** between them. **No perichondrium.** Found where compressive and tensile forces meet: **intervertebral disc, symphysis pubis, menisci, temporomandibular joint**. The disc is the perfect example, with an inner **nucleus pulposus** (notochord remnant, rich in hyaluronic acid) and an outer **annulus fibrosus** of concentric fibrocartilage lamellae.

Feature	Hyaline	Elastic	Fibrocartilage
Collagen	**Type II**	Type II + **elastic fibers**	**Type I + II**
Cells	Isogenous groups	More numerous	**Rows** between bundles
Perichondrium	Present (not on articular surfaces)	Present	**Absent**
Locations	Trachea, bronchi, nose, ribs, epiphyseal plate, joints	Epiglottis, pinna, auditory canal	Disc, symphysis pubis, menisci
Function	Smooth bearing surface, template for bone	Flexible support	Shock absorption, tensile strength

Cartilage Formation, Growth, & Repair

- **Appositional growth** — new cartilage added at the periphery by chondroblasts of the perichondrium. Increases width.
- **Interstitial growth** — chondrocytes divide within lacunae and secrete matrix, pushing cells apart. Increases length; possible only in young cartilage.

Because cartilage is avascular, nutrition depends on diffusion from the perichondrium or synovial fluid. This limits repair: defects fill with **fibrocartilage** rather than hyaline cartilage, and that patch is mechanically inferior.

CLINICAL CORRELATION

Osteoarthritis is a failure of the chondrocyte to maintain matrix: proteoglycans are depleted, water content rises, collagen fibrils fragment, and the surface fissures. Because hyaline cartilage has no vessels and no nerves, pain comes late and from the underlying bone, capsule, and synovium. **Rheumatoid arthritis** is different: an autoimmune synovitis destroys cartilage from the surface inward, producing a pannus.

UNDER THE MICROSCOPE

Identify cartilage by the **lacuna**: a cell alone in a clear space within a solid matrix. Then the matrix: **glassy purple with cells in clusters = hyaline; wavy dark fibers = elastic; thick parallel pink bundles with cells in rows = fibrocartilage.**

EXAM TIP

Two commonly confused points: **hyaline cartilage contains type II collagen, not type I**, and **fibrocartilage has no perichondrium**, which is why disc injuries persist.

Chapter 7 — summary

- Cartilage = chondrocytes in lacunae + solid matrix; no blood vessels, no nerves.
- Hyaline cartilage = **type II collagen** + aggrecan; glassy; perichondrium present except at articular surfaces.
- Elastic cartilage = type II collagen + **elastic fibers**; perichondrium present; epiglottis, auricle, auditory tube.
- Fibrocartilage = **type I + type II collagen**, rows of chondrocytes, **no perichondrium**; discs, menisci, pubic symphysis.
- Territorial matrix surrounds each lacuna; interterritorial matrix lies between groups.

- Isogenous groups = daughter cells from recent division, still in one lacuna.
- Appositional growth = from perichondrium, adds width.
- Interstitial growth = division inside lacunae, adds length, young cartilage only.
- Articular cartilage is hyaline **without perichondrium** and heals poorly.
- Cartilage repair fills with fibrocartilage, not hyaline.
- Osteoarthritis = matrix failure; rheumatoid arthritis = synovial inflammation with pannus.
- Chondroblasts make matrix; chondrocytes maintain it.

Active recall

1. **Which cartilage has no perichondrium, and why does that matter?** — Fibrocartilage; without a perichondrial source of new chondroblasts, disc and meniscal injuries heal poorly.
2. **Which collagen types are in hyaline and fibrocartilage?** — Hyaline: type II. Fibrocartilage: type I and type II.
3. **What is an isogenous group?** — Daughter chondrocytes from a recent division, still sharing one lacuna.
4. **Appositional versus interstitial growth?** — Appositional adds cartilage at the periphery and increases width; interstitial growth divides chondrocytes inside lacunae, increases length, and occurs only in young cartilage.
5. **Why is articular cartilage a poor repairer?** — It is avascular, aneural, and lacks perichondrium, so nutrition depends on synovial fluid diffusion.

Exam-style question

A tissue shows rows of chondrocytes between thick parallel bundles of type I collagen and has no perichondrium. Which is it?

A. Hyaline cartilage
B. Elastic cartilage
C. Fibrocartilage
D. Decalcified bone

Answer: C — Rows of chondrocytes plus type I bundles and no perichondrium identify fibrocartilage (intervertebral disc, meniscus, pubic symphysis).

Practical identification

Compare hyaline, elastic and fibrocartilage in one session. Hyaline has a glassy matrix, chondrocytes in lacunae often in isogenous groups, and a perichondrium except at articular surfaces; elastic cartilage has visible elastic fibers in the matrix; fibrocartilage has thick collagen bundles and no perichondrium. Locate the territorial and interterritorial matrix, and explain why the staining differs between them.

8 · Bone

In this chapter: Bone Cells · Bone Matrix · Periosteum & Endosteum · Types of Bone · Osteogenesis · Bone Remodeling & Repair · Metabolic Role of Bone · Joints

Learning objectives

By the end of this chapter you should be able to:

1. Describe the cells and matrix of bone and how each contributes to strength and turnover.
2. Distinguish woven from lamellar bone, and compact from spongy bone.
3. Sequence endochondral and intramembranous ossification.
4. Explain remodeling, calcium homeostasis, and fracture repair.

The core idea

Bone is the main constituent of the adult skeleton: it gives **solid support**, protects vital organs of the cranial and thoracic cavities, encloses **medullary cavities** where blood cells form, and acts as a **reservoir of calcium and phosphate**, storing or releasing them to keep body fluid concentrations constant. It also forms a system of **levers** that multiply the force of muscle contraction into movement. Its stiffness comes from mineral, its toughness from collagen, and its capacity to adapt from four cell types that continuously demolish and rebuild it — and most skeletal disease reflects failure of one of those cells, or of the balance between them.

Bone Cells

Cell	Origin	Function	Marker / feature
Osteoprogenitor	Mesenchyme	Divide and differentiate into osteoblasts	Periosteum (inner layer), endosteum
Osteoblast	Osteoprogenitor (periosteum, endosteum)	Secrete **osteoid**, rich in **type I collagen** plus proteoglycans; mineralize it; regulate osteoclasts via **RANKL**	Cuboidal, basophilic, cover existing bone surfaces; **alkaline phosphatase** high
Osteocyte	Osteoblast that becomes embedded	Maintain matrix; sense mechanical load	In **lacunae**, connected by **canaliculi** via gap junctions
Osteoclast	**Monocyte/macrophage lineage**	**Resorb bone**	Large, multinucleated, **ruffled border**, resorption bay (**Howship lacuna**); **TRAP**, carbonic anhydrase II, H^+-ATPase

As osteoid calcifies and hardens it **entraps some osteoblasts**, which then differentiate into osteocytes sitting in **lacunae** between the lamellae, with cytoplasmic processes in **canaliculi**. Osteocytes maintain the calcified matrix and are fed from microvasculature in the **central canals of osteons** through those canaliculi.

The osteoclast is the exception. Every other bone cell is mesenchymal; the osteoclast is **monocyte-derived** and is the cell required for bone remodeling. It dissolves mineral by pumping H^+ and digests collagen with cathepsin K.

The RANK–RANKL–OPG axis: osteoblasts express **RANKL**, which binds **RANK** on osteoclast precursors and drives differentiation; they also secrete **osteoprotegerin (OPG)**, a decoy receptor that blocks RANKL. **PTH** shifts the balance toward resorption; **estrogen** suppresses it.

Bone Matrix

- **Inorganic (~50% of dry weight)** — calcium phosphate as **hydroxyapatite** crystals, with bicarbonate, citrate, magnesium, potassium, and sodium; compressive strength.
- **Organic (~50% of dry weight)** — **type I collagen** (90% of the organic matter), osteocalcin, osteonectin, osteopontin, proteoglycans, growth factors; tensile strength. Some references give a 65:35 ratio instead; both conventions are in current use, so state which one you are using.

Remove the mineral and bone becomes flexible; remove the collagen and it becomes brittle. **Bone is strong because it is both at once.**

Periosteum & Endosteum

Periosteum — dense connective tissue with a mainly **fibrous layer** over a more **cellular** layer; small vessels penetrate the matrix from it. Absent on articular surfaces and at tendon insertions. **Endosteum covers all trabeculae around the marrow cavities** — not merely the shaft. Both are sources of osteoprogenitor cells for growth and repair.

Types of Bone

Compact (cortical) bone — dense, on the surface; built of **osteons (Haversian systems)**: concentric lamellae around a central **Haversian canal** containing vessels and nerves, with **osteocytes** in lacunae linked by canaliculi, and **Volkmann canals** running transversely.

Spongy (cancellous) bone — a network of **trabeculae** with marrow between them, oriented along lines of mechanical stress. No true osteons.

Woven versus lamellar. Woven bone is disorganized and produced rapidly in the fetus, fracture callus, and disease (Paget, osteosarcoma). Lamellar bone is ordered and mechanically superior; adult bone is lamellar but always being remodelled through **basic multicellular units (BMUs)**.

Osteogenesis

Intramembranous ossification — mesenchyme condenses and differentiates directly into osteoblasts, depositing bone in a fibrous membrane. Produces the **flat bones of the skull, mandible, and clavicle**. No cartilage template.

Endochondral ossification — a hyaline cartilage model is replaced by bone. This is how **long bones** form and grow in length. The **epiphyseal plate** has **five zones**:

Zone (from epiphysis toward diaphysis)	What happens
1. Reserve (resting)	Small, inactive chondrocytes
2. Proliferative	Chondrocytes divide in columns — the actual growth in length
3. Hypertrophic	Chondrocytes enlarge and the matrix calcifies
4. Calcified cartilage	Chondrocytes die; matrix calcifies
5. Ossification	Osteoblasts deposit bone on the calcified spicules

Growth in **width** is **appositional**, by periosteal osteoblasts. The **primary ossification center** appears in the diaphysis; **secondary centers** in the epiphyses. When the plate ossifies, an **epiphyseal line** remains and length growth stops.

Bone Remodeling & Repair

Fracture → **hematoma** → inflammation → **soft callus** (fibrocartilage) → **hard callus** (woven bone) → **remodeling** to lamellar bone. Immobilization and blood supply decide the outcome. **Wolff's law** — bone is laid down where it is loaded and resorbed where it is not — explains both trabecular architecture and the bone loss of bed rest.

Metabolic Role of Bone

Bone is the body's calcium and phosphate reservoir.

Hormone	Source	Effect on bone	Net serum calcium
PTH	Parathyroid chief cells	Increases resorption, increases renal reabsorption, activates vitamin D	**Raises**
Calcitriol (1,25-(OH)$_2$D)	Kidney	Increases intestinal absorption and bone resorption	**Raises**
Calcitonin	Thyroid **C cells**	Inhibits osteoclasts	**Lowers**

Joints

Fibrous joints (synarthroses) — bones united by dense connective tissue, no cavity (sutures of the skull, gomphosis of a tooth).

Cartilaginous joints (amphiarthroses) — bones united by cartilage: **primary** by hyaline cartilage (epiphyseal plate, first sternocostal joint); **secondary** by fibrocartilage (intervertebral disc, symphysis pubis).

Synovial joints (diarthroses) — a joint cavity between **hyaline articular cartilage** on each bone, enclosed by a **capsule** of dense connective tissue lined by **synovial membrane**, which secretes **synovial fluid** for lubrication and nutrition of the avascular cartilage. Some joints add a **meniscus** (fibrocartilage), a **labrum**, or a **fat pad**. Because articular cartilage has no perichondrium and no vessels, it heals hardly at all — the anatomical basis of degenerative joint disease.

CLINICAL CORRELATION

Six to learn as a set. **Osteoporosis** — reduced bone mass with normal mineralization; estrogen deficiency drives it, trabeculae are lost first, fragility fractures follow. **Rickets/osteomalacia** — unmineralized osteoid; in children the growth plate widens and deforms (rachitic rosary, bowed legs). **Paget disease** — chaotic excessive remodeling with mosaic woven bone. **Osteopetrosis** — osteoclast failure, dense but brittle bone with a crowded marrow cavity. **Achondroplasia** — failure of chondrocyte proliferation in the plate, short limbs with normal trunk. **Osteogenesis imperfecta** — defective type I collagen, brittle bones, blue sclerae.

UNDER THE MICROSCOPE

On a ground section, **concentric lamellae around a central canal** = an osteon = compact bone. On a decalcified H&E section, osteoblasts are the row of cuboidal cells on the surface and osteocytes sit in lacunae. A **multinucleated cell in a scalloped pit** is an osteoclast resorbing.

MEMORY HOOK

"OsteoBlast Builds, OsteoClast Chews, OsteoCytes Communicate." For the epiphyseal plate, **"Rest, Proliferate, Hypertrophy, Calcify, Ossify"** — read from the joint outward.

EXAM TIP

The commonest error is saying osteoclasts are mesenchymal. They are **hematopoietic**, from the monocyte–macrophage line. The second is reversing the epiphyseal zones: growth happens at the **epiphyseal** side, ossification at the **diaphyseal** side.

Chapter 8 — summary

- Bone matrix is type I collagen plus hydroxyapatite; osteoid must mineralize before it is rigid.
- Osteoblasts make osteoid; osteocytes maintain matrix; osteoclasts resorb bone with a ruffled border.
- Osteoclasts are multinucleated and derive from the monocyte–macrophage lineage, not from osteoblasts.
- Compact bone is built of osteons; spongy bone is a trabecular lattice without osteons.

- Endochondral ossification replaces a cartilage model; intramembranous ossification forms directly in mesenchyme.
- The growth plate runs reserve → proliferative → hypertrophic → calcified → ossification zones.
- Osteogenesis imperfecta = type I collagen; achondroplasia = FGFR3; Marfan = fibrillin.
- PTH raises serum calcium; calcitonin lowers it; vitamin D increases absorption.
- Fracture repair: hematoma → fibrocartilaginous callus → bony callus → remodeling.

Active recall

1. **Which cell makes bone matrix, and which resorbs it?** — Osteoblasts make osteoid; osteoclasts resorb bone using a ruffled border and acid.
2. **What does osteoid contain, and what mineralizes it?** — Type I collagen and proteoglycans; hydroxyapatite crystals deposit along the collagen.
3. **How do osteocytes stay alive inside lacunae?** — Their processes run through canaliculi and contact processes of neighboring cells and the central canal, allowing diffusion of nutrients.
4. **Endochondral versus intramembranous ossification?** — Endochondral replaces a cartilage model (long bones, base of skull); intramembranous forms directly in mesenchyme (flat skull bones, mandible).
5. **What happens at a growth plate, zone by zone?** — Reserve → proliferative → hypertrophic → calcified matrix → ossification, with chondrocyte death and vascular invasion.

Exam-style question

A child has brittle bones, blue sclerae, and recurrent fractures. Which protein is defective?

A. Type II collagen
B. Type I collagen
C. Fibrillin
D. Dystrophin

Answer: B — Osteogenesis imperfecta is a type I collagen defect. Type II defects cause chondrodysplasias; fibrillin causes Marfan syndrome.

Practical identification

In a ground or decalcified section, find osteons (concentric lamellae around a central canal), interstitial and circumferential lamellae, and Volkmann canals. Identify osteocytes in lacunae with canaliculi, osteoblasts lining a surface, osteoclasts in Howship lacunae, and the endosteum and periosteum. In a developing long bone, follow the epiphyseal plate from resting cartilage to ossification and name the change at each zone.

9 · Nerve Tissue & the Nervous System

In this chapter: Development of Nerve Tissue · Neurons · Glial Cells & Neuronal Activity · Central Nervous System · Peripheral Nervous System · Neural Plasticity & Regeneration

Learning objectives

By the end of this chapter you should be able to:

1. Draw a neuron and label its parts, and classify neurons by shape and function.
2. Name the six glial cell types and give each one's job.
3. Explain how a synapse transmits a signal, and how myelin speeds conduction.
4. Compare regeneration in the peripheral and central nervous systems.

The core idea

Nervous tissue is built for **rapid signaling over distance**. The neuron is a polarized cell with a receiving end (dendrites and soma), a conducting cable (axon), and a transmitting end (synapse). Everything else — myelin, glia, barriers — exists to make that signaling faster, safer, and better supplied.

Development of Nerve Tissue

The nervous system arises from the **neural plate**, which folds into the **neural tube** (giving the CNS) while **neural crest** cells migrate away to form the peripheral neurons, Schwann cells, satellite cells, adrenal medulla, melanocytes, and much of the facial skeleton. This embryology explains three exam facts: the CNS is lined by **ependyma** derived from the tube; Schwann cells are neural crest; and the adrenal medulla is neural crest while the cortex is mesoderm.

Neurons

- **Perikaryon (soma)** — large, pale nucleus with a prominent nucleolus and abundant **Nissl bodies** (stacked RER and polysomes). Nissl substance is absent from the **axon hillock** — a useful landmark.
- **Dendrites** — short, branched, receptive; contain Nissl bodies.
- **Axon** — one per neuron, arises at the axon hillock, contains no Nissl bodies, specialized for conduction.
- **Axonal transport** — **kinesin** anterograde; **dynein** retrograde. Rabies, herpes, and tetanus toxin exploit retrograde transport.

Classification: multipolar (most CNS neurons and motor neurons), bipolar (retina, olfactory epithelium), pseudounipolar (sensory ganglia), anaxonic. Also by function — sensory, motor, interneuron.

The synapse. A chemical synapse has a presynaptic terminal with **synaptic vesicles** and mitochondria, a **synaptic cleft** (~20–30 nm), and a postsynaptic membrane with receptors. Transmission is unidirectional and modifiable. **Electrical synapses (gap junctions)** are fast and bidirectional.

Glial Cells & Neuronal Activity

Cell	Location	Function	Key point
Astrocyte	CNS	**The most numerous cell of the CNS**; hundreds of processes enclose neuronal perikarya, synapses, and capillaries, providing regulated microenvironments	**GFAP**; foot processes on capillaries; glial scar
Oligodendrocyte	CNS	Wraps flattened processes around portions of axons, forming **myelin sheaths**	One cell, many segments
Microglia	CNS	Resident phagocyte, immune surveillance	**Mesodermal** origin — the odd one out
Ependymal cell	CNS	Lines ventricles; cilia move CSF; choroid plexus	Produces CSF
Schwann cell (neurolemmocyte)	PNS	**Myelinates one segment of one axon**	Neural crest origin
Satellite cell	PNS ganglia	Supports sensory and autonomic neurons	—

There are **six major glial types**: oligodendrocytes, astrocytes, ependymal cells, and microglia in the CNS; Schwann cells and satellite cells in the PNS. **Myelination:** in the PNS a Schwann cell wraps one axon segment; in the CNS an oligodendrocyte wraps several. The gaps are **nodes of Ranvier**, where voltage-gated Na^+ channels cluster, so the impulse jumps by **saltatory conduction**. Velocity rises with axon diameter and myelin thickness.

Central Nervous System

- **Cerebral cortex** — six layers; **pyramidal cells** are the output neurons; gray matter outside.
- **Cerebellar cortex** — three layers: **molecular**, **Purkinje** (large flask-shaped cells, the only output), and **granular**.
- **Spinal cord** — gray matter in an H-shaped core, white matter outside; **anterior horn** motor, posterior horn sensory. This is the reverse of the cerebral cortex.
- **Meninges** — dura, arachnoid, pia; CSF in the subarachnoid space.
- **Choroid plexus** — produces CSF.
- **Blood–brain barrier** — tight junctions between endothelial cells, reinforced by astrocyte end-feet and pericytes. Absent in the circumventricular organs (area postrema, median eminence, pineal, posterior pituitary), which is how circulating hormones reach the brain.

Peripheral Nervous System

- **Sensory ganglia** — pseudounipolar neurons with satellite cells; **no synapses inside the ganglion**.
- **Autonomic ganglia** — multipolar neurons, synapses present.
- **Peripheral nerve** — axons in fascicles surrounded by **perineurium**, the whole nerve by **epineurium**, individual fibers by **endoneurium**.

Neural Plasticity & Regeneration

After axotomy the distal stump undergoes **Wallerian degeneration**; the soma shows **central chromatolysis**; Schwann cells form **bands of Büngner** to guide regrowth; regeneration proceeds at about **1 mm/day**. CNS axons regenerate poorly because of inhibitory molecules (Nogo, MAG) and glial scarring. **Plasticity** — the remodeling of synapses with use — is the basis of learning and of rehabilitation after injury.

CLINICAL CORRELATION

Multiple sclerosis is autoimmune demyelination in the CNS. **Guillain–Barré syndrome** is demyelination in the PNS, ascending from the legs. **Parkinson disease** is loss of dopaminergic neurons in the substantia nigra with Lewy bodies. **Alzheimer disease** shows neurofibrillary tangles and amyloid plaques. **Poliomyelitis** destroys anterior horn motor neurons and produces flaccid paralysis.

UNDER THE MICROSCOPE

Identify neurons by the **large pale nucleus, prominent nucleolus, and clumped Nissl bodies**. The axon hillock is the pale cone where Nissl substance stops. In peripheral nerve, look for **wavy bundles of axons with myelin rings** and connective tissue sheaths. In the cerebellar cortex, the **single row of enormous Purkinje cells** is unmistakable.

EXAM TIP

Oligodendrocyte = CNS, several axons. Schwann cell = PNS, one segment of one axon. And **microglia are the only glia of mesodermal origin.**

Chapter 9 — summary

- Neurons are classified by shape (multipolar, bipolar, pseudounipolar) and by function (sensory, motor, interneuron).
- Nissl bodies are rough ER plus polysomes; their position marks the soma and proximal dendrites.
- Oligodendrocytes myelinate the CNS (several axons each); Schwann cells myelinate the PNS (one segment each).
- Astrocytes form end-feet, contribute to the blood–brain barrier, and scar the CNS.
- Microglia are the resident macrophages, derived from monocytes.
- Ependymal cells line the ventricles and produce cerebrospinal fluid.
- Myelin allows saltatory conduction between nodes of Ranvier.
- The blood–brain barrier is sealed by tight junctions between capillary endothelial cells.
- PNS axons regenerate through bands of Büngner; CNS axons do not regenerate.

Active recall

1. **Which glial cell myelinates the CNS, and which myelinates the PNS?** — Oligodendrocyte (CNS, several axons); Schwann cell (PNS, one segment of one axon).
2. **Which glial cell is the macrophage of the CNS?** — Microglia, derived from monocytes.
3. **What forms the blood-brain barrier?** — Tight junctions between capillary endothelial cells, reinforced by astrocyte end-feet and pericytes.
4. **How is a signal carried along an axon?** — An action potential; myelin allows saltatory conduction between nodes of Ranvier.
5. **Why does CNS injury scar rather than regenerate?** — Oligodendrocytes and myelin inhibit axon growth, astrocytes form a glial scar, and there is no permissive sheath like the PNS bands of Büngner.

Exam-style question

A peripheral nerve is cut and surgically reapproximated. Which cell guides regrowing axons?

A. Oligodendrocyte
B. Schwann cell
C. Astrocyte
D. Microglial cell

Answer: B — Schwann cells dedifferentiate, form bands of Büngner, and provide a physical and trophic guide for the regenerating axon.

Practical identification

Identify a neuron by its large euchromatic nucleus, prominent nucleolus and Nissl bodies; find its dendrites and its single axon hillock. Distinguish astrocytes (star-shaped), oligodendrocytes (small, round, myelinating in the CNS), microglia (small, elongated, perivascular), ependymal cells (ciliated, lining ventricles), and, in peripheral nerve, Schwann cells and satellite cells. In a nerve trunk, find the myelin sheath, node of Ranvier, endoneurium, perineurium and epineurium.

10 · Muscle Tissue

In this chapter: Skeletal Muscle · Cardiac Muscle · Smooth Muscle · Regeneration of Muscle Tissue

Learning objectives

By the end of this chapter you should be able to:

1. Distinguish skeletal, cardiac, and smooth muscle by light microscopy.
2. Explain the sarcomere and the sliding-filament mechanism.
3. Compare excitation–contraction coupling in the three muscle types.
4. Predict which muscle regenerates and which is replaced by scar.

The core idea

Muscle tissue consists of cells that optimize the universal cell property of **contractility**. **Most muscle cells originate from embryonic mesoderm**; the intrinsic muscles of the iris and some head muscles are the standard exceptions, deriving from neural crest and other sources. Muscle differentiates by elongation and abundant synthesis of the myofibrillar proteins **actin and myosin**. All three types use the same actin–myosin interaction but arrange it differently: **striated and voluntary** (skeletal), **striated and involuntary** (cardiac), **unstriated and involuntary** (smooth).

Skeletal Muscle

Organization, large to small: muscle → fascicles → fibers → myofibrils → myofilaments. Each fiber is a long, cylindrical, **multinucleated** syncytium with **peripheral nuclei**, produced by fusion of myoblasts. Satellite cells persist beneath the basal lamina and mediate limited regeneration.

The sarcomere, bounded by two **Z lines**:

Band	Contents	Behavior in contraction
A band	Thick filaments (**myosin**)	Constant length
I band	Thin filaments (**actin**) only	Shortens
H band	Thick filaments only, no overlap	Shortens
Z line	Anchor for actin	Moves closer
M line	Center of the A band	—

Sliding filament mechanism: myosin heads bind actin, hydrolyze ATP, and pivot, pulling thin filaments toward the M line. **The A band never changes length.**

Control. The **sarcoplasmic reticulum** stores Ca^{2+}; **T-tubules** are invaginations of the sarcolemma that encircle each myofibril **near the aligned A- and I-band boundaries**, where each T-tubule associates with **two terminal cisternae** — a **triad**. Depolarization spreads along the T-tubules to trigger simultaneous Ca^{2+} release throughout the fiber. An action potential releases Ca^{2+}, which binds **troponin C**, moves **tropomyosin** off the myosin-binding site, and contraction begins. Relaxation requires Ca^{2+} reuptake.

Innervation. The **motor end plate** — acetylcholine binds nicotinic receptors, depolarizes the sarcolemma, and triggers contraction. One motor neuron plus all the fibers it supplies is a **motor unit**.

Fiber types:

Type	Speed	Metabolism	Fatigue	Color
I (slow, red)	Slow	**Oxidative**	Resistant	Red
IIa (fast, oxidative-glycolytic)	Fast	Mixed	Intermediate	Intermediate

Type	Speed	Metabolism	Fatigue	Color
IIx/IIb (fast, glycolytic)	Fast	**Glycolytic**	Fast fatigue	White

Sensory receptors. Muscle spindles (intrafusal fibers, Ia afferents, gamma motor neurons) sense length and stretch; **Golgi tendon organs** sense tension.

Cardiac Muscle

- **Striated**, branched fibers, **one or two central nuclei**.
- **Intercalated discs** — the diagnostic feature — contain **fascia adherens** (anchors actin), **desmosomes** (mechanical strength), and **gap junctions** (electrical coupling), making the heart a functional syncytium.
- **Diads** at the **Z line** (one T-tubule + one cisterna), rather than the **triads** at the A–I junction of skeletal muscle.
- Abundant mitochondria and lipid droplets; highly oxidative; resistant to fatigue.
- **Purkinje fibers** — specialized conducting cells, larger and paler, with fewer myofibrils; they conduct rapidly through the ventricles.
- **Regeneration is essentially absent**; damage is repaired by fibrous scar.

Smooth Muscle

- **Spindle-shaped** cells with a single **central** nucleus and no striations.
- Contractile apparatus: **dense bodies** (analogous to Z lines) linked by desmin and actin.
- **No troponin.** Calcium binds **calmodulin**, which activates **myosin light-chain kinase (MLCK)**.
- **Caveolae** replace T-tubules; the SR is sparse and Ca^{2+} entry across the membrane matters more.
- **Unitary (single-unit)** smooth muscle — gap junctions, spontaneous rhythmicity, in gut, uterus, bladder. **Multiunit** — each cell separately innervated, in the iris and vas deferens.
- Capable of **hypertrophy and hyperplasia** (uterine myometrium in pregnancy).

Regeneration of Muscle Tissue

Skeletal muscle regenerates weakly, from **satellite cells**, and only if the basal lamina survives; otherwise the defect fills with fibrous tissue. **Cardiac muscle does not regenerate** — infarcted myocardium is replaced by scar. **Smooth muscle** can both hypertrophy and, in the uterus, proliferate.

Feature	Skeletal	Cardiac	Smooth
Striation	**Yes**	**Yes**	No
Nuclei	**Many, peripheral**	1–2, **central**	1, **central**
Cell shape	Long cylinder	Branched	Spindle
T-tubules	A–I junction, **triads**	Z line, **diads**	None (caveolae)
Ca^{2+} sensor	**Troponin C**	**Troponin C**	**Calmodulin**
Junctions	None between fibers	**Intercalated discs**	Gap junctions (unitary)
Control	Somatic, voluntary	Autonomic, involuntary	Autonomic, involuntary
Regeneration	Satellite cells (limited)	None	Hypertrophy/hyperplasia

CLINICAL CORRELATION

Duchenne muscular dystrophy is an X-linked defect of **dystrophin**, which links the sarcolemma to the extracellular matrix; without it, contraction tears the membrane, fibers die, and fat and fibrous tissue replace muscle (pseudohypertrophy). **Myasthenia gravis** is autoantibody blockade of the nicotinic acetylcholine receptor, producing fatigable weakness. **Cardiac hypertrophy** is an adaptation to pressure overload; beyond a point the myocytes outgrow their blood supply and fail.

UNDER THE MICROSCOPE

Look first for **striations**; if present, look at the **nuclei**. **Many nuclei at the edge = skeletal. One or two central nuclei with dark transverse bands = cardiac (intercalated discs).** No striations and central nuclei in spindle-shaped cells = smooth.

MEMORY HOOK

"Skeletal = Surface nuclei. Cardiac = Central nucleus, Cross-striations, interCalated discs. Smooth = Spindle, Single central nucleus, no Striations."

EXAM TIP

The A band does not shorten. This is the muscle fact to keep straight: I and H shorten, A stays constant, Z lines approach.

Chapter 10 — summary

- Skeletal muscle: long cylinders, **many peripheral nuclei**, triads at the **A–I junction**.
- Cardiac muscle: branched cells, **one or two central nuclei**, **intercalated discs**, diads at the **Z line**.
- Smooth muscle: spindle cells, **one central nucleus**, no striations, no T-tubules (caveolae).
- Sarcomere = Z line to Z line; **A band constant**, I band and H zone shorten.
- Thin filament = actin + troponin + tropomyosin; thick filament = myosin.
- Ca^{2+} binds **troponin C** in striated muscle and **calmodulin** in smooth muscle.
- Smooth muscle contraction uses **myosin light-chain kinase**; it has no troponin.
- Intercalated discs contain **fasciae adherentes, desmosomes, and gap junctions** — not tight junctions.
- Skeletal muscle regenerates from **satellite cells**, and only if the basal lamina survives.
- Cardiac muscle does not regenerate; infarction heals by scar.
- Type I fibers = slow, oxidative, fatigue-resistant. Type II = fast, glycolytic.
- Duchenne dystrophy = **dystrophin** defect; myasthenia gravis = antibody to the nicotinic receptor.
- Muscle spindle = stretch receptor; Golgi tendon organ = tension receptor.

Active recall

1. **Which band does not change length during contraction?** — The A band, because it is the length of the thick filaments.
2. **What hides the myosin-binding site at rest, and what moves it?** — Tropomyosin; calcium binds troponin C and shifts tropomyosin.
3. **Cardiac versus skeletal muscle at the T-tubule level?** — Cardiac diads lie at the Z line; skeletal triads lie at the A–I junction.
4. **What is the calcium sensor in smooth muscle?** — Calmodulin, which activates myosin light-chain kinase; smooth muscle has no troponin.
5. **Which muscle types regenerate?** — Skeletal (limited, from satellite cells), smooth (hyperplasia/hypertrophy); cardiac does not.

Exam-style question

A muscle biopsy shows branched fibers with one central nucleus and dark transverse bands. What is the diagnosis?

A. Skeletal muscle
B. Cardiac muscle
C. Smooth muscle
D. Dense regular connective tissue

Answer: B — Branching, central nuclei, and intercalated discs are the triad of cardiac muscle.

Practical identification

Compare the three muscle types at the same magnification. Skeletal fibers are long, cylindrical and multinucleated with peripheral nuclei and striations; cardiac fibers branch, have central nuclei, intercalated discs and perinuclear halos; smooth muscle cells are spindle-shaped with central cigar-shaped nuclei and no striations. In skeletal muscle, identify the A band, I band, Z line and triad; in cardiac muscle, the diad at the Z line.

PART III — TRANSPORT, DEFENSE, AND BLOOD

Chapters 11–14 · circulation, blood, hemopoiesis, and immunity

11 · The Circulatory System

In this chapter: Heart · Tissues of the Vascular Wall · Vasculature · Lymphatic Vascular System

Learning objectives

By the end of this chapter you should be able to:

1. Name the three tunics of the heart and of a blood vessel, and describe what each contains.
2. Distinguish elastic artery, muscular artery, arteriole, capillary, venule, and vein in section.
3. Match each capillary type to the organ that needs it.
4. Trace the lymphatic system from blind-ended capillary to thoracic duct.

The core idea

The circulatory system pumps and directs blood cells and substances to all tissues. In an adult the capillaries alone have a total length of **more than 100,000 km** and a surface area of about **5,000 m²**. Almost every vessel is a variation on one three-layered tube. What changes between an aorta and a capillary is **how much muscle and elastic tissue each layer contains** — decided entirely by the pressure the vessel must withstand and the job it must do.

Heart

- **Endocardium** — endothelium + subendothelial connective tissue + a layer of Purkinje fibers; continuous with the intima of vessels.
- **Myocardium** — the thickest layer; cardiac muscle in complex spirals, atria thinner than ventricles, left ventricle thickest.
- **Epicardium** — connective tissue containing many **adipocytes**, covered by mesothelium; the visceral layer of the pericardium.
- **Cardiac skeleton** — dense irregular connective tissue rings that anchor the valves and electrically insulate atria from ventricles.
- **Valves** — endocardium folded over a core of dense connective tissue; **avascular**. The mitral and tricuspid have chordae tendineae and papillary muscles.

Tissues of the Vascular Wall

Layer	Contents	Function
Intima	Endothelium, basal lamina, subendothelial connective tissue, **internal elastic lamina** (in arteries)	Non-thrombogenic surface, exchange, barrier
Media	**Smooth muscle** circularly arranged, elastic lamellae (in elastic arteries), **external elastic lamina**	Tone and compliance
Adventitia	Loose connective tissue, **vasa vasorum**, **nervi vascularis**	Anchorage and supply of the wall

The rule of the wall: the thicker the media, the higher the pressure the vessel handles. Arteries have a thick media; veins a thin media and a thick adventitia. That single ratio lets you identify a vessel on any slide.

Vasculature

Vessel	Intima	Media	Adventitia	Distinguishing feature
Elastic artery (aorta, pulmonary, common carotid)	Endothelium + subendothelial	**Many elastic lamellae** alternating with smooth muscle	Thin, with vasa vasorum	**Elastic lamellae**
Muscular artery	Prominent **internal elastic lamina**	Thick, 3–40 layers of smooth muscle	Moderate	**Well-defined IEL**
Arteriole	Endothelium + thin subendothelium	1–2 layers of smooth muscle	Minimal	**The resistance vessel**
Capillary	Endothelium + basal lamina only	None	None	One cell thick
Venule	Endothelium	None or sparse pericytes	Thin	Site of **leukocyte diapedesis**
Small/medium vein	Endothelium	Thin	**Thickest layer**	Often collapsed on section
Large vein (vena cava)	Endothelium	Thin	**Thick, with longitudinal smooth muscle**	Valve-like intimal folds

Capillaries:

Type	Structure	Location	Function
Continuous	Uninterrupted endothelium, tight junctions	Muscle, skin, lung, CNS	Regulated exchange; the **blood–brain barrier** is this type plus tight junctions and astrocyte feet
Fenestrated	Pores with diaphragms	Kidney glomerulus, intestine, endocrine glands	Rapid fluid and solute exchange
Sinusoidal	Discontinuous, large gaps, incomplete basal lamina	**Liver, spleen, bone marrow**	Free passage of cells and large molecules

Microcirculation. Arteriole → metarteriole → capillary bed → venule, with **precapillary sphincters** regulating flow. **Arteriovenous anastomoses** shunt blood directly, important in skin thermoregulation.

Lymphatic Vascular System

Begins as blind-ended **lymphatic capillaries** with overlapping endothelial flaps and **anchoring filaments**, which open when interstitial pressure rises. They converge into collecting vessels with valves, then nodes, trunks, and the **thoracic duct** and **right lymphatic duct**. Lymphatics have **no continuous basal lamina** and no fenestrations — which is exactly why tumor cells and bacteria enter them so easily.

CLINICAL CORRELATION

Atherosclerosis begins with endothelial dysfunction: LDL crosses the endothelium, is oxidized, and provokes monocyte recruitment. Macrophages become **foam cells**, a fatty streak forms, smooth muscle migrates into the intima, and a fibrous cap covers a necrotic core. Rupture of a thin cap exposes thrombogenic material and produces the acute coronary syndrome. **Aneurysm** follows destruction of medial elastic fibers. **Varicose veins** follow valve failure. **Lymphedema** follows lymphatic obstruction — classically after axillary node clearance.

UNDER THE MICROSCOPE

Arteries: **round lumen, thick wall, corrugated internal elastic lamina**. Veins: **irregular collapsed lumen, thin media**. In elastic arteries the media shows wavy pink bands with no orderly muscle coat; in muscular arteries the media is uniform smooth muscle.

EXAM TIP

The three capillaries are distinguished by **continuity, not size**: continuous (tight), fenestrated (pores), sinusoidal (gaps wide enough for a red cell). Liver, spleen, marrow = sinusoidal. Kidney glomerulus and intestine = fenestrated. Brain = continuous with tight junctions.

Chapter 11 — summary

- Three tunics: intima (endothelium + subendothelial tissue), media (smooth muscle, elastic lamellae), adventitia (collagen).
- Elastic arteries have many concentric elastic lamellae; muscular arteries have a prominent internal elastic lamina.
- Arterioles have one or two smooth-muscle layers and control peripheral resistance.
- Continuous capillaries are the default; fenestrated ones serve endocrine glands, gut, and kidney; sinusoids serve liver, spleen, and marrow.
- Veins have thinner walls, larger lumens, and valves in medium-sized veins.
- The heart has endocardium, myocardium, and epicardium; Purkinje fibers conduct the impulse.
- Lymphatic capillaries are blind-ended and more permeable than blood capillaries; they drain to the thoracic duct.
- The thoracic duct empties into the left subclavian vein.

Active recall

1. **Three layers of a vessel wall, outside in?** — Tunica adventitia, tunica media, tunica intima.
2. **Which vessel has a media of only one or two smooth-muscle layers and controls peripheral resistance?** — The arteriole.
3. **Which capillary type is fenestrated, and where is it found?** — Fenestrated capillaries are in endocrine glands, gut mucosa, and kidney; they allow rapid exchange.
4. **What distinguishes a vein from an artery in section?** — Veins have a larger, often collapsed lumen and a thinner wall with less muscle; valves are present in medium veins.
5. **Where does lymph begin and end?** — Blind-ended lymphatic capillaries in tissue; it drains through nodes to the thoracic duct and into the subclavian vein.

Exam-style question

A section shows a vessel with a thick media containing many elastic lamellae and a wide lumen. Which vessel is it?

A. Large vein
B. Elastic artery
C. Arteriole
D. Capillary

Answer: B — Numerous concentric elastic lamellae in a thick media define an elastic artery (aorta, pulmonary artery).

Practical identification

Compare artery, vein and capillary in one section. Name the three tunics, and judge the artery by a thick media with concentric smooth-muscle layers and an internal elastic lamina; judge the vein by a thin media, a wide lumen and a thick adventitia. Distinguish elastic from muscular arteries by the amount of elastin in the media. In a capillary bed, classify the endothelium as continuous, fenestrated or sinusoidal and relate that to the organ.

12 · Blood

In this chapter: Composition of Plasma · Blood Cells

Learning objectives

By the end of this chapter you should be able to:

1. Describe plasma and explain how serum differs from it.
2. Relate erythrocyte shape, size, and lifespan to gas transport.
3. Identify each leukocyte by nucleus and granules, and state its function.
4. Explain platelet structure and the steps of hemostasis.

The core idea

Blood is connective tissue with a liquid matrix. Its cells all derive from one stem cell, and their morphology is a direct readout of their job: **red for oxygen, granular white for killing, agranular white for immunity, and fragments for clotting.**

Composition of Plasma

About **55% of blood volume**. Water plus plasma proteins — **albumin, α- and β-globulins, complement proteins, and fibrinogen**, all secreted in the **liver**, plus the **immunoglobulins** — and electrolytes, nutrients, hormones, and waste.

The composition of plasma is summarized in the following table:

Component	Share	Job
Water	~92% of plasma	Solvent in which formed elements are suspended
Plasma proteins	~7% of plasma	Buffer pH; maintain osmotic pressure
— **Albumin**	~58% of plasma protein	Osmotic pull (oncotic pressure); carries fatty acids, hormones, drugs
— **Globulins**	~37% of plasma protein	α- and β-globulins transport lipids, metals, and iron; γ-globulins are antibodies
— **Fibrinogen**	~4% of plasma protein	Precursor of fibrin for clotting
— **Regulatory proteins**	**>1%** of plasma protein	Enzymes, proenzymes, hormones, complement
Other solutes (electrolytes and nutrients)	**~1%** of plasma	Electrolytes maintain membrane potentials, pH, and osmotic control; nutrients (glucose, amino acids, cholesterol, vitamins) supply energy and building blocks

On clotting, some proteins are consumed and others are released from platelets, leaving **serum**, whose biological properties differ from plasma. **Serum is plasma minus fibrinogen** (i.e., after clotting).

Blood Cells

Erythrocytes

- **Biconcave disc**, **7.5 µm** in diameter, **2.6 µm** thick at the rim but only **0.75 µm** at the center — flexible enough to pass small capillaries; **no nucleus, no organelles**, so the cell is packed with hemoglobin.
- Lifespan about **120 days**; removed by splenic macrophages; the globin is recycled and the heme converted to bilirubin.
- Shape maintained by **spectrin and ankyrin**; defects produce hereditary spherocytosis.

- **Hemoglobin** — four globin chains, each with a heme group carrying Fe^{2+}. Adult HbA ($\alpha_2\beta_2$) and fetal HbF ($\alpha_2\gamma_2$, higher affinity for O_2).
- **Reticulocytes** are young red cells with residual RNA, appearing polychromatic; a raised count signals active marrow output.
- **Hematocrit** — the sedimented erythrocyte column — is about **44% of total blood volume** in healthy adults; the thin layer between cells and plasma is the **buffy coat** of leukocytes and platelets. An average adult has **5–6 L** of blood.

The leukocytes — broadly grouped as **granulocytes** (neutrophils, eosinophils, basophils) and **agranulocytes** (lymphocytes, monocytes). The granulocytes are **12–15 µm**, with eosinophils about the same size as neutrophils or slightly larger. Lymphocytes range from **6 to 15 µm** — small, medium, and large forms — and monocytes are **12–20 µm**, the largest of the leukocytes in a smear. (Estimates differ: 12–15 µm in some descriptions, 12–20 µm in measurements made on smears. Both are recorded here rather than one being silently preferred.)

Cell	% of WBC	Size	Nucleus	Cytoplasm	Main function
Neutrophil	**50–70%**	**12–15 µm**	**3–5 lobes**, dense	Pale; fine granules	**Phagocytosis of bacteria**; the acute responder
Eosinophil	**1–4%**	**12–15 µm** (neutrophil-sized or slightly larger)	**Bilobed**	**Bright orange-red** granules	**Parasites and allergy**; major basic protein
Basophil	**0.5–1%**	**12–15 µm**	**S-shaped**, obscured	**Large dark blue** granules	**Histamine and heparin**
Lymphocyte	**20–40%**	**6–15 µm** (small, medium, large)	Round, dense	Pale rim	**T cells**, **B cells**, **NK cells**
Monocyte	**2–8%**	**12–20 µm** (largest; some descriptions give 12–15 µm)	**Kidney-shaped**	Abundant blue-gray	Become **macrophages**

Neutrophil granules: azurophilic (primary) granules contain myeloperoxidase and defensins; **specific (secondary)** granules contain lysozyme and lactoferrin. The neutrophil dies after phagocytosis; the accumulation of dead neutrophils and debris is **pus**. Each **phagocyte** kills ingested bacteria inside a phagolysosome.

Platelets are **2–4 µm** anucleate fragments of **megakaryocyte** cytoplasm, with a peripheral clear zone (hyalomere) and a central granular zone (granulomere). They number **150,000–400,000/µL** and circulate for about **10 days**. They adhere, activate, and aggregate to form the primary hemostatic plug; **von Willebrand factor** links them to collagen.

Hemostasis in four steps (the standard physiology sequence; this chapter describes only the platelet plug and clotting): (1) **Vascular spasm** — the injured vessel constricts, slowing flow. (2) **Primary hemostasis** — platelets bind exposed collagen (directly and through von Willebrand factor), degranulate, and aggregate into a soft plug. (3) **Coagulation** — exposed tissue factor and platelet phospholipid trigger the clotting cascade, converting prothrombin to **thrombin**, which cleaves fibrinogen into **fibrin**; the fibrin mesh stabilizes the plug as a clot. (4) **Retraction and repair** — the clot contracts, and endothelial cells and fibroblasts rebuild the wall.

CLINICAL CORRELATION

The differential count is a diagnostic instrument. **Neutrophilia** suggests bacterial infection or steroid use; **eosinophilia** parasites or allergy; **lymphocytosis** viral infection or chronic lymphocytic leukemia; **pancytopenia** marrow failure. In **sickle cell disease** a single glutamic acid → valine substitution in β-globin allows deoxygenated HbS to polymerize; the cell sickles, occludes small vessels, and is removed by the spleen.

UNDER THE MICROSCOPE

Count lobes, not granules. **Neutrophil: 3–5 lobes. Eosinophil: exactly two, orange granules. Basophil: granules so dark the nucleus is hidden. Monocyte: kidney-shaped nucleus, largest cell. Lymphocyte: round dense nucleus, almost no cytoplasm.**

MEMORY HOOK

"Never Let Monkeys Eat Bananas" = **N**eutrophil, **L**ymphocyte, **M**onocyte, **E**osinophil, **B**asophil — in descending order of abundance.

EXAM TIP

Eosinophil granules stain with eosin because they are **basic proteins** — the naming is the reverse of what students expect. The basophil's granules contain **heparin and histamine**, so the "basic" cell is the one full of allergy mediators.

Chapter 12 — summary

- Plasma is the fluid phase; serum is plasma after clotting and lacks fibrinogen.
- Erythrocytes are biconcave, 7.5 µm, anucleate, and live about 120 days.
- Hemoglobin is the gas carrier; the biconcave shape maximizes surface-to-volume ratio.
- Neutrophils are the most numerous leukocyte and the first responder to acute inflammation.
- Eosinophils have bilobed nuclei and orange-red granules (major basic protein); they attack parasites.
- Basophils carry histamine and heparin and mediate immediate hypersensitivity.
- Lymphocytes are the second most numerous; monocytes are the largest and become macrophages.
- Platelets are anucleate fragments of megakaryocytes and form the primary hemostatic plug.
- Hemostasis = vascular spasm → platelet plug → coagulation → clot retraction and repair.

Active recall

1. **How does serum differ from plasma?** — Serum is plasma after clotting, so it lacks fibrinogen and consumed clotting factors.
2. **Normal erythrocyte dimensions and lifespan?** — About 7.5 µm diameter, biconcave with a 2.6-µm rim, circulating about 120 days.
3. **Which leukocyte is most numerous, and which is largest?** — Neutrophils (~50–70%); monocytes are the largest.
4. **What do eosinophil granules contain?** — Major basic protein and eosinophil cationic protein, active against parasites and in allergy.
5. **What are the four steps of hemostasis?** — Vascular spasm, primary hemostasis (platelet plug), coagulation (fibrin clot), and clot retraction with repair.

Exam-style question

A blood smear shows a cell with a bilobed nucleus and bright orange-red granules. What is it?

A. Neutrophil
B. Eosinophil
C. Basophil
D. Monocyte

Answer: B — Bilobed nucleus plus large eosinophilic granules is the eosinophil.

Practical identification

On a stained blood smear, identify erythrocytes (biconcave, anucleate, pink), platelets (small purple fragments) and each leukocyte: neutrophil (segmented nucleus, fine granules), eosinophil (bilobed, orange-red granules), basophil (dark granules obscuring the nucleus), lymphocyte (round dense nucleus, thin cytoplasm) and monocyte (kidney-shaped nucleus, abundant blue-gray cytoplasm). Perform a differential count and explain what each cell does.

13 · Hemopoiesis

In this chapter: Stem Cells, Growth Factors, & Differentiation · Bone Marrow · Maturation of Erythrocytes · Maturation of Granulocytes · Maturation of Agranulocytes · Origin of Platelets

Learning objectives

By the end of this chapter you should be able to:

1. Trace the shift of hemopoietic sites from embryo to adult.
2. Describe the marrow niche and the stromal cells that support it.
3. Follow each lineage from stem cell to mature cell, naming the recognizable stages.
4. Match growth factors and colony-stimulating factors to the lineages they drive.

The core idea

All blood cells descend from one **hematopoietic stem cell**, and the pathway from that cell to a mature erythrocyte, granulocyte, or platelet is a sequence of named stages you can recognize by **decreasing nucleus, increasing cytoplasm, and changing color**. It is the lineage with the most named stages in histology, so it rewards systematic learning.

Sites of hemopoiesis

Period	Site
First trimester	Yolk sac (mesoblastic phase)
Second trimester	**Liver and spleen** (hepatic phase)
Third trimester	**Bone marrow** — marrow becomes the major organ in the third trimester as skeletal elements ossify

In the adult, active **red marrow** persists in the **sternum, ribs, vertebrae, pelvis, skull, and proximal femur**. The rest is **yellow (fatty) marrow**, which can revert to red marrow under demand — why chronic anemia produces marrow expansion and frontal bossing.

Stem Cells, Growth Factors, & Differentiation

The **hematopoietic stem cell** is $CD34^+$, self-renewing, and pluripotent, sitting in a **niche** of stromal cells, other cells, and ECM. Committed **progenitor cells** — also called **CFUs** (colony-forming units) — proliferate and differentiate under growth factors, which are also called **CSFs** (colony-stimulating factors) or cytokines. Differentiation is driven by **growth factors**, listed here by name:

Factor	Target
Erythropoietin (EPO)	Erythroid progenitors (kidney, some liver)
Thrombopoietin (TPO)	Megakaryocytes (liver)
G-CSF	Granulocyte production
GM-CSF, M-CSF	Granulocyte–monocyte and monocyte lineages
IL-3	Early multipotent progenitors
IL-5	Eosinophils
SCF (stem cell factor)	Stem cell maintenance

Bone Marrow

Between the trabeculae lie **hemopoietic cords** and **sinusoids** with a discontinuous endothelium, so mature cells can enter the circulation. The **myeloid:erythroid (M:E) ratio** is normally **2:1 to 4:1**; it shifts toward erythroid in hemolytic anemia and toward myeloid in infection or leukemia.

Maturation of Erythrocytes

Proerythroblast → basophilic → polychromatophilic → orthochromatophilic erythroblast → reticulocyte → erythrocyte, developing in **erythropoietic islands or cords**. The names track the cytoplasm's transition from **RNA-rich to hemoglobin-filled**; at the last stage the nucleus is **extruded**, producing a **reticulocyte** that still contains some polyribosomes when released into the circulation.

The cell **shrinks**, the nucleus **condenses and is extruded**, hemoglobin accumulates so the cytoplasm shifts from blue to pink, and the nucleolus disappears. The sequence takes about **a week**. EPO drives it; iron, vitamin B_{12}, and folate are required.

Maturation of Granulocytes

Myeloblast → promyelocyte → myelocyte → metamyelocyte → band cell → segmented neutrophil (or eosinophil/basophil).

The suffixes carry the meaning: **-blast** = dividing precursor (**myeloblast**, **lymphoblast**, **proerythroblast**), **-cyte** = the later, maturing cell (**myelocyte**, **lymphocyte**, **reticulocyte**); the suffix does not by itself mean post-mitotic, because **myelocytes still divide**; **thrombocyte** is another name for the platelet.

Primary (azurophilic) granules appear at the promyelocyte stage, **specific granules** at the myelocyte stage, the nucleus indents at the metamyelocyte stage, and becomes a **C-shaped band** before segmenting. A **left shift** — more band forms and younger cells in the peripheral blood — indicates accelerated production: immature neutrophils, chiefly **band (stab) cells**, are released prematurely when the circulating neutrophil pool is depleted by bacterial infection.

Maturation of Agranulocytes

Lymphopoiesis begins in the marrow for B cells and in the thymus for T cells. **Monopoiesis** produces monocytes that become tissue macrophages. Both pass through stages in which the nucleus becomes less lobated, the cytoplasm more abundant, and the chromatin pattern characteristic of the mature cell.

Origin of Platelets

Megakaryoblasts → megakaryocytes, which become polyploid and develop **demarcation membranes** that delineate future platelet territories; platelets bud from the cytoplasm under TPO control.

CLINICAL CORRELATION

Aplastic anemia is marrow failure with an empty, fatty marrow and pancytopenia. **Leukemia** is the opposite: marrow packed with malignant blasts, crowding out normal lineages and producing anemia, infection, and bleeding. **Megaloblastic anemia** (B_{12} or folate deficiency) shows nuclear–cytoplasmic asynchrony — the nucleus matures more slowly than the cytoplasm, so cells are large and hypersegmented. **Iron deficiency** produces microcytic hypochromic cells because hemoglobin synthesis lags behind cell division.

UNDER THE MICROSCOPE

Identify the lineage by **nucleus-to-cytoplasm ratio and granule color**. Early cells have a large nucleus, a thin rim of basophilic cytoplasm, and a nucleolus. As cells mature, the ratio falls, granules appear, and the nucleus condenses. **Maturation moves in one direction: nucleus shrinks, cytoplasm grows, color shifts from blue to pink or from blue to granular.**

EXAM TIP
Two facts that are easy to invert: **EPO comes from the kidney** (peritubular interstitial cells), and **TPO comes mostly from the liver**. Students routinely invert these.

Chapter 13 — summary

- Hemopoiesis shifts from yolk sac → liver and spleen → bone marrow; the adult site is red marrow of the axial skeleton.
- The marrow stroma (reticular cells, macrophages, adipocytes, endothelial cells) forms the hematopoietic niche.
- Erythropoiesis: proerythroblast → basophilic → polychromatophilic → orthochromatophilic normoblast → reticulocyte → erythrocyte.
- The nucleus is extruded at the orthochromatophilic erythroblast stage, producing the reticulocyte, which still contains residual RNA.
- Granulopoiesis: myeloblast → promyelocyte → myelocyte → metamyelocyte → band → segmented cell.
- Specific granules appear at the myelocyte stage; azurophilic granules at the promyelocyte stage.
- Megakaryocytes are polyploid and shed platelets from their cytoplasm.
- Erythropoietin from the kidney drives erythroid production; G-CSF drives granulocytes.
- The normal myeloid-to-erythroid ratio is about 2:1 to 4:1.

Active recall

1. **Where does hemopoiesis occur in the adult?** — Red marrow of the axial skeleton (vertebrae, ribs, sternum, pelvis, proximal femur).
2. **What is the first recognizable erythroid precursor, and when is the nucleus lost?** — The proerythroblast; the nucleus is extruded at the orthochromatophilic erythroblast stage, leaving a reticulocyte that still contains residual RNA.
3. **How do you recognize a megakaryocyte?** — A huge polyploid cell with a lobulated nucleus whose cytoplasm fragments into platelets.
4. **Which growth factor drives erythroid production?** — Erythropoietin, made by peritubular interstitial cells of the kidney.
5. **What does the M:E ratio describe?** — The ratio of myeloid (granulocytic) to erythroid precursors in marrow, normally about 2:1 to 4:1.

Exam-style question

A bone marrow smear shows a cell with a large, lobulated nucleus and abundant granular cytoplasm producing platelets. What is it?

A. Megakaryocyte
B. Osteoclast
C. Macrophage
D. Proerythroblast

Answer: A — The megakaryocyte is the polyploid platelet factory; osteoclasts are multinucleated but resorb bone, not make platelets.

Practical identification

In a marrow smear, identify the erythroid series by its round nucleus, progressively condensed chromatin and pink cytoplasm, and the granulocytic series by its granules and nuclear indentation. Name each stage by nuclear shape and cytoplasmic color rather than by size alone. Find a megakaryocyte by its large, lobulated nucleus, and explain how platelets are shed from it.

14 · The Immune System & Lymphoid Organs

In this chapter: Innate & Adaptive Immunity · Cytokines · Antigens & Antibodies · Antigen Presentation · Cells of Adaptive Immunity · Thymus · Mucosa-Associated Lymphoid Tissue · Lymph Nodes · Spleen

Learning objectives

By the end of this chapter you should be able to:

1. Contrast innate and adaptive immunity in cells, speed, and memory.
2. Explain antigen presentation by MHC I and MHC II and the T-cell classes they activate.
3. Compare the architecture and function of thymus, lymph node, MALT, and spleen.
4. Explain how T and B cells are selected and activated, and what happens when selection fails.

The core idea

The immune system is a set of organs arranged to **bring antigen to lymphocytes**. Each organ is a different solution to the same problem: how to filter a fluid (blood, lymph, or air/food), trap antigen, and present it to the right cell.

Innate & Adaptive Immunity

Feature	Innate	Adaptive
Timing	**Present from birth**, acts within minutes	**Develops more slowly**, improves with exposure
Specificity	Broad molecular patterns	**Exact antigen**
Memory	None	**Yes**
Cells	**Granulocytes** (mainly neutrophils), macrophages, dendritic cells, NK cells	**T and B lymphocytes**
Mediators	**Defensins, complement, lysozyme, interferons**, cytokines	Antibodies, cytotoxic granules

A working definition is worth memorizing: innate immunity is present from birth and uses leukocytes (mainly granulocytes) plus soluble proteins; adaptive immunity develops more slowly and is **based on antigen presentation to lymphocytes**.

Cytokines

Polypeptide hormones of the immune system, secreted by immune cells to regulate one another. The ones worth naming: **interleukins** (IL-1, IL-2 for T-cell proliferation, IL-4 for IgE and Th2 responses, IL-5 for eosinophils, IL-6 for acute-phase response), **interferons** (antiviral; IFN-γ activates macrophages), **TNF-α** (inflammation, fever, cachexia), **chemokines** (directed leukocyte migration), and **colony-stimulating factors** (hemopoiesis, Chapter 13). Cytokines act locally (paracrine) or on the secreting cell (autocrine), and their excessive release produces a cytokine storm and the systemic inflammatory response.

Antigens & Antibodies

Antigens are molecules recognized by the adaptive immune system; **epitopes** are the specific parts bound by a receptor. **Antibodies** are immunoglobulins:

Class	Structure	Main role
IgG	Monomer	Most abundant; crosses the placenta; opsonizes
IgM	Pentamer	First responder; complement activation

Class	Structure	Main role
IgA	Dimer with secretory component	Mucosal secretions
IgE	Monomer	Mast-cell binding; allergy and parasites
IgD	Monomer	B-cell receptor

Antigen Presentation

T cells cannot see free antigen. They recognize peptide displayed on **MHC** molecules:

- **MHC class I** — on the surface of **all nucleated cells**, bearing fragments of that cell's own proteins (and viral proteins if infected); recognized by **CD8⁺ cytotoxic T cells**.
- **MHC class II** — only on **antigen-presenting cells (APCs)**, mostly derived from monocytes; presents fragments of **endocytosed foreign** proteins to **CD4⁺ helper T cells**. This is the defining difference: MHC I says "here is what I am making"; MHC II says "here is what I have eaten".

This pairing — **CD4/MHC II, CD8/MHC I** — is the one to know cold.

Cells of Adaptive Immunity

T lymphocytes mature in the thymus and are responsible for cell-mediated immunity. Their **T-cell receptor (TCR)** is usually an **αβ heterodimer**; each T cell also carries one CD marker that fixes its class:

T-cell class	Marker	Job
Helper T cell	**CD4⁺**	Recognizes MHC II; secretes cytokines that activate B cells and macrophages
Cytotoxic T cell	**CD8⁺**	Recognizes MHC I; kills infected or tumor cells
Regulatory T cell	**CD4⁺ CD25⁺**	Suppresses immune responses and prevents autoimmunity
γδ T cell	γδ TCR	Mainly in **epithelia**; early defense at body surfaces

B lymphocytes mature in bone marrow and carry a **B-cell receptor (BCR)** — surface **IgM or IgD** — that binds antigen directly, without MHC. On activation they become **plasma cells** (antibody factories) and **memory cells**. **NK cells** kill without prior sensitization and without MHC restriction.

How a Lymphoid Nodule Forms

This sequence explains every secondary lymphoid organ in the chapter:

1. Lymphocytes are packed in a meshwork of **reticulin** made by **fibroblastic reticular cells**; most APCs among them are **dendritic cells** with many processes.
2. In a secondary organ, **follicular dendritic cells (FDCs)** display antigen–antibody complexes on their large surface. A B cell whose BCR binds that antigen is held there.
3. With cytokines from helper T cells, that B cell **proliferates clonally**, producing a large transient nodule with a pale **germinal center**.
4. The cells disperse as **plasma cells, various T cells, and B and T memory cells** — which is why the same nodule can look "active" (large germinal center) or "resting" (dense primary follicle).

Thymus

- **Capsule** with trabeculae dividing it into lobules, each with a **cortex** and a **medulla**.
- **No follicles.** The cortex is densely packed with developing **thymocytes** and epithelial reticular cells; the medulla is paler and contains **Hassall corpuscles** — concentric whorls of epithelial cells, unique to the thymus.
- **Blood–thymus barrier** — formed by **squamous thymic epithelial cells** joined by desmosomes and occluding junctions, together with vascular endothelium and pericytes — prevents

unregulated exposure of thymocytes to antigens.

- **Cortex** is packed with **T lymphoblasts (thymocytes)**, macrophages, and **thymic epithelial cells (TECs)**. Stellate TECs form a **cytoreticulum** to which lymphocytes attach instead of reticulin fibers; these TECs are **antigen-presenting cells** expressing **MHC class II** and secreting cytokines — which is why the thymus is also counted among endocrine glands.
- **Hassall corpuscles** — aggregates of medullary TECs, concentrically arranged, **up to 100 µm**, **unique to the medulla**; their cells secrete cytokines that control local dendritic cells and promote regulatory T cells for peripheral tolerance.
- Function: **T-cell maturation and selection**. **Positive selection occurs in the cortex**: only thymocytes whose TCR recognizes the individual's own MHC molecules survive. **Negative selection occurs in the medulla**: thymocytes that bind self-antigen too tightly are deleted, leaving tolerance. Medullary TECs express **Aire**, which lets them display a wide sample of tissue self-antigens for this test.
- **Involutes** after puberty, replaced by adipose tissue.

Mucosa-Associated Lymphoid Tissue

Tonsils (palatine, lingual, pharyngeal/adenoid, tubal) — a partial capsule (hemicapsule), no afferent lymphatics, epithelium invaginates into crypts. **Peyer's patches** in the ileum; **appendix**; **MALT** throughout the gut, respiratory tract, and genitourinary tract.

Lymph Nodes

Region	Contents	Function
Cortex	**Lymphoid follicles** — primary (dense) or secondary (with pale **germinal center** and dark mantle)	**B-cell** zone
Paracortex	Diffuse lymphoid tissue, **high endothelial venules (HEV)**	**T-cell** zone; lymphocytes enter from blood here
Medulla	Cords and sinuses, macrophages and plasma cells	Filtration; antibody secretion

Flow: afferent lymphatics → subcapsular sinus → trabecular sinuses → medullary sinuses → **efferent lymphatics at the hilum**. The one-way flow explains why nodal metastases lodge first in the subcapsular sinus.

Spleen

- **No afferent lymphatics.** It filters **blood**.
- **White pulp** — lymphocytes around a central arteriole: the **periarteriolar lymphoid sheath (PALS)** is the T-cell zone; follicles at its periphery are the B-cell zone.
- **Red pulp** — **cords of Billroth** and **sinusoids** whose elongated endothelial **stave cells** are separated by narrow slits.
- **Two routes through the red pulp.** In the **closed circulation**, capillaries from the penicillar arterioles connect directly to sinusoids. In the **open circulation** (about half of the penicillar arterioles), capillaries are open-ended and dump blood into the splenic cords, so plasma and cells must squeeze between stave cells to re-enter the vasculature. Platelets, motile leukocytes, and thin flexible erythrocytes pass; **stiff or swollen erythrocytes at the end of their 120-day life are blocked and removed by macrophages**.
- **Iron recycling.** Iron released from **hemoglobin** during erythrocyte degradation is stored by macrophages in **ferritin**, released bound to **transferrin**, and reused for erythropoiesis. Heme is carried by **hemopexin** and degraded to **bilirubin**, which is excreted in bile.
- **Functions:** blood filtration, removal of old erythrocytes, immune response to blood-borne antigen, and **extramedullary hemopoiesis** in fetal life and marrow failure.

Organ	Capsule	Afferent lymphatics	Follicles	Unique feature
Thymus	Yes	None	**No**	**Hassall corpuscles**
Lymph node	Yes	**Yes**	Yes	Cortex/paracortex/medulla; HEV
Spleen	Yes	**None**	Yes	**White and red pulp**
Tonsil	Partial	None	Yes	Crypts

CLINICAL CORRELATION

DiGeorge syndrome is thymic aplasia (third and fourth pharyngeal pouches) with T-cell deficiency and hypocalcemia. **Severe combined immunodeficiency** is failure of both arms. **Hodgkin lymphoma** spreads contiguously through nodes and shows Reed–Sternberg cells. **Splenectomy** increases susceptibility to encapsulated organisms — pneumococcus, meningococcus, Haemophilus influenzae — because the spleen's marginal zone is the first line against polysaccharide capsules.

UNDER THE MICROSCOPE

Ask **"what is at the center of the nodule?"** Thymus: lobules with dark cortex and pale medulla plus pink whorls (Hassall corpuscles). Lymph node: capsule, subcapsular sinus, hilum. Spleen: **white pulp islands inside a red sea** of sinusoids and cords.

MEMORY HOOK

"Thymus Teaches T cells; Nodes Filter lymph; Spleen Filters blood." And **"CD4 sees MHC II (2 × 2); CD8 sees MHC I (8 × 1)."**

EXAM TIP

Two negative facts are worth more than most positive ones: the **thymus has no follicles**, and the **spleen has no afferent lymphatics**.

Chapter 14 — summary

- Innate immunity = present at birth, granulocytes and soluble proteins; adaptive = antigen presentation to lymphocytes.
- MHC I is on **all nucleated cells** and presents endogenous peptide to **CD8⁺** cells.
- MHC II is on **APCs** (mostly monocyte-derived) and presents endocytosed peptide to **CD4⁺** cells.
- BCR = surface **IgM or IgD**; TCR is usually **αβ** with a CD co-receptor.
- T cells: CD4 helper, CD8 cytotoxic, $CD4^+CD25^+$ regulatory, γδ in epithelia.
- B cells make antibodies (humoral); T cells mediate cell-mediated immunity.
- Plasma cells secrete antibody; memory B and T cells give the rapid second response.
- Primary lymphoid organs: **bone marrow (B)** and **thymus (T)**.
- Secondary: lymph nodes, MALT, spleen — where lymphocytes are activated.
- FDCs display antigen–antibody complexes; activated B cells form germinal centers.
- Thymus: cortex = positive selection, medulla = negative selection; **Aire** displays self-antigens.
- Thymus has **no follicles** and Hassall corpuscles in the medulla.
- Lymph node: cortex B cells, paracortex T cells and HEVs, medulla cords and sinuses.
- Spleen: white pulp (PALS = T zone, follicles = B zone) and red pulp (cords + sinusoids).
- Spleen filters blood; **open circulation** lets macrophages remove stiff 120-day erythrocytes.
- Iron is stored as **ferritin**, carried by **transferrin**, and heme iron binds **hemopexin**; the rest becomes bilirubin.

Active recall

1. **Which MHC class is on all nucleated cells, and which T cell reads it?** — MHC I; $CD8^+$ cytotoxic T cells.

2. **Which cells present on MHC II?** — Antigen-presenting cells, mostly monocyte-derived: dendritic cells, macrophages, B cells.
3. **Where do T cells undergo positive and negative selection?** — Positive selection in the cortex; negative selection in the medulla.
4. **What is the difference between the thymus and the spleen in one line?** — The thymus makes and selects T cells and has no follicles; the spleen filters blood and has no afferent lymphatics.
5. **What does an FDC do?** — Follicular dendritic cells display antigen–antibody complexes to B cells and drive germinal-center formation.

Exam-style question

A child has thymic aplasia with absent T cells and hypocalcemia. Which structure failed to develop?

A. First pharyngeal arch
B. Third and fourth pharyngeal pouches
C. Neural crest
D. Spleen

Answer: B — DiGeorge syndrome is a third/fourth pharyngeal pouch defect affecting the thymus and parathyroids.

Practical identification

Compare the four lymphoid organs. In the thymus, find lobules with a dark cortex, pale medulla and Hassall corpuscles; in a lymph node, the capsule, subcapsular sinus, cortical follicles, paracortex with high endothelial venules, and medullary cords; in the spleen, white pulp around a central arteriole and red pulp with sinusoids and cords; in a tonsil, crypts and a partial capsule. Use the presence or absence of afferent lymphatics to separate them.

PART IV — DIGESTION AND RESPIRATION

Chapters 15–18 · gut, liver, pancreas, lung, and skin

15 · Digestive Tract

In this chapter: General Structure of the Digestive Tract · Oral Cavity · Esophagus · Stomach · Small Intestine · Large Intestine

Learning objectives

By the end of this chapter you should be able to:

1. Draw and name the four layers of the gut wall and their contents.
2. Identify each segment of the digestive tract from its mucosa and muscularis.
3. Match each gastric and intestinal cell type to its product.
4. Explain how the small intestine amplifies surface area and how each nutrient is absorbed.

The core idea

The gut is one tube with four concentric layers, and every region is a modification of those four layers for a specific job: **grinding, acidifying, absorbing, or dehydrating**. Learn the four layers and their regional variations and you can describe any segment from esophagus to anus.

General Structure of the Digestive Tract

Layer	Contents	Function
Mucosa	Epithelium + **lamina propria** + **muscularis mucosae**	Digestion, absorption, secretion, barrier
Submucosa	Loose connective tissue, vessels, lymphatics, **Meissner (submucosal) plexus**, glands where present	Supply and control of secretion
Muscularis	Inner circular + outer longitudinal smooth muscle, **Auerbach (myenteric) plexus** between them	Peristalsis
Serosa or adventitia	Mesothelium + connective tissue (intraperitoneal) or connective tissue only (retroperitoneal)	Suspension and protection

This is the skeleton of every answer about the gut. Two rules organize it:

- In the digestive tract the **mucosa has three components** — a lining epithelium, a lamina propria of loose connective tissue, and smooth muscle from the **muscularis mucosae**. The epithelium changes; the other two do not.
- **Serosa = suspended in the abdomen; adventitia = embedded in surrounding tissue.** The esophagus develops within the mediastinum and therefore has an adventitia along almost its whole length.

Both plexuses belong to the **enteric nervous system**: Meissner's controls secretion and blood flow, Auerbach's controls motility. The lymphocytes and nodules scattered through mucosa and submucosa are **MALT** (Chapter 14).

Oral Cavity

The lining is **stratified squamous** everywhere, but the degree of keratinization tracks wear: **keratinized on the hard palate and gingiva**, partly keratinized on the lips, and **nonkeratinized** over most of the soft oral mucosa.

Lips — three zones: cutaneous (skin with hair and sebaceous glands), **vermilion** (thin keratinized epithelium over vascular connective tissue — the red of the lip), and mucosal (nonkeratinized, wet). A core of **striated muscle** makes the lips mobile.

Tongue — skeletal muscle in bundles, covered by stratified squamous epithelium with four papilla types:

Papilla	Shape	Taste buds	Notes
Filiform	Cone-shaped, **heavily keratinized**	**No**	Most numerous; mechanical; gives the whitish dorsal appearance
Fungiform	Mushroom-shaped, lightly keratinized	Yes	Scattered among filiform
Foliate	Leaf-like folds	Yes	Lateral margins
Vallate (circumvallate)	Largest, **1–3 mm** wide, in a trench	**Yes, ~250 per papilla**	**8–12** in a V in front of the terminal sulcus

Taste buds are ovoid intraepithelial organs of **50–100 cells**, about half of them elongated **gustatory cells** with a **7–10-day** turnover. Their microvilli project toward a **taste pore** in the epithelial surface; support cells and basal stem cells complete the bud. **Filiform papillae never carry taste buds** — a standard trap.

Teeth. An adult has **32 permanent teeth**. Each has a crown, a constricted neck, and roots in dental alveoli.

- **Enamel** — hardest material in the body: **96% calcium hydroxyapatite**, only **2–3% organic**, **no collagen**. It is built of **enamel rods ~5 µm wide** running the full **~2 mm** thickness. **Ameloblasts** secrete a collagen-free matrix rich in **amelogenin (20 kDa)**, which initiates and guides crystal growth. Enamel is **acellular**; ameloblasts disappear at eruption, so enamel cannot regenerate, and it is lost from decalcified sections.
- **Dentin** — **70% hydroxyapatite** in a **type I collagen** matrix, harder than bone. **Odontoblasts** (cranial neural crest) secrete **predentin**, which mineralizes within about a day. Odontoblasts persist for life, and their processes occupy **dentinal tubules** across the full thickness of the dentin, accompanied by unmyelinated nerves — the basis of dentin sensitivity.
- **Cementum** — bone-like covering of the roots. **Pulp** — vascular, innervated loose connective tissue entering through apical foramina.
- **Periodontium = cementum + periodontal ligament + alveolar bone + gingiva**. The ligament's collagen bundles (**Sharpey fibers**) bind cementum to bone; it is **150–350 µm** thick.

Esophagus

A muscular tube about **25 cm** long. **Nonkeratinized stratified squamous** epithelium, a thick muscularis mucosae, **esophageal glands in the submucosa** (mucus-secreting, lubricating), and small **cardiac glands** in the lamina propria near the junction.

The muscularis shows the progression to know: **skeletal muscle at the superior end → mixed in the middle → exclusively smooth in the lower third**. The outer layer is **adventitia** except for the distal **1–2 cm**, which lies in the peritoneal cavity and has a serosa. At the **esophagogastric junction** the epithelium changes abruptly from stratified squamous to simple columnar, and the adventitia gives way to serosa.

Stomach

Four regions: the superior **cardia** and inferior **pylorus** resemble each other histologically (branching mucous glands, few parietal or chief cells), while the intervening **fundus and body** contain the main gastric glands. The cardia and pylorus are **histologically similar** to each other and **lack parietal and chief cells**, secreting mucus instead. The surface is perforated by **gastric pits**, which are lined by the same **surface mucous cells** as the stomach lumen and lead into branching glands. **No goblet cells, no villi.**

The gland epithelium contains surface mucous cells, mucous neck cells, parietal cells, chief cells, and enteroendocrine cells, plus pluripotent stem cells in the neck:

Cell	Location	Product	Function
Parietal (oxyntic)	Upper half of gland	**HCl + intrinsic factor**	Large cell, many mitochondria, **intracellular canaliculi**; intrinsic factor is required for **vitamin B_{12}** uptake; **eosinophilic**
Chief (zymogenic)	Deeper half of gland	**Pepsinogen**	Activated to **pepsin** by luminal acid; **basophilic** with apical zymogen granules
Mucous neck	Neck	Mucus (less alkaline than surface mucus)	Secretion; the proliferative stem-cell zone lies among them
Surface mucous	Surface and pits	**Viscous mucus + bicarbonate**	The barrier that protects the mucosa
Enteroendocrine	Throughout	**Gastrin, histamine**, others	Paracrine and endocrine control of secretion
Stem cells	Narrow neck region	—	Pluripotent source of all the above

Parietal cells are controlled both by **parasympathetic innervation** and by paracrine **histamine** and **gastrin** from enteroendocrine cells — the pharmacological basis of proton-pump inhibitors and H_2 blockers. The **muscularis has three layers** (oblique, circular, longitudinal), an exception to the four-layer template that allows churning.

Small Intestine

The small intestine is about **5 m long**. Its absorptive surface is amplified at three levels: **plicae circulares** (mucosa + submucosa, about a 3-fold increase), **villi** (mucosa, about 10-fold), and **microvilli** (cell surface, a further 20-fold), for a total absorptive area of **over 200 m²** — the clearest "structure follows function" example in the body.

Each **villus** is a finger of lamina propria covered by simple columnar epithelium, containing a **lacteal** and smooth muscle. Between villi, simple tubular **crypts (of Lieberkühn)** are the proliferative compartment. Smooth muscle of the lamina propria and muscularis mucosae, controlled by **Meissner's plexus**, moves the villi and propels lymph; the muscularis, controlled by **Auerbach's plexus**, produces peristalsis.

Cell types and what each absorbs or secretes:

- **Enterocytes** — brush border; **disaccharidases and peptidases in the glycocalyx** complete digestion, and the resulting monosaccharides and amino acids are absorbed across the brush border and pass to capillaries.
- **Goblet cells** — mucus.
- **M cells** — specialized epithelium over the lymphoid nodules, which takes up antigen from the lumen and hands it to the lymphocytes below.
- **Paneth cells** — base of crypts, **eosinophilic** granules, secrete **defensins** (and lysozyme).
- **Enteroendocrine cells** — CCK, secretin, GIP, motilin.
- **Stem cells** — crypt base, replacing the epithelium every few days.

Lipid absorption is different and worth knowing: lipid products associate with **bile salts**, enter enterocytes, are re-esterified to **triglycerides** and packaged as **chylomicrons**, and leave through the **lacteal** rather than the capillary.

Region	Defining feature
Duodenum	**Duodenal (Brunner) glands in the submucosa**; leaf-like villi
Jejunum	Tallest villi; no Brunner glands; Peyer's patches belong to the ileum
Ileum	**Peyer's patches** (mucosal and submucosal); shortest villi

Large Intestine

Three regions: the short **cecum** with the **appendix**; the long **colon** (ascending, transverse, descending, **sigmoid**); and the **rectum**.

No villi, straight tubular crypts lined by **colonocytes** (absorptive, water and electrolyte uptake) and **abundant goblet cells**. The outer longitudinal muscle is gathered into three **teniae coli**, which produce the bulging **haustra**; the serosa carries hanging fat masses, the **omental appendages** (appendices epiploicae). Function: water and electrolyte absorption, mucus secretion, bacterial fermentation. The **appendix** is a blind diverticulum of the cecum with abundant lymphoid tissue in its lamina propria and submucosa — part of MALT.

The **rectum** continues as the **anal canal**. There the mucosa is highly vascular, folded into **anal columns (of Morgagni)** with intervening anal sinuses, and the columnar lining changes to **stratified squamous** epithelium at the rectoanal junction. Swollen mucosal or submucosal veins here are **hemorrhoids**.

CLINICAL CORRELATION

Peptic ulcer — H. pylori breaches the mucous barrier; acid and pepsin digest the wall, classically in the duodenal bulb or gastric antrum. **Pernicious anemia** — loss of parietal cells means no intrinsic factor, so B_{12} cannot be absorbed in the terminal ileum. **Celiac disease** — villous atrophy, crypt hyperplasia, and intraepithelial lymphocytes in the proximal small intestine, driven by gluten. **Barrett esophagus** — chronic acid reflux causes metaplasia from stratified squamous to **intestinal-type columnar epithelium with goblet cells**, a premalignant change. **Diverticulosis** — mucosa and submucosa herniate between the teniae coli; trapped feces cause diverticulitis.

UNDER THE MICROSCOPE

Identify the segment by **the surface, not the glands**. **Villi + crypts = small intestine. No villi + deep pits + thick gland mass = stomach. No villi + straight crypts + many goblet cells = colon.** Then find the regional marker: Brunner glands = duodenum; Peyer's patches = ileum; three muscle layers = stomach.

MEMORY HOOK

"Duodenum has Brunner Below; Ileum has Peyer's Patches." And **"Filiform = Feel only; Vallate = Carry most taste."**

HIGH-YIELD PEARL

The numbers that appear in questions: enamel **96%** mineral and **2 mm** thick; dentin **70%**; esophageal length **25 cm**; **8–12** vallate papillae with about **250** taste buds each. The stomach has **no goblet cells** and **no villi**; the intestine has **both**.

Chapter 15 — summary

- Four gut layers, outside in: mucosa, submucosa, muscularis, serosa or adventitia.
- Gut mucosa = epithelium + lamina propria + muscularis mucosae.
- Serosa where the gut is suspended; adventitia where it is embedded.
- Auerbach lies between the muscle layers (motility); Meissner lies in the submucosa (secretion, blood flow).
- Keratinized oral epithelium: hard palate and gingiva. Everywhere else, mostly nonkeratinized.
- Filiform papillae are the only type without taste buds.
- Vallate papillae: 8–12, largest, ~250 taste buds each.
- Enamel is acellular and cannot regenerate; ameloblasts disappear at eruption.
- Dentin is 70% hydroxyapatite in type I collagen; odontoblast processes stay in dentinal tubules.
- Esophageal muscle: skeletal above, mixed in the middle, smooth below.
- Esophagogastric junction: squamous → columnar, and adventitia → serosa.
- Parietal cells: HCl + intrinsic factor, intracellular canaliculi. Chief cells: pepsinogen.

- Stomach has three muscle layers and no goblet cells; the colon has teniae coli and no villi.
- Duodenum = Brunner glands; ileum = Peyer's patches; colon = teniae coli.
- Lipids leave the villus as chylomicrons in the lacteal; sugars and amino acids enter capillaries.

Active recall

1. **Name the four layers of the gut wall in order.** — Mucosa, submucosa, muscularis, serosa or adventitia.
2. **Which cells secrete HCl and intrinsic factor?** — Parietal cells; they have intracellular canaliculi and many mitochondria.
3. **What are the three surface amplifiers of the small intestine?** — Plicae circulares, villi, and microvilli.
4. **How do lipids leave the villus?** — As chylomicrons through the lacteal, not through capillaries.
5. **Which segment has teniae coli and no villi?** — The colon (large intestine).

Exam-style question

A biopsy of the small intestine shows submucosal mucous glands. Which segment is it?

A. Duodenum
B. Jejunum
C. Ileum
D. Colon

Answer: A — Brunner (duodenal) glands lie in the submucosa of the duodenum only.

Practical identification

Walk the gut from esophagus to anal canal on the sections available. At each level, name the epithelium, the shape of the glands and whether villi are present. Identify the four wall layers and the two plexuses. In the stomach, distinguish parietal cells (eosinophilic, with canaliculi) from chief cells (basophilic); in the small intestine, find a villus with a lacteal and crypts with Paneth cells; in the colon, find straight crypts and teniae coli.

16 · Organs Associated with the Digestive Tract

In this chapter: Salivary Glands · Pancreas · Liver · Biliary Tract & Gallbladder

Learning objectives

By the end of this chapter you should be able to:

1. Compare the secretory units and ducts of the three major salivary glands.
2. Explain how the exocrine pancreas makes and delivers digestive enzymes and bicarbonate.
3. Trace blood and bile through a hepatic lobule in opposite directions.
4. Explain how the gallbladder concentrates bile and what controls its release.

The core idea

Four gland systems share one duct system and one embryological origin: the salivary glands **lubricate and begin digestion**, the pancreas **digests**, the liver **processes**, and the gallbladder **stores**. Each is built around its job, and each has one histologic signature that identifies it.

Salivary Glands

The three major pairs produce about **90%** of saliva; minor glands scattered through the oral mucosa contribute the rest. Total output is **0.75–1.5 L/day**.

Gland	Secretory units	Secretion
Parotid	**Serous acini only**	Watery, enzyme-rich
Submandibular	Mixed, **mainly serous** acini	Mixed
Sublingual	Mixed, **mainly mucous tubules**, some with **serous demilunes** (partly a fixation artifact)	Viscous

Serous cells are protein-secreting and form round or oval acini; **mucous cells** secrete mucin and form elongated tubules. **Myoepithelial cells** sit between the secretory cells and the basal lamina and squeeze the secretion onward.

The duct system is a three-step sequence, and each step has a different epithelium:

1. **Intercalated ducts** — simple cuboidal; the smallest.
2. **Striated ducts** — simple columnar, with **mitochondria-lined basolateral folds** that reabsorb **Na^+ and Cl^-** and add **HCO_3^-**. This is why saliva is **hypotonic** and why the striated duct is the histologic signature of a salivary gland.
3. **Interlobular (excretory) ducts** — unusually, lined by **stratified cuboidal or columnar** epithelium.

Pancreas

A compound acinar gland, entirely **exocrine** except for the scattered **islets of Langerhans**. It secretes about **1.5 L/day** of alkaline juice.

- **Acinar cells** — **pyramidal**, with a basal nucleus, abundant **rough ER and Golgi** in the basal cytoplasm, and **eosinophilic zymogen granules** apically. The cell is polarized for export: make protein at the base, package it at the apex.
- **Centroacinar cells** — pale cells inserted into the acinar lumen; they begin the duct system inside the acinus.
- **Ducts** — intercalated → intralobular → interlobular → main pancreatic duct. **Centroacinar and intercalated duct cells secrete HCO_3^-** under **secretin**, neutralizing chyme and establishing the pH at which pancreatic enzymes work. The high pH also keeps the enzymes inactive until they reach the duodenum.

- **Endocrine islets** — pale, round clusters between acini (details in Chapter 20).

The pancreas has only a **very thin capsule**, whose septa carry the larger vessels and ducts and separate the parenchyma into lobules; each acinus is surrounded by a basal lamina and a delicate reticular sheath — a useful contrast with the parotid, which has striated ducts and no islets.

Liver

The liver is both an **endocrine** gland (plasma proteins into blood) and an **exocrine** gland (bile into ducts). That dual output explains its unusual architecture: two fluid streams run in opposite directions through the same lobule.

Blood supply. About **75%** of hepatic blood is nutrient-rich, O_2-poor **portal vein** blood; **25%** is oxygenated **hepatic artery** blood. Both enter at the portal areas and flow **toward the central vein**.

- **Classic lobule** — a polygon of **plates of hepatocytes radiating toward a central vein**, with **portal areas** at the corners. Connective tissue between lobules is sparse, becoming more abundant in the portal areas.
- **Portal area (triad + one)** — a **portal venule**, a **hepatic arteriole**, a **bile ductule**, and a small **lymphatic**.
- **Hepatocytes** — large polygonal epithelial cells with large central nuclei; about **50% are polyploid** and many are **binucleated**. They contain abundant **rough and smooth ER**, many small **Golgi complexes**, glycogen, and peroxisomes.
- **Sinusoids** — discontinuous, **fenestrated** endothelium. Between endothelium and hepatocytes is the **perisinusoidal space (of Disse)**, where plasma exchanges with hepatocytes.
- **Kupffer cells** — stellate macrophages in the sinusoidal lining; remove effete erythrocytes and release iron and bilirubin.
- **Hepatic stellate (Ito) cells** — in the perisinusoidal space, storing **vitamin A** and other fat-soluble vitamins in lipid droplets; they become myofibroblasts in fibrosis.
- **Bile canaliculi** — grooves **between adjacent hepatocytes**, sealed by tight junctions. Bile flows **toward** the portal area while blood flows **away** from it.
- **Canals of Hering** — where canaliculi join the bile ductules; all ducts eventually form the left and right hepatic ducts.

Functions: carbohydrate, lipid, and protein metabolism; **albumin** and clotting factor synthesis; detoxification by SER and peroxisomes; bile production; bilirubin conjugation; storage of glycogen, iron, and vitamins A, D, and B_{12}.

Zonation. Zone I (periportal) receives the most oxygen and nutrients and performs oxidative metabolism, such as protein synthesis; **zone III** (pericentral) receives the least, and its cells are the first to accumulate fat and undergo ischemic necrosis. Zone III is also where drug biotransformation predominates — the basis for many patterns of hepatic injury.

Biliary Tract & Gallbladder

Every bile-conducting structure after the canaliculi is lined by **cholangiocytes** — simple cuboidal or columnar epithelium.

Route: canaliculi → canals of Hering → bile ductules → interlobular ducts → left and right hepatic ducts → **common hepatic duct** → **cystic duct** → gallbladder → **cystic duct** → **common bile duct** → **hepatopancreatic ampulla (of Vater)** → duodenum.

The **gallbladder** is a sac with a **highly folded mucosa**, a well-vascularized lamina propria, **no muscularis mucosae or submucosa**, and a **muscularis of interwoven bundles**. Its columnar cholangiocytes **absorb water**, concentrating bile; the muscle contracts under **cholecystokinin (CCK)** released by duodenal enteroendocrine cells when food arrives.

CLINICAL CORRELATION

Cirrhosis is the end result of hepatocyte injury: fibrosis with regenerative nodules, portal hypertension, and synthetic failure. **Acute pancreatitis** is autodigestion by prematurely activated enzymes; **chronic pancreatitis** destroys acini and islets, producing diabetes and steatorrhea. **Carcinoma of the head of the pancreas** obstructs the common bile duct and produces painless obstructive jaundice. **Gallstones** form when bile is supersaturated with cholesterol or pigment; impaction in the cystic or common bile duct produces biliary colic and obstructive jaundice.

UNDER THE MICROSCOPE

The liver is recognized by **cords of eosinophilic cells with central veins and portal areas**. The pancreas by **dark purple acini with pink apical granules and pale islets**. Pancreas versus parotid: the pancreas has **islets** and **no striated ducts**; the parotid has **striated ducts** and **no islets**.

EXAM TIP

Bile flows **toward** the portal triad; blood flows **away** from it. The common error is to memorize "everything flows to the center".

Chapter 16 — summary

- Major salivary glands make ~90% of saliva; minor glands ~10%.
- Parotid = serous only. Submandibular = mixed, mainly serous. Sublingual = mixed, mainly mucous with demilunes.
- Intercalated duct = simple cuboidal; striated duct = simple columnar with basal folds; excretory duct = stratified cuboidal or columnar.
- Striated ducts reabsorb Na^+ and Cl^- and add HCO_3^-, making saliva hypotonic.
- Myoepithelial cells contract to expel secretion.
- Pancreatic acinar cell: basal RER and Golgi, apical zymogen granules — polarized for export.
- Centroacinar cells sit inside the acinus and begin the duct.
- Secretin drives HCO_3^- secretion from centroacinar and intercalated duct cells.
- Liver blood: 75% portal vein, 25% hepatic artery.
- Portal area = portal venule + hepatic arteriole + bile ductule + lymphatic.
- About half of hepatocytes are polyploid; many are binucleated.
- Kupffer cells = sinusoidal macrophages. Ito cells = vitamin A storage.
- Bile canaliculi lie between hepatocytes, sealed by tight junctions, and drain to the canals of Hering.
- Hepatic zone 1 = periportal, most oxygen; zone 3 = pericentral, most biotransformation and first to die in ischemia.
- Gallbladder: folded mucosa, no muscularis mucosae, no submucosa; cholangiocytes absorb water; CCK contracts the muscle.

Active recall

1. **Which salivary gland is purely serous?** — The parotid. Submandibular is mixed, mainly serous; sublingual is mixed, mainly mucous.
2. **What do striated ducts do?** — Reabsorb Na^+ and Cl^- and add HCO_3^-, making saliva hypotonic.
3. **What do centroacinar cells secrete, and under what hormone?** — Bicarbonate-rich fluid, under secretin.
4. **Trace bile from canaliculus to duodenum.** — Canaliculi → canals of Hering → bile ductules → interlobular ducts → hepatic ducts → common hepatic duct → cystic duct → gallbladder → cystic duct → common bile duct → hepatopancreatic ampulla.
5. **Which way does blood flow in a hepatic lobule?** — From portal areas at the periphery toward the central vein.

Exam-style question

A liver biopsy shows a portal area containing a venule, an arteriole, a bile ductule, and a lymphatic. Which structure is missing from the center of the lobule?

A. Central vein
B. Portal vein
C. Hepatic artery
D. Bile duct

Answer: A — The portal triad (plus lymphatic) sits at the lobule corner; the central vein sits at the center.

Practical identification

In a salivary gland, distinguish serous acini, mucous tubules and demilunes, and identify intercalated and striated ducts. In the pancreas, find acini with basal basophilia and apical zymogen granules, plus pale islets. In the liver, identify the lobule, central vein, portal area, plates of hepatocytes and sinusoids, and trace the direction of blood and bile. In the gallbladder, note the folded mucosa and muscle bundles with no muscularis mucosae.

17 · The Respiratory System

In this chapter: Nasal Cavities · Pharynx · Larynx · Trachea · Bronchial Tree & Lung · Lung Vasculature & Nerves · Pleural Membranes · Respiratory Movements

Learning objectives

By the end of this chapter you should be able to:

1. Follow the airway from nose to alveolus and name what changes at each level.
2. Identify respiratory epithelium and its cell types.
3. Explain the blood–air barrier, surfactant, and the mechanics of ventilation.
4. Distinguish conducting from respiratory airways in a micrograph.

The core idea

The respiratory tract is a **conducting** series (nose to terminal bronchiole) followed by a **respiratory** series (respiratory bronchiole to alveolus). The transition is marked by the disappearance of cartilage and the appearance of alveoli. Walking down the tree, each level loses one structural support and gains a thinner epithelium.

Nasal Cavities

Each nasal cavity has a **vestibule** where air enters, and **three conchae** projecting from the medial wall that create turbulence and force air over a large mucosal surface.

- **Vestibule** — stratified squamous epithelium with **vibrissae** (stiff hairs) that filter large particles.
- **Respiratory region** — **respiratory epithelium**: pseudostratified ciliated columnar with goblet cells, over a highly vascular lamina propria containing seromucous glands. It warms, humidifies, and cleans air; the cilia sweep mucus toward the pharynx.
- **Olfactory region** — the roof and part of the superior concha: **olfactory epithelium**, about **10 cm²** and up to **100 µm** thick, containing **bipolar olfactory neurons**, supporting cells, and basal stem cells. Each neuron has about **20 long nonmotile cilia** bearing odorant receptors. Olfactory neurons are replaced every **1–2 months** — unusual for neurons. **No goblet cells**, but **Bowman glands** in the lamina propria.

The respiratory epithelium is a mixed population, and the proportions are worth remembering:

Cell	Share	Function
Ciliated columnar cells	Majority	Move the mucus blanket
Goblet cells	—	Secrete mucus
Brush cells	~3%	Chemosensory; short blunt microvilli
Small granule (Kulchitsky) cells	~3%	**DNES**; dense-core granules 100–300 nm
Basal cells	~30%	Stem cells

Pharynx

Nasopharynx — respiratory epithelium with **lymphoid tissue** (pharyngeal tonsil) in the lamina propria. **Oropharynx** and **laryngopharynx** — stratified squamous epithelium, continuous with the oral cavity and esophagus. The change from respiratory to stratified squamous epithelium marks where the airway is exposed to abrasion from food.

Larynx

A short air passage between pharynx and trachea, with skeletal cartilages and intrinsic muscles.

- **Epiglottis** — elastic cartilage, with stratified squamous epithelium on its lingual surface and respiratory epithelium on its laryngeal surface; the switch is abrupt.
- **Vocal folds (true cords)** — **stratified squamous** epithelium, avascular, with the **vocal ligament** and the **vocalis muscle**. Tension and vibration produce sound.
- **False cords (vestibular folds)** — respiratory epithelium and glands.

Trachea

- **Epithelium** — respiratory epithelium (pseudostratified ciliated columnar with goblet cells); the trachea is **10–12 cm** long and supported by **C-shaped rings** of hyaline cartilage.
- **Lamina propria** — elastic fibers, lymphocytes, seromucous glands.
- **Cartilage** — **C-shaped hyaline rings**, incomplete posteriorly.
- **Trachealis muscle** — smooth muscle bridging the posterior gap, allowing the esophagus to expand.

Bronchial Tree & Lung

Bronchi — the primary bronchus enters each lung and branches repeatedly. Cartilage breaks into **irregular plates** rather than rings, a complete ring of smooth muscle appears, and elastic fibers and glands persist. Each **tertiary (segmental) bronchus** with its branches forms a **bronchopulmonary segment** — a pyramidal portion of lung with its own blood supply, separated from its neighbors by connective tissue septa.

Bronchioles — airways of **1 mm or less**, formed after about the **tenth generation** of branching. They have **no cartilage and no glands**, a simple ciliated columnar or cuboidal epithelium, circular smooth muscle, and elastic fibers.

- **Club cells** (formerly Clara cells) replace goblet cells as the main secretory cell; they secrete **surfactant-like material**, help innate immunity, and their **smooth ER P450 system detoxifies inhaled compounds**. They are the progenitor cells of the bronchiolar epithelium.
- **Terminal bronchiole** — the last conducting airway, lined mainly by club cells, with **no alveoli**.

Respiratory region: terminal bronchiole → **respiratory bronchiole** (simple cuboidal, interrupted by alveoli) → **alveolar duct** (a linear series of alveoli) → **alveolar sac** (a cluster). Gas exchange begins at the respiratory bronchiole.

Cell	Shape	Function	Share of alveolar lining
Type I pneumocyte	Squamous, extremely thin	**Gas exchange**; forms the blood–air barrier	Lines almost the **entire** alveolar surface
Type II pneumocyte	Cuboidal, **lamellar bodies**	**Surfactant**; innate immunity; progenitor for type I	A **small part** of each alveolus
Alveolar macrophage	Large, free ("dust cell")	Phagocytosis	—

Numbers to know:

- Each alveolus is about **200 µm** across. Each adult lung has about **200 million alveoli**, with a total internal surface area of about **75 m²**.
- The **blood–air barrier** = surfactant film + **type I pneumocyte** + fused basal laminae + capillary endothelium. Its total thickness is **0.1–1.5 µm**, and the attenuated type I cytoplasm can be as thin as **25 nm**.
- **Surfactant** is an oily mixture of **cholesterol, phospholipids, and surfactant proteins** secreted by **type II cells and club cells**. It lowers surface tension and prevents alveolar collapse at end-expiration.
- **Pores of Kohn** — **10–15 µm** openings that connect adjacent alveoli and equalize air pressure.

Lung Vasculature & Nerves

The lung has a **double blood supply**: the **pulmonary artery** carries deoxygenated blood for gas exchange and forms a dense capillary network around alveoli, while the **bronchial arteries** (from the aorta) supply the conducting airways and pleura. Pulmonary arteries are unusual in carrying **deoxygenated** blood and in having thin walls suited to low pressure. Autonomic nerves control bronchial smooth muscle and gland secretion — parasympathetic (vagal) constriction, sympathetic relaxation.

Pleural Membranes

A **serous membrane** of simple squamous mesothelium over connective tissue: **visceral pleura** covers the lung and is continuous with **parietal pleura** lining the thoracic wall. A thin fluid film lets the layers slide while surface tension holds them together, so the lung follows the chest wall.

Respiratory Movements

Ventilation is pressure-driven: the **diaphragm** and **external intercostal muscles** enlarge the thoracic cavity, pleural pressure falls, the lung expands passively, and air enters. **Elastic recoil** of the lung and surface tension at the alveolar air–liquid interface drive expiration — precisely why surfactant deficiency causes collapse.

CLINICAL CORRELATION

Respiratory distress syndrome of the newborn — immature type II pneumocytes produce insufficient surfactant, so alveoli collapse; treatment is exogenous surfactant and antenatal steroids. **Emphysema** — destruction of alveolar walls enlarges air spaces and reduces surface area. **Asthma** — bronchoconstriction, mucosal edema, eosinophilic inflammation, thickened basement membrane, and smooth muscle hypertrophy. **Chronic bronchitis** — mucus hypersecretion and gland hypertrophy. **Cystic fibrosis** — defective CFTR thickens secretions, causing bronchiectasis and recurrent infection. **Lung cancer** — three main types are described here: **squamous cell carcinoma** (smoking-related, arises in segmental bronchi), **adenocarcinoma** (commonest in non-smokers, arises peripherally in bronchioles and alveoli), and **small cell carcinoma** (less common, highly malignant, arising from small granule Kulchitsky cells of the bronchial epithelium).

UNDER THE MICROSCOPE

Walk down the tree and count what disappears. **Cartilage rings + glands = trachea. Cartilage plates + no glands = bronchus. No cartilage, no glands, muscular wall = bronchiole.** Then look for alveoli in the wall: if present, it is a respiratory bronchiole.

MEMORY HOOK

"Rings, Plates, None" for cartilage from trachea → bronchus → bronchiole. And **"Type I for gas, Type II for surfactant, macrophage for garbage."**

EXAM TIP

Club cells are found in **bronchioles**, not alveoli, and are the progenitor cells of the bronchiolar epithelium. A question that places them in the alveolus is testing whether you know the level of the airway.

Chapter 17 — summary

- The conducting airway is nose → pharynx → larynx → trachea → bronchi → bronchioles → terminal bronchioles.
- Respiratory epithelium is pseudostratified ciliated columnar with goblet cells.
- Cartilage changes down the tree: C-shaped rings → irregular plates → none.
- The respiratory bronchiole is the first airway with alveoli in its wall.
- Type I pneumocytes line almost the entire alveolar surface and form the thin gas-exchange barrier.
- Type II pneumocytes secrete surfactant and divide to replace type I cells.

- The blood–air barrier is surfactant, type I cell, fused basal laminae, and capillary endothelium (0.1–1.5 µm).
- Alveolar macrophages (dust cells) clear particles and debris.
- The pleura is a serous membrane: visceral and parietal layers with a fluid-filled cavity.

Active recall

1. **What marks the transition from conducting to respiratory airway?** — The appearance of alveoli in the wall — the respiratory bronchiole.
2. **Which cells line almost the entire alveolar surface?** — Type I pneumocytes.
3. **What do type II pneumocytes secrete, and why does it matter?** — Surfactant, which lowers surface tension and prevents alveolar collapse.
4. **Name the components of the blood-air barrier.** — Surfactant film, type I pneumocyte, fused basal laminae, capillary endothelium.
5. **How does cartilage change down the tree?** — C-shaped rings (trachea) → irregular plates (bronchi) → absent (bronchioles).

Exam-style question

A premature infant develops respiratory distress because alveoli collapse at end-expiration. Which cell is functionally immature?

A. Type I pneumocyte
B. Type II pneumocyte
C. Alveolar macrophage
D. Club cell

Answer: B — Type II cells produce surfactant; their immaturity causes neonatal respiratory distress syndrome.

Practical identification

Follow the airway from trachea to alveolus. At each level, record the epithelium, the cartilage (ring, plate or none), the glands (present or absent) and the smooth muscle. Identify type I and type II pneumocytes, an alveolar macrophage, the blood–air barrier and a pore of Kohn. Distinguish a bronchiole from a tangentially cut vessel and from an alveolar duct.

18 · Skin

In this chapter: Epidermis · Dermis · Subcutaneous Tissue · Sensory Receptors · Hair · Nails · Skin Glands · Skin Repair

Learning objectives

By the end of this chapter you should be able to:

1. Name the epidermal layers in order and the defining feature of each.
2. Match each epidermal cell type to its origin and function.
3. Identify the dermis, hypodermis, and each skin appendage.
4. Explain the stages of skin repair, and recognize healing by first and second intention.

The core idea

Skin is a **stratified, keratinizing, self-renewing barrier** with appendages embedded in it. The epidermis is a factory that moves cells outward and turns them into keratin; the dermis provides mechanical and nutritional support; the appendages provide temperature control, sensation, and protection. Every fact in the chapter is a consequence of one of those three jobs.

Epidermis

Thickness: 75–150 µm in thin skin, **400–600 µm** in thick skin. Full skin (epidermis + dermis) is about **4 mm** on the back and **1.5 mm** on the scalp. The epidermis is **avascular** — nutrients and O_2 arrive by diffusion from the dermal papillae. It turns over about every **15–30 days**.

Layer	Cells	Key feature
Stratum basale	Mitotic stem cells, melanocytes, Merkel cells	Single layer on the basement membrane, attached by **hemidesmosomes and integrins**; **keratin filaments ~10 nm** first appear here
Stratum spinosum	Polyhedral cells with **desmosomes**	"Spiny" because shrinkage pulls apart desmosome-bound bundles (**tonofibrils**); **Langerhans cells** live here
Stratum granulosum	Flattened cells with **keratohyalin granules** and **lamellar granules**	Keratohyalin contains **filaggrin**, which aggregates keratin; lamellar granules discharge lipid into the intercellular space
Stratum lucidum	Clear, anucleate, refractile	**Only in thick skin** of palms and soles
Stratum corneum	Anucleate **squames**	The barrier; keratin packed in an electron-dense matrix with a lipid envelope

Keratin filaments contain at least **six polypeptides of 40–70 kDa**, synthesized as the cell differentiates. The **lipid envelope** between squames is what makes epidermis impermeable to water.

Cell	Origin	Function	Identifying feature
Keratinocyte	Ectoderm	Barrier, keratin production	90% of epidermal cells; desmosomes
Melanocyte	**Neural crest**	**Melanin** via tyrosinase; transferred in melanosomes	Dark, dendritic; basal layer
Langerhans cell	**Monocyte/bone marrow**	Antigen presentation	Dendritic, stratum spinosum; **2–8%** of epidermal cells
Merkel cell	Neural crest	**Light touch** (mechanoreception)	Basal, with a sensory nerve ending

Melanocytes — about **one per five or six keratinocytes** (**600–1200/mm²** of thin skin), with processes reaching **10–30 keratinocytes**. **Tyrosinase** converts tyrosine to DOPA and then to **eumelanin** (brown-black) or **pheomelanin** (red-yellow). Mature melanosomes are ellipsoid, **0.5 × 1 µm**, and are transferred to keratinocytes, which hold more pigment than the cells that make it. The pigment sits over the nucleus and **protects nuclear DNA from UV damage** — the reason tanning exists and the reason melanoma risk tracks sun exposure. One melanocyte plus its keratinocytes is an **epidermal-melanin unit**. Melanocytes are about **four times more numerous and active in thin skin** than in thick palmar and plantar skin.

The melanin rule: densities of epidermal-melanin units are **similar across populations**; what differs is how rapidly melanin is made and how abundantly it accumulates in keratinocytes. Dark skin is not skin with more melanocytes.

Dermis

- **Papillary dermis** — loose connective tissue forming **dermal papillae** that interdigitate with the epidermis; contains the **subpapillary microvascular plexus**, which sends capillary loops into the papillae.
- **Reticular dermis** — dense irregular connective tissue with thick collagen bundles and elastic fibers; up to **4 mm** thick on the back.

Dermal vasculature has two jobs: **nutrition** (two plexuses, subpapillary and deep) and **thermoregulation** through **arteriovenous anastomoses** between them.

Thick versus thin skin. Thick skin (palms, soles) has a thick stratum corneum, a **stratum lucidum**, and **no hair follicles**; it has many sweat glands. Thin skin covers most of the body, has hair follicles and sebaceous glands, and a thinner epidermis.

Subcutaneous Tissue

The **hypodermis** — loose connective tissue and adipose tissue (**panniculus adiposus**) binding skin loosely to underlying organs, so the skin can slide. It insulates, cushions, stores energy, and determines how a wound behaves and how an injection is absorbed.

Cutaneous Sensory Receptors

Receptor	Location	Structure	Modality
Free nerve endings	Epidermis and dermis	Unencapsulated	Pain, temperature extremes, mechanical movement
Merkel cell	Basal epidermis	Cell + disc-like nerve ending	**Light touch**, sustained pressure
Meissner corpuscle	Dermal papilla	Elliptical, **~150 µm**, stacked Schwann cells	**Light touch**, fine discrimination
Pacinian corpuscle	Deep dermis and hypodermis	Oval, **~1 mm**, **15–50 concentric lamellae**	**Pressure and high-frequency vibration**
Krause end bulb	Penis and clitoris mainly	Thin collagenous capsule	Low-frequency vibration
Ruffini corpuscle	Dermis	Fusiform capsule anchored to collagen	**Tissue distortion and sustained stretch**

Hair

Hairs are keratinized structures formed in **hair follicles**, epidermal invaginations extending into the dermis.

- The **hair bulb** at the base contains the **matrix** (rapidly proliferating keratinocytes) and is invaginated by a vascular **dermal papilla** that supplies it.

- The hair root has three concentric layers — **medulla, cortex, cuticle** — surrounded by **internal and external root sheaths** continuous with the epidermis, a **glassy membrane** (partly basal lamina), and a connective tissue sheath.
- The **arrector pili** smooth muscle raises the hair; a **root plexus** provides sensation.
- Density: about **600 hairs/cm²** on the face and **60/cm²** elsewhere.
- Growth cycles: **anagen** (growth), **catagen** (involution), **telogen** (rest).

Nails

Hard keratin plates on the dorsal surface of the distal phalanx. The **nail root** lies under a skin fold whose cornified layer forms the **cuticle (eponychium)**; keratinocytes proliferate in the **matrix** and differentiate into the **nail plate**, which slides forward over the **nail bed** (which does not contribute). The **lunula** is the visible part of the matrix.

Skin Glands

Sebaceous glands — holocrine: sebocytes accumulate lipid, die, and discharge **sebum** into the hair follicle (pilosebaceous unit). Absent from palms and soles.

Eccrine sweat glands — merocrine, coiled tubular, with a secretory portion of cuboidal cells and a duct of **stratified cuboidal** cells; abundant on palms and soles; watery sweat for thermoregulation.

Apocrine sweat glands — axillae and perineum, much **wider lumens**, develop after **puberty**, and secrete protein-rich sweat onto hair follicles; odor follows bacterial action.

Skin Repair

Repair runs through overlapping stages: a **clot** forms and platelets release growth factors; **neutrophils and macrophages** remove bacteria and debris (inflammation lasts **2–3 days**); fibroblasts proliferate under growth-factor and macrophage signals and produce new collagen while capillaries grow into the site, forming **granulation tissue**; **keratinocytes** from the epidermal basal layer migrate beneath and through the clot and restore epithelial continuity; and excess collagen is left as **scar**. Epithelialization is complete before the collagen of the scar remodels.

Standard clinical terminology: healing by **first intention** — clean, apposed edges needing minimal granulation tissue; **second intention** — an open wound with abundant granulation tissue, wound contraction by myofibroblasts, and a larger scar. **Burns** are graded by depth: first-degree (epidermis), second-degree (into dermis, blistering), third-degree (full thickness, including appendages, so it cannot re-epithelialize).

CLINICAL CORRELATION

Psoriasis — T-cell-driven accelerated production and differentiation of keratinocytes, with thickening, scaling, and a defective barrier. **Skin cancer** — about **one-third of all cancers** begin in skin; most arise from basal or spinous cells (basal cell and squamous cell carcinoma), are diagnosed early, and are rarely lethal; incidence rises with fair skin and solar exposure. **Melanoma** — moles can show dysplasia and progress to malignant melanoma, whose cells penetrate the basal lamina and metastasize through blood and lymph. **Merkel cell carcinoma** is **40 times less common** than melanoma but has **twice its mortality**. **Albinism** — defective tyrosinase or another melanin-pathway component; **vitiligo** — loss or reduced activity of melanocytes. **Pemphigus** (intercellular junctions) and **bullous pemphigoid** (dermal–epidermal junction) are the blistering diseases. **Cystic fibrosis** — salty sweat, because CFTR fails to reabsorb salt in the sweat duct.

UNDER THE MICROSCOPE

Count layers and look for appendages. **Thick skin: five layers, thick keratin, no hair. Thin skin: four layers, hair follicles and sebaceous glands.** A dark condensed nucleus at the basal layer with a clear halo suggests a melanocyte.

EXAM TIP

"Melanocytes are more numerous in dark skin" is false. **Unit density is similar in all people; melanin production and transfer differ.** Also remember that the **nail bed makes no contribution** to the nail plate — the matrix does.

MEMORY HOOK

"Come Let's Get Some Beer" — **C**orneum, **L**ucidum, **G**ranulosum, **S**pinosum, **B**asale (superficial to deep) — and remember **lucidum exists only in thick skin**.

Chapter 18 — summary

- Epidermal layers, deep to superficial: basale, spinosum, granulosum, lucidum (thick skin only), corneum.
- Keratinocytes make keratin; they are the most numerous epidermal cells.
- Melanocytes are neural crest cells that transfer melanin to keratinocytes.
- Langerhans cells are antigen-presenting cells of monocyte origin.
- Merkel cells are mechanoreceptors at the basal layer.
- Dermis = papillary (loose) + reticular (dense irregular); hypodermis is adipose and loose connective tissue.
- Eccrine sweat glands are merocrine; apocrine glands open into hair follicles; sebaceous glands are holocrine.
- Meissner corpuscles sense light touch; Pacinian corpuscles sense pressure and vibration.
- Thick skin has a stratum lucidum and no hair follicles; thin skin has hair and sebaceous glands.

Active recall

1. **Name the epidermal layers from deep to superficial.** — Basale, spinosum, granulosum, lucidum (thick skin only), corneum.
2. **Which epidermal cells are neural crest derived?** — Melanocytes and Merkel cells; Langerhans cells come from monocytes.
3. **What is the function of lamellar bodies?** — They discharge lipid into the intercellular space, forming the water barrier.
4. **Where are Meissner and Pacinian corpuscles?** — Meissner in the dermal papillae (light touch); Pacinian deep in the dermis/hypodermis (pressure, vibration).
5. **How do sebaceous and eccrine glands differ in secretion?** — Sebaceous: holocrine. Eccrine: merocrine.

Exam-style question

A skin section shows a thick stratum corneum, a clear layer, and no hair follicles. Which site is it from?

A. Scalp
B. Palm
C. Back
D. Axilla

Answer: B — Thick skin with a stratum lucidum and no hair follicles is palmar or plantar skin.

Practical identification

Identify the five epidermal layers in thick skin and the four in thin skin, and find the dermal papilla, the reticular dermis and the hypodermis. Name the appendages: hair follicle with its root sheaths and arrector pili, sebaceous gland (holocrine), eccrine gland (merocrine, coiled) and apocrine gland (wide lumen). Distinguish Meissner from Pacinian corpuscles by location, size and capsule.

PART V — EXCRETION AND REPRODUCTION

Chapters 19–23 · kidney, endocrine glands, and the reproductive systems

19 · The Urinary System

In this chapter: Kidneys · Blood Circulation · Renal Function: Filtration, Secretion, & Reabsorption · Ureters, Bladder, & Urethra

Learning objectives

By the end of this chapter you should be able to:

1. Trace filtrate from glomerulus to ureter, naming each segment and its epithelium.
2. Explain the three layers of the filtration barrier and how each contributes to selectivity.
3. Describe the juxtaglomerular apparatus and its role in blood-pressure control.
4. Identify urothelium and explain how it protects against hypertonic urine.

The core idea

The kidney is a **countercurrent filtration and reabsorption machine** built from millions of identical tubes. Each segment of the tubule has a distinct epithelium matched to a distinct transport job, so identifying the segment from its histology is the same as naming its function.

Kidneys

Each kidney is bean-shaped, about **12 cm long, 6 cm wide, and 2.5 cm thick** in adults, with a convex lateral border and a concave medial **hilum**, which admits nerves and vessels and lets the ureter and lymphatics exit. Inside the hilum the ureter expands as the **renal pelvis**, which divides into **two or three major calyces**, then **minor calyces**.

- **Capsule → cortex** (renal corpuscles + convoluted tubules) → **medulla** (loops of Henle, collecting ducts, vasa recta) organized into **8–15 pyramids** (some descriptions give 8–12; both counts are in use).
- **Renal columns (of Bertin)** — cortical tissue between pyramids.
- **Lobe** = pyramid + overlying cortex; **lobule** = a medullary ray + the cortical tissue around it.
- **Papilla** → minor calyx → major calyx → renal pelvis → ureter.
- The cortex receives **more than 10 times** the blood flow of the medulla.

The nephron — about **1.25 million per kidney** — begins with the **renal corpuscle**: the **glomerulus** (a tuft of fenestrated capillaries) inside the **glomerular (Bowman) capsule**.

- **Parietal layer** — simple squamous epithelium, continuous with the proximal tubule.
- **Visceral layer** — **podocytes** with primary processes and interdigitating **pedicels** that embrace the capillaries. Between pedicels are **filtration slit pores**, bridged by a zipper-like **slit diaphragm** of P-cadherin, **nephrins**, and the negatively charged glycoprotein **podocalyxin**.
- **Mesangial cells** — between capillaries, covering surfaces not covered by podocytes; **contractile** (like pericytes) and produce external lamina.

The filtration barrier has three layers: fenestrated endothelium → glomerular basement membrane (GBM), about **330 nm** thick, of **type IV collagen** and other proteins made by endothelial cells and podocytes → **podocyte slit diaphragms**. The barrier is both a size and a charge filter.

Functional numbers to know:

- Total filtration area ≈ **500 cm²**.
- **GFR ≈ 125 mL/min**, or about **180 L/day**.
- Plasma volume ≈ **3 L**, so the kidneys filter a volume equal to the whole plasma volume about **60 times a day**.

Juxtaglomerular apparatus (JGA) — at the vascular pole: **juxtaglomerular (JG) cells** in the afferent arteriole (modified smooth muscle, contain **renin**), the **macula densa** (a plaque of tall, closely packed cells in the distal straight tubule sensing NaCl), and **extraglomerular mesangial (lacis) cells**.

Blood Circulation

Renal artery → segmental → interlobar (between pyramids) → arcuate (at the corticomedullary junction) → interlobular (radiating into the cortex) → **afferent arteriole** → glomerulus → **efferent arteriole** → peritubular capillaries in the cortex and **vasa recta** in the medulla. The **efferent arteriole**, not a venule, drains the glomerulus — an arrangement that keeps glomerular hydrostatic pressure high. Efferent arterioles of **cortical** glomeruli branch into a diffuse peritubular network; those of **juxtamedullary** glomeruli form the long hairpin **vasa recta** that run alongside the loops of Henle and are the anatomical basis of the **countercurrent multiplier**.

Renal Function: Filtration, Secretion, & Reabsorption

Segment	Epithelium	Key features	Function
Proximal convoluted tubule (PCT)	Simple cuboidal, **eosinophilic**	**Long microvilli (brush border)**, abundant mitochondria, large interdigitating **basolateral folds**	**Major reabsorption**: all glucose and organic nutrients, all small proteins and peptides (degraded to amino acids), much water and electrolyte
Thin descending limb	Simple squamous	Thin	Water permeability
Thin ascending limb	Simple squamous	Thin	Passive NaCl reabsorption; much less water-permeable than the descending limb
Thick ascending limb (TAL = distal straight tubule)	Simple cuboidal, low brush border	Ends as the **macula densa** at the vascular pole	**NaCl reabsorption, impermeable to water** — the diluting segment and loop-diuretic site
Distal convoluted tubule (DCT)	Simple cuboidal, **pale**, no brush border	Many mitochondria, basal infoldings	Further adjustment of electrolytes; thiazide site; leads to **connecting tubules**
Collecting duct	Simple cuboidal → columnar in medulla	**Principal cells** (aquaporins, ADH) and **intercalated cells** (acid–base transport)	Final water and acid–base regulation

Size contrast: the straight proximal tubule is about **60 µm** in outer diameter and narrows abruptly to about **30 µm** in the thin limbs of the loop.

The visual contrast that answers most questions: the **proximal tubule is eosinophilic with a ragged brush border and a star-shaped lumen; the distal tubule is pale with a clean lumen and no brush border.**

Collecting ducts. Connecting tubules from several nephrons join to form cortical collecting ducts, which enter the medulla parallel to the loops and vasa recta and become larger with more columnar cells. The largest (papillary) ducts — **ducts of Bellini** — open into a minor calyx, where filtrate undergoes no further modification and is called **urine**.

Ureters, Bladder, & Urethra

Ureter — **urothelium (transitional epithelium)**, a folded mucosa, and a **thick muscularis** that moves urine by peristalsis; inner longitudinal and outer circular layers, with a third layer added near the bladder. **Bladder** — urothelium, a highly folded mucosa, and the **detrusor** muscle in three poorly delineated interlacing layers. It holds **400–600 mL**, with the urge to void at about **150–200 mL**.

Urothelium is a specialized barrier, not just a stratified epithelium:

- **Umbrella cells** — large superficial cells up to **100 µm** in diameter, with **uroplakin plaques ~16 nm** across, hinged to allow unfolding as the bladder fills.
- The epithelium is organized in **three layers**: superficial **umbrella cells**, an intermediate layer of cuboidal or low columnar cells (one to several deep), and a basal layer. Emptying folds the mucosa and lets umbrella cells retract folded plaques; filling flattens the cells and thins the epithelium.
- Tight junctions plus plaques make it an osmotic barrier against hypertonic urine.

Urethra — in both sexes lined initially by urothelium. In men there are three parts: **prostatic** (3–4 cm, urothelium), **membranous** (short, through the urogenital diaphragm; stratified or pseudostratified columnar), and **penile** (long; stratified columnar/pseudostratified proximally, stratified squamous distally). In women the urethra is **3–5 cm**, urothelium proximally, then nonkeratinized stratified squamous; the middle part is surrounded by the **external striated sphincter**. The short female urethra explains the frequency of cystitis.

CLINICAL CORRELATION

Polycystic kidney disease — inherited cysts arising from nephron epithelium destroy cortical organization. **Glomerulonephritis** — immune-complex or anti-glomerular antibody injury; **proteinuria** means the filter has become permeable to protein. **Diabetic glomerulosclerosis** — thickening and loss of function of the GBM; the leading cause of end-stage kidney disease in the United States. **Sickle cell nephropathy** — sickling in the vasa recta causes papillary infarcts. **Kidney stones** — calculi form in the pelvis or calyces, usually from calcium oxalate/phosphate or uric acid. **Pyelonephritis** — bacteria reach the papilla from a calyx; **cystitis** — the commonest bladder problem, favored in women by the short urethra.

UNDER THE MICROSCOPE

Start at the renal corpuscle: the **glomerulus** is a tuft of capillaries inside a capsule. Then compare tubules: **eosinophilic + brush border = proximal; pale + clean = distal; large pale cells in a row = collecting duct.** The **macula densa** is the patch of tall cells on the distal tubule facing the vascular pole.

EXAM TIP

The proximal tubule is the **most eosinophilic** structure in the renal cortex, packed with mitochondria for active transport. Students often call it "basophilic" because the brush border looks fuzzy — the cytoplasm is pink.

Chapter 19 — summary

- Kidney: **8–15 pyramids**; about **1.25 million nephrons**.
- Lobe = pyramid + cortex; lobule = medullary ray + surrounding cortex.
- Cortex receives **more than 10×** the blood flow of the medulla.
- Afferent arteriole → glomerulus → **efferent arteriole** (not a venule), keeping glomerular pressure high.
- Filtration area ≈ **500 cm²**; GFR ≈ **125 mL/min ≈ 180 L/day**.
- Filtration barrier = fenestrated endothelium + **GBM (type IV collagen, ~330 nm)** + slit diaphragms (nephrin, podocalyxin).
- Mesangial cells are contractile and support the capillary tuft.
- **PCT**: brush border, eosinophilic, reabsorbs all glucose and small proteins.
- **TAL**: water-impermeable, reabsorbs NaCl, ends as the **macula densa**.
- **JG cells** in the afferent arteriole secrete **renin**; macula densa senses NaCl.
- Collecting duct: **principal cells** (aquaporins, ADH) and **intercalated cells** (acid–base transport).
- Urothelium: three layers — umbrella cells with uroplakin plaques above intermediate and basal cells; it thins as the bladder fills.
- Bladder holds **400–600 mL**; urge at **150–200 mL**.
- Male urethra: prostatic, membranous, penile; female urethra 3–5 cm — short, hence cystitis.

Active recall

1. **Which nephron segment is most eosinophilic, and why?** — The proximal convoluted tubule: abundant mitochondria and a brush border for active reabsorption.
2. **What are the three layers of the filtration barrier?** — Fenestrated endothelium, glomerular basement membrane, and podocyte slit diaphragms.
3. **What does the macula densa sense, and what does the JGA release?** — NaCl concentration; juxtaglomerular cells release renin.
4. **Which cells of the collecting duct respond to ADH?** — Principal cells, by inserting aquaporins into the apical membrane.
5. **What makes urothelium a barrier?** — Umbrella cells with uroplakin plaques and tight junctions.

Exam-style question

A patient has proteinuria from podocyte injury. Which layer of the filtration barrier is primarily affected?

A. Fenestrated endothelium
B. Glomerular basement membrane
C. Podocyte slit diaphragms
D. Mesangial matrix

Answer: C — Podocyte foot processes and their slit diaphragms form the final size filter; their loss produces proteinuria.

Practical identification

In the cortex, distinguish proximal from distal convoluted tubules by cytoplasm, brush border and lumen; find a renal corpuscle and identify its parietal layer, podocytes and mesangial cells. In the medulla, identify thin and thick limbs and collecting ducts. Trace the vascular route from interlobular artery to peritubular capillary or vasa recta. In the bladder, identify urothelium with umbrella cells and the detrusor muscle.

20 · Endocrine Glands

In this chapter: Pituitary Gland (Hypophysis) · Adrenal Glands · Pancreatic Islets · Diffuse Neuroendocrine System · Thyroid Gland · Parathyroid Glands · Pineal Gland

Learning objectives

By the end of this chapter you should be able to:

1. Classify endocrine organs by their architectural pattern (cords, follicles, clusters).
2. Map each pituitary hormone to its cell type and hypothalamic control.
3. Match each adrenal zone to its hormone and trophic control.
4. Distinguish thyroid, parathyroid, and pineal tissue in a micrograph.

The core idea

Endocrine glands release hormone into the blood, so their histology is organized around **storage and release** rather than ducts. The three patterns to recognize are **cords** (pituitary, adrenal, parathyroid), **follicles** (thyroid), and **clusters** (pancreatic islets). Each pattern is dictated by how the hormone is stored.

Pituitary Gland (Hypophysis)

Two embryologically distinct parts in one sella turcica.

Adenohypophysis (oral ectoderm — **Rathke's pouch**). The **pars distalis accounts for 75%** of the adenohypophysis and is covered by a thin fibrous capsule:

Region	Cells	Hormones
Pars distalis	**Acidophils**	**GH, prolactin**
	Basophils	**ACTH, TSH, FSH, LH**
	Chromophobes	Inactive or degranulated
Pars intermedia	Basophils, colloid cysts	MSH (rudimentary in humans)
Pars tuberalis	—	Wraps the infundibulum

Reading the stain. Routine H&E divides the pars distalis into **acidophils** (pink), **basophils** (blue), and **chromophobes** (poorly stained, the least numerous). The modern names describe the hormone, not the color: **somatotrophs** (GH), **lactotrophs** (prolactin), **corticotrophs** (ACTH), **thyrotrophs** (TSH), and **gonadotrophs** (FSH, LH). Acidophils are the somatotrophs and lactotrophs; basophils are the corticotrophs, thyrotrophs, and gonadotrophs.

Neurohypophysis (neuroectoderm): the **pars nervosa** stores **oxytocin and ADH (arginine vasopressin)** — both **9-amino-acid peptides** — synthesized in the **supraoptic and paraventricular nuclei** of the hypothalamus and carried down axons. In the axon terminals they are bound to **10-kDa carrier proteins, neurophysin I and II**, and accumulate in dilatations called **neurosecretory (Herring) bodies**, visible as faintly eosinophilic structures. **Pituicytes** are the glial cells of the lobe. The neurohypophysis **makes no hormone of its own** — it stores and releases hypothalamic hormones.

Control: the hypothalamus releases **releasing and inhibiting hormones** into the **hypophyseal portal system**, carrying them directly to the pars distalis — which is why a tiny hypothalamic signal controls a large pituitary output.

Adrenal Glands

Two glands in one organ, with entirely different embryology and function:

Region	Cells	Hormone	Control
Zona glomerulosa	Rounded or arched cords under the capsule, **~15% of the cortex**	**Aldosterone**	**Angiotensin II, K^+**
Zona fasciculata	Long cords of pale, foamy cells	**Cortisol**	**ACTH**
Zona reticularis	Irregular cords, pigmented cells	**Androgens (DHEA)**	ACTH
Medulla	**Chromaffin cells** (neural crest)	**Epinephrine, norepinephrine**	Sympathetic splanchnic nerve

Each adrenal gland is a flattened half-moon about **4–6 cm long, 1–2 cm wide, and 4–6 mm thick**, weighing about **8 g** in total. The **fasciculata** is the **thickest** zone, and its cells are pale and vacuolated because they are full of **lipid droplets (cholesterol esters)**, **tubular mitochondrial cristae**, and **smooth ER** for steroid synthesis. The **medulla** is functionally a sympathetic ganglion without axons: **chromaffin cells** contain electron-dense granules **150–350 nm** in diameter, storing catecholamines with Ca^{2+}, ATP, and **49-kDa chromogranins**. About **80%** of the catecholamine released is **epinephrine**. Mnemonic for the zones: **"GFR — Salt, Sugar, Sex"** from outside in.

Pancreatic Islets

Pale, round clusters between the acini of the exocrine pancreas (Chapter 16). The pancreas has **more than 1 million islets**, mostly in the tail, but they occupy only **1–2%** of the organ's volume. Most are **100–200 µm** in diameter and contain several hundred cells. There are **four major cell types**:

Cell	Product	Notes
β (beta)	**Insulin**	About **twice as abundant** as α cells; scattered in human islets
α (alpha)	**Glucagon**	Abundant and scattered through most islets
δ (delta)	**Somatostatin**	Paracrine inhibition
PP	Pancreatic polypeptide	—

Islets are richly vascularized, and their hormones pass directly into the bloodstream rather than into ducts.

Diffuse Neuroendocrine System

Scattered hormone-secreting cells throughout the body — gut enteroendocrine cells, respiratory neuroendocrine cells, and similar — sharing the ability to take up amine precursors (**APUD** cells). They are the reason a "pure" endocrine organ is not the only source of hormones: the gut alone produces more regulatory peptides than any single gland.

Thyroid Gland

- **Follicles** of varying size filled with **colloid** (thyroglobulin), lined by a simple epithelium whose height ranges from **squamous to low columnar**.
- **Thyrocytes (follicular cells)** — synthesize **T4 and T3**. Activity is read from the epithelium: **active gland = more follicles of tall columnar cells with scalloped colloid; hypoactive gland = squamous cells with dense colloid.**
- **Parafollicular (C) cells** — large, pale cells in the follicular epithelium or between follicles, of **neural crest** origin, secreting **calcitonin**.
- **Folliculogenesis:** thyroglobulin is synthesized, iodinated at the apical surface, stored as colloid, taken back by endocytosis, and digested to release T3/T4.

Parathyroid Glands

Four small ovoid glands, each about **3 × 6 mm**, with a total weight of about **0.4 g**, usually embedded in the thyroid capsule.

- **Principal (chief) cells** — small, pale cells with round central nuclei, arranged in dense cords around capillaries; they secrete **PTH**.
- **Oxyphil cells** — much larger, intensely eosinophilic cells packed with abnormally shaped mitochondria. They are **degenerated derivatives of principal cells**, some still showing low-level PTH synthesis, and they accumulate with age.
- With age, adipocytes replace secretory cells and may constitute **more than 50%** of the gland.
- Cords and capillaries only — **no follicles**.

Pineal Gland

A pine-cone-shaped organ about **5–8 × 3–5 mm**, attached to the third ventricle and covered by pia mater.

- **Pinealocytes** — modified neurons in cords and lobules, with pale nuclei and long processes ending near capillaries.
- **Corpora arenacea (brain sand)** — concretions of calcium and magnesium salts formed on extracellular protein deposits; they appear in childhood and increase with age without affecting function. Their density is why the gland is radio-opaque and easy to identify on imaging.
- **Melatonin** — a tryptophan derivative released in **darkness** under **sympathetic** control; it sets diurnal rhythms.
- Interstitial (glial) cells lie between pinealocytes; unmyelinated sympathetic fibers run in the septa.

CLINICAL CORRELATION

Pituitary adenoma — benign tumors of acidophils or basophils; a somatotroph adenoma causes **gigantism** before epiphyseal closure and **acromegaly** afterwards. **Diabetes insipidus** — deficient ADH (mutation, tumor compression, or head trauma) produces copious dilute urine. **Addison disease** — autoimmune destruction of the adrenal cortex with loss of glucocorticoid, mineralocorticoid, and androgen output. **Pheochromocytoma** — a benign chromaffin-cell tumor that discharges catecholamines episodically, causing swings between hypertension and hypotension. **Diabetes mellitus** — type 1 from autoimmune β-cell loss; type 2 from insulin resistance with relative insulin deficiency. **Goiter** — dietary iodine deficiency lowers thyroid hormone output, so TSH rises and the gland enlarges. **Graves disease** — autoantibodies chronically stimulate follicular cells, producing hyperthyroidism. **Hypoparathyroidism** — low PTH leaves bone over-mineralized and causes tetanic muscle contraction; **hyperparathyroidism** — excess PTH drives osteoclasts and hypercalcemia. **Pineal tumors** are rare but may be benign or highly malignant.

UNDER THE MICROSCOPE

Three patterns, three organs. **Follicles of colloid = thyroid. Cords of pale foamy cells under a capsule = adrenal cortex; the medulla is paler and central. Small pale chief cells in dense cords = parathyroid.** Pituitary: **acidophils and basophils in sheets (anterior); pale fibrillary tissue without epithelial cells (posterior).**

EXAM TIP

The adrenal cortex has **three** zones and the medulla has **none** of the cortex's function. Catecholamines are **medullary**.

Chapter 20 — summary

- Pars distalis = **75%** of the adenohypophysis; acidophils, basophils, chromophobes.
- Acidophils = somatotrophs (GH) and lactotrophs (prolactin); basophils = corticotrophs, thyrotrophs, gonadotrophs.
- Hypothalamic releasing and inhibiting hormones reach the pars distalis through the **hypophyseal portal system**.
- Neurohypophysis stores **oxytocin and ADH** made in the supraoptic and paraventricular nuclei.

- ADH and oxytocin are **9-amino-acid peptides** carried by **neurophysin I and II**; Herring bodies are their axonal dilatations.
- Pituicytes are the glial cells of the pars nervosa.
- Adrenal cortex: glomerulosa (**aldosterone**) → fasciculata (**cortisol**) → reticularis (**androgens**).
- Zona fasciculata is the **thickest** zone; glomerulosa is ~15% of the cortex.
- Adrenal medulla **= chromaffin cells**, neural crest, **80% epinephrine**.
- Pancreatic islets: **β insulin, α glucagon, δ somatostatin, PP pancreatic polypeptide**; β about twice as abundant as α.
- Thyroid: follicles of **colloid (thyroglobulin)**; active epithelium is columnar with scalloped colloid, inactive is squamous.
- **Parafollicular C cells** secrete **calcitonin**.
- Parathyroid: **chief cells** (PTH) and **oxyphil cells** (function uncertain).
- Pineal: **pinealocytes** and **corpora arenacea**; melatonin under sympathetic control.

Active recall

1. **Which pituitary cells are acidophils?** — Somatotrophs (GH) and lactotrophs (prolactin).
2. **Where are oxytocin and ADH made and stored?** — Made in the supraoptic and paraventricular nuclei; stored and released in the pars nervosa.
3. **Name the adrenal zones and their main hormones.** — Glomerulosa: aldosterone; fasciculata: cortisol; reticularis: androgens.
4. **Which islet cells secrete insulin, and how abundant are they?** — β cells; about twice as abundant as α cells.
5. **What are the two cell types of the parathyroid, and which secretes PTH?** — Principal (chief) cells secrete PTH; oxyphil cells are larger, acidophilic, and increase with age.

Exam-style question

A patient has episodic hypertension and palpitations. A tumor is found in the adrenal medulla. Which cells are proliferating?

A. Zona glomerulosa cells
B. Chromaffin cells
C. Zona fasciculata cells
D. Pituitary corticotrophs

Answer: B — Pheochromocytoma is a chromaffin-cell tumor releasing catecholamines.

Practical identification

Compare the endocrine patterns: cords (pituitary, adrenal, parathyroid), follicles (thyroid) and clusters (pancreatic islets). In the pituitary, distinguish acidophils, basophils and chromophobes, and find the pars nervosa; in the adrenal, name the three zones and the medulla; in the thyroid, find colloid, follicular cells and parafollicular cells; in the parathyroid, chief and oxyphil cells; in the pineal, pinealocytes and corpora arenacea.

21 · The Male Reproductive System

In this chapter: Testes · Intratesticular Ducts · Excretory Genital Ducts · Accessory Glands · Penis

Learning objectives

By the end of this chapter you should be able to:

1. Describe the compartments of the seminiferous tubule and the blood–testis barrier.
2. Sequence spermatogenesis and spermiogenesis and identify each stage.
3. Trace sperm from tubule to urethra, naming the epithelium at each level.
4. Explain the vascular mechanism of erection and ejaculation.

The core idea

The testis does two jobs that require opposite conditions: **producing sperm**, which needs a temperature below core (hence the scrotum), and **producing testosterone**, which needs a rich blood supply. The architecture keeps them apart by separating the **seminiferous tubules** from the **interstitium**.

Testes

- **Tunica albuginea** — thick capsule; the **mediastinum testis** sends septa dividing the gland into about **250 lobules**, each holding **1–4** convoluted seminiferous tubules. Each testis has **250–1000** tubules; each tubule is **150–250 µm** wide and **30–70 cm** long, for a total of about **250 m** per testis.
- **Tunica vaginalis** — the serous sac the testis drags with it during descent: parietal layer lines the scrotum, visceral layer covers the albuginea.
- **Seminiferous tubules** — lined by a **stratified epithelium of germ cells** plus **Sertoli cells**, on a basal lamina, surrounded by **peritubular myoid cells**.
- **Interstitium** — **Leydig cells**, blood vessels, connective tissue.

Temperature is part of the anatomy. Sperm formation cannot proceed at 37 °C, so the scrotum holds the testis at about **34 °C** by three mechanisms: evaporation of sweat from scrotal skin; a **countercurrent heat exchanger** in the spermatic cord, where the **pampiniform plexus** of veins cools the incoming testicular artery; and the **dartos** and **cremaster** muscles, which move the testis toward or away from the body.

Sertoli cells extend from base to lumen; their nuclei are pale, oval, and have a prominent nucleolus. They are **not** germ-line cells, and each one supports **30–50** developing germ cells. Functions:

- **Physical support and nutrition** of developing germ cells.
- **Blood-testis barrier** — tight junctions between adjacent Sertoli cells near the base, creating basal and adluminal compartments. This is why the immune system does not reject sperm.
- **Phagocytosis** of residual bodies.
- **Secretion** of **androgen-binding protein (ABP)**, **inhibin** (a 39-kDa glycoprotein that suppresses FSH), **Müllerian-inhibiting substance** (MIS, 140 kDa, which regresses the paramesonephric ducts in the fetus), **transferrin**, and fluid that carries sperm out of the testis.
- **Response to FSH** — which acts on Sertoli cells, not on germ cells.

Spermatogenesis: spermatogonia → primary spermatocyte → secondary spermatocyte → spermatid → spermatozoon. Meiosis I halves the chromosome number and **meiosis II** separates chromatids, giving four spermatids per primary spermatocyte. The germ cells do not divide in isolation: cytokinesis is incomplete, so clones stay joined by **cytoplasmic bridges** — a syncytium that lets a haploid cell use products of the full diploid genome, including genes on X or Y. The whole sequence from the last spermatogonial mitosis to spermatid formation takes **about 2 months**, and

the testis produces about **200 million sperm per day**.

Spermiogenesis is the metamorphosis of a round spermatid into a sperm, and it is conventionally divided into **four phases**:

1. **Golgi phase** — proacrosomal vesicles from the Golgi coalesce into a single **acrosomal cap**; centrioles migrate to the opposite pole and one becomes the basal body of the flagellum.
2. **Cap phase** — the acrosome spreads over about half the nucleus. It is a specialized **lysosome** containing **hyaluronidase** and the trypsin-like protease **acrosin**.
3. **Acrosome phase** — the nucleus condenses and elongates, its histones replaced by **protamines**; a microtubule **manchette** shapes the head; mitochondria gather around the proximal flagellum to form the **middle piece**.
4. **Maturation phase** — excess cytoplasm is shed as a **residual body**, intercellular bridges are lost, and the sperm is released into the lumen.

Because the barrier makes the adluminal compartment immunologically privileged, sperm antigens are not tolerized — which is why vasectomy can produce anti-sperm antibodies.

Leydig (interstitial) cells — large, polygonal, **eosinophilic** cells with central nuclei, abundant SER, mitochondria, and lipid droplets, lying singly or in small groups near capillaries. They secrete **testosterone** under **LH**, also called **interstitial cell-stimulating hormone (ICSH)** — the same hormone named LH in females. Fetal Leydig cells are active in the third and fourth months of pregnancy, then become quiescent until puberty.

Intratesticular Ducts

Sperm leave each tubule through a short **straight tubule**, lined at first only by Sertoli cells and then by simple cuboidal epithelium. It enters the **rete testis**, a labyrinth of cuboidal-lined channels in the mediastinum testis. From there **10–20 efferent ductules** carry sperm to the head of the epididymis. Their epithelium has a characteristic **scalloped** profile in section — patches of absorptive, nonciliated cuboidal cells with microvilli alternating with patches of taller ciliated cells. Absorption of Sertoli-cell fluid plus ciliary beating creates the flow that moves sperm out of the testis.

Excretory Genital Ducts

Segment	Epithelium	Function
Epididymis	**Pseudostratified columnar with stereocilia**	**Maturation and storage of sperm**; about **6 m** long, and transit takes **2–4 weeks**
Ductus (vas) deferens	Pseudostratified columnar	Transport; **very thick muscular wall** (inner longitudinal, middle circular, outer longitudinal)
Ejaculatory duct	Columnar	Delivers into the prostatic urethra

Accessory Glands

- **Seminal vesicles** — paired, highly tortuous tubes about **15 cm** long, with thin complex mucosal folds and inner circular plus outer longitudinal smooth muscle. Their viscid yellowish secretion makes up about **70%** of the ejaculate and supplies **fructose** (sperm energy), inositol, citrate, **prostaglandins** (which stimulate the female tract), and **fibrinogen** (which lets semen coagulate).
- **Prostate** — a **20-g** organ of **30–50 branched tubuloacinar glands** in a dense **fibromuscular stroma**; **corpora amylacea** are small spherical concretions (up to **2 mm**, partly calcified, made of glycoproteins and keratan sulfate) that become more numerous with age but have no clinical significance. The gland has three zones around the urethra:
- **Peripheral zone** — about **70%** of the tissue, with the longest ducts; the site of most **carcinomas**.
- **Central zone** — about **25%**, with submucosal glands.
- **Transition zone** — about **5%**, with mucosal glands around the upper urethra; the site of **benign prostatic hyperplasia**.

- **Prostate-specific antigen (PSA)** — a **34-kDa serine protease** that liquefies the coagulated ejaculate; it leaks into blood, so a raised serum PSA points to carcinoma or inflammation.
- **Bulbourethral (Cowper) glands** — paired, **3–5 mm** mucous glands in the urogenital diaphragm; their clear secretion lubricates the urethra before ejaculation.

Penis

Three erectile bodies: two dorsal **corpora cavernosa** and one ventral **corpus spongiosum** containing the urethra and expanding distally as the **glans**. Each is a sponge of **endothelium-lined cavernous spaces** separated by trabeculae of smooth muscle and connective tissue, with a dense **tunica albuginea** around the corpora cavernosa. Most of the penile urethra is **pseudostratified columnar**, becoming **stratified squamous** in the glans.

Erection is a vascular event with a molecular explanation worth memorizing: parasympathetic acetylcholine makes the endothelium release **nitric oxide**, which activates **guanylate cyclase** and raises **cGMP** in smooth muscle; the muscle relaxes, **helicine arteries** dilate, cavernous spaces fill, and the swollen corpora compress the dorsal veins against the tunica albuginea. **Sildenafil** blocks the phosphodiesterase that degrades cGMP and therefore sustains an erection. Sympathetic discharge at ejaculation constricts the helicine arteries and trabecular muscle, emptying the spaces.

CLINICAL CORRELATION

Cryptorchidism — failure of descent, present in about **4%** of male neonates; most descend in the first year, and bilateral cases cause infertility if not corrected by **2–3 years**. An undescended testis is exposed to core temperature, so spermatogenesis fails and malignancy risk rises. **Hydrocele** — serous fluid in the tunica vaginalis, the commonest cause of scrotal swelling. **Varicocele** — dilation of the pampiniform plexus raises testicular temperature and impairs spermatogenesis. **Prostatic carcinoma** — arises in the peripheral zone, is palpable on rectal examination, and metastasizes to bone. **BPH** — nodular hyperplasia of the transition zone, compressing the urethra. **Testicular tumors** — **95%** are germ-cell tumors, appearing after puberty and more often in untreated cryptorchidism; a painless mass is malignant until proven otherwise.

UNDER THE MICROSCOPE

Identify the two compartments instantly. **Inside the tubule: Sertoli cells (pale, tall) and germ cells at various stages. Outside the tubule: Leydig cells (eosinophilic, polygonal, in clusters).** If asked for the cell that forms the blood–testis barrier, the answer is Sertoli.

EXAM TIP

FSH acts on **Sertoli** cells; LH acts on **Leydig** cells. One sentence, and the two are hard to confuse.

Chapter 21 — summary

- About 250 lobules per testis, each with 1–4 seminiferous tubules, 150–250 µm wide and 30–70 cm long.
- Spermatogenesis takes about 2 months and yields about 200 million sperm per day.
- Sertoli tight junctions form the blood–testis barrier and divide basal from adluminal compartments.
- Sertoli cells secrete ABP (FSH-dependent), inhibin, MIS, and transferrin, and phagocytose residual bodies.
- Leydig cells secrete testosterone under LH (ICSH).
- Spermiogenesis has four phases: Golgi, cap, acrosome, maturation.
- The acrosome is a lysosome containing hyaluronidase and acrosin; protamines replace histones.
- Straight tubules → rete testis → efferent ductules → epididymis → ductus deferens → ejaculatory duct.
- Seminal vesicles contribute about 70% of semen; the prostate contributes PSA and corpora amylacea.

- Prostate zones: peripheral 70% (cancer), central 25%, transition 5% (BPH).
- Erection is parasympathetic, NO–cGMP mediated; ejaculation is sympathetic.

Active recall

1. **What forms the blood-testis barrier, and what does it separate?** — Tight junctions between Sertoli cells; basal and adluminal compartments.
2. **Which hormone acts on Sertoli cells, and which on Leydig cells?** — FSH on Sertoli; LH on Leydig.
3. **Name the four phases of spermiogenesis.** — Golgi, cap, acrosome, maturation.
4. **What epithelium lines the epididymis, and what is its hallmark?** — Pseudostratified columnar with stereocilia.
5. **What are corpora amylacea, and where are they found?** — Calcified glycoprotein concretions in prostatic glands.
6. **Which prostatic zone gives rise to cancer, and which to benign hyperplasia?** — Cancer: peripheral (70% of the gland). Benign hyperplasia: transition (5%).
7. **What cools the testicular arterial blood, and how?** — The pampiniform plexus, by countercurrent heat exchange in the spermatic cord.

Exam-style question

A 25-year-old man has a painless testicular mass. Which cell type is most likely the origin of a germ-cell tumor?

A. Leydig cell
B. Sertoli cell
C. Spermatogenic cell
D. Peritubular myoid cell

Answer: C — Germ-cell tumors arise from the spermatogenic lineage; Leydig and Sertoli tumors are sex-cord stromal tumors and less common.

Practical identification

In the testis, identify seminiferous tubules, Sertoli cells, germ cells at each stage, peritubular myoid cells and Leydig cells. Follow sperm from tubule to rete testis, efferent ductules (scalloped epithelium), epididymis (pseudostratified with stereocilia), vas deferens (thick muscle) and prostate (tubuloacinar glands, corpora amylacea). Distinguish the prostatic zones by gland architecture and stroma.

22 · The Female Reproductive System

In this chapter: Ovaries · Uterine Tubes · Major Events of Fertilization · Uterus · Embryonic Implantation, Decidua, & the Placenta · Cervix · Vagina · External Genitalia · Mammary Glands

Learning objectives

By the end of this chapter you should be able to:

1. Sequence follicular development from primordial follicle to corpus albicans.
2. Explain the two-cell, two-gonadotropin mechanism of estrogen production.
3. Correlate ovarian and endometrial cycles and identify each phase histologically.
4. Describe implantation, placental structure, and the hormonal control of pregnancy.

The core idea

The ovary and the endometrium are **rebuilt through a monthly cycle**. Every histological feature of the ovary, tube, uterus, and vagina is a stage in one of two cycles — **ovarian** (follicle to corpus luteum) and **endometrial** (proliferative to secretory) — locked together by the same four hormones.

Ovaries

Each ovary is an almond-shaped body about **3 cm long, 1.5 cm wide, and 1 cm thick**.

- **Germinal (surface) epithelium** — simple cuboidal mesothelium (a misnomer; it does not produce germ cells).
- **Tunica albuginea** — dense connective tissue.
- **Cortex** — follicles at all stages, plus corpus luteum and corpus albicans.
- **Medulla** — loose connective tissue with the vessels entering at the hilum. Cortex and medulla have no sharp border.

The oocyte supply is fixed before birth and only falls. Primordial germ cells reach the gonads in the first month; oogonia peak at about **7 million** in the fifth month of gestation. About **2–2.5 million** primordial follicles are present at birth and about **400,000** at puberty; only about **450** are ovulated in a lifetime, and the rest undergo atresia. Menopause arrives when the reserve falls below about **1000** follicles.

Stage	Structure	Hormonal influence
Primordial follicle	Oocyte + single layer of flat granulosa cells	None; present from birth
Primary follicle	Oocyte enlarges (to ~120 µm); granulosa cells become **cuboidal** (unilaminar), then stratified (multilaminar); **zona pellucida** appears	FSH-dependent from this stage
Secondary (antral) follicle	Antrum forms among the granulosa cells; **theca interna** (steroidogenic, vascular) and **theca externa** (fibrous, contractile) differentiate	FSH and LH
Graafian (mature) follicle	Single large antrum up to **2 cm**; oocyte on the **cumulus oophorus**, wrapped by the **corona radiata**	LH surge triggers ovulation
Atretic follicle	Degenerating; **glassy membrane** remains	—
Corpus luteum	**Granulosa-lutein cells (progesterone, ~80% of the gland)** and **theca-lutein cells (estrogens)**	LH maintains it
Corpus albicans	White fibrous scar	—

The corpus luteum is a temporary gland with two cell populations:

- **Granulosa-lutein cells** — the old granulosa cells; they enlarge to **20–35 µm** without dividing, make up about **80%** of the gland, and secrete **progesterone** (their SER expands, and they retain

some aromatase).

- **Theca-lutein cells** — from the theca interna; about **half the size**, darker staining, lying in the peripheral folds, and secreting **androstenedione** and **estrogens**.

The LH surge sustains progesterone secretion for **10–12 days**. (A common error is to reverse the two populations: the granulosa-lutein cells make progesterone, and the theca-lutein cells make androgens and estrogens.) Without pregnancy, both populations undergo apoptosis, macrophages clear the debris, and fibroblasts lay down a collagen scar — the **corpus albicans**. Note that **involution is not atresia**: atresia is the death of an unovulated follicle.

The zona pellucida is a **5–10 µm** glycoprotein coat between the oocyte and the granulosa cells, secreted by the oocyte. Its glycoproteins **ZP1, ZP2, and ZP3**, all made by the oocyte, form the sperm-binding receptor; binding of a capacitated sperm to this coat begins the acrosomal reaction. In humans the family also includes ZP4.

Ovulation, step by step. The LH surge triggers: completion of **meiosis I** with extrusion of the **first polar body**; granulosa production of fluid rich in prostaglandins, proteoglycans, and proteases; **hyaluronan** release that loosens the cumulus; activation of **plasmin**, which digests collagen at the **stigma** (the ischemic bulge on the ovarian surface); and contraction of **theca externa** smooth muscle. The follicle ruptures, releasing the **secondary oocyte, arrested in metaphase II**, wrapped in the **corona radiata** — viable for about **24 hours**. If a sperm arrives, the oocyte completes meiosis II and emits the **second polar body**; the two pronuclei fuse to form the **zygote**.

A mature follicle takes about **90 days** to develop from a primordial follicle, and only one per cycle reaches that stage while its neighbors become atretic.

Two-cell, two-gonadotropin theory: LH stimulates **theca interna** cells to make **androstenedione**; it crosses the basement membrane (the granulosa layer is avascular) and **FSH** stimulates **granulosa** cells to **aromatize** it to **estradiol**. The estrogen returns to the theca capillaries and reaches the whole body. This division of labor explains why both gonadotropins are needed for normal estrogen production — and why the granulosa layer must stay avascular.

Uterine Tubes

About **10–12 cm** long, in four parts — **infundibulum with fimbriae → ampulla (longest; fertilization happens here) → isthmus → intramural portion**. The mucosa has complex branching folds, tallest in the ampulla, and is lined by **ciliated columnar cells** that sweep fluid toward the uterus and **peg (secretory) cells** that make the nutritive fluid. The muscularis has interwoven circular and longitudinal layers and provides most of the transport — women with immotile cilia still carry the embryo to the uterus.

Major Events of Fertilization

Fertilization normally occurs in the **ampulla**, and only a few hundred of the millions of ejaculated sperm reach it. The sequence:

1. **Capacitation** in the female tract makes sperm able to fertilize.
2. The sperm releases **hyaluronidase** to pass between the corona radiata cells.
3. At the zona, binding to the **ZP1–ZP3** complex triggers the **acrosomal reaction**: **acrosin** and other proteases digest a path through the zona.
4. The first sperm to cross fuses with the oolemma. Membrane depolarization and a **Ca^{2+} wave** trigger the **cortical reaction** — proteases from cortical granules modify the zona glycoproteins and convert the zona into the impenetrable **perivitelline barrier**, a permanent block to polyspermy.
5. The oocyte completes **meiosis II**, emits the **second polar body**, and forms the female pronucleus; the sperm head decondenses into the male pronucleus. Fusion gives the diploid **zygote**.

Cleavage follows as the embryo travels to the uterus — about **5 days** from fertilization to the blastocyst in the uterine cavity.

Uterus

The uterus is pear-shaped: **body**, **fundus**, **isthmus**, and **cervix**, with the cervical canal opening at the internal and external os.

- **Perimetrium (serosa)**, **myometrium** (thick smooth muscle in interlacing bundles), **endometrium**.
- **Endometrium** — **stratum functionalis** (shed each month) and **stratum basalis** (retained, regenerates it). The basal layer has a distinctive stroma: fibroblasts, ground substance, **fine type III collagen**, and little or no adipose tissue. **Spiral arteries** supply the functionalis and constrict to cause shedding.

Phase	Days	Hormone	Histology
Menstrual	1–4	Falling progesterone	Shedding, hemorrhage, neutrophilic infiltrate
Proliferative	5–14	**Estrogen**	Glands straight and narrow, mitoses, dense stroma
Secretory	15–28	**Progesterone**	Glands **tortuous and dilated with secretion**, edematous stroma, coiled spiral arteries

The single best discriminator: **straight, narrow glands = proliferative; saw-toothed, dilated, secreting glands = secretory.**

Thickness is a second discriminator. At the end of the proliferative phase the endometrium is **2–3 mm**; in the secretory phase it reaches **5–6 mm**, and the functional layer becomes about **four times** thicker than the basal layer.

How menstruation happens. Progesterone and estrogen fall **8–10 days after ovulation**. The spiral arteries contract and relax intermittently, then constrict under local **prostaglandins**, producing ischemia. Hypoxic cells release cytokines that attract leukocytes, whose **collagenases and matrix metalloproteinases** digest the matrix. The functional layer — epithelium, most of each gland, stroma, and blood-filled lacunae — separates from the basal layer and is shed. The basal layer, which does not depend on the spiral arteries, survives and regenerates the next functionalis under estrogen.

Embryonic Implantation, Decidua, & the Placenta

The blastocyst reaches the uterine cavity about **5 days** after fertilization and floats in the glandular secretion for about **2 days**. Implantation (nidation) takes about **3 days** and is complete by about **day 9**: trophoblast receptors bind glycoproteins on the endometrial epithelium, the **syncytiotrophoblast** digests the stroma, and the embryo embeds. The trophoblast has two layers — **cytotrophoblast** (mitotic, inner) and **syncytiotrophoblast** (invasive, multinucleated, outer, and the source of **hCG**).

Chorionic villi project into maternal blood in the **intervillous space** and develop in three stages: **primary** villi (cytotrophoblast covered by syncytiotrophoblast) about 2 days after implantation; **secondary** villi (mesenchymal cores) about day 15; **tertiary** villi (with capillary loops) a few days later. By the end of the first month there are thousands of tertiary villi.

Decidua — the modified endometrium of pregnancy: **decidua basalis** (under the embryo, forms the maternal placenta), **decidua capsularis** (over it), **decidua parietalis** (the rest).

Placental hormones: hCG (maintains the corpus luteum early; the basis of pregnancy tests), a **lactogen**, **relaxin**, growth factors, progesterone, and estrogen. hCG keeps the corpus luteum of pregnancy alive for **4–5 months**, by which time the placenta itself makes enough progesterone.

Cervix

The cervix is the lower cylindrical part of the uterus and differs from the body: its mucosa is **2–3 mm** thick, has **no spiral arteries**, and **does not shed** with the cycle.

- **Endocervix** — simple columnar mucus-secreting cells with large branched **cervical glands**.
- **Ectocervix** — **stratified squamous** nonkeratinized epithelium, continuous with the vagina.
- **Transformation zone** — the area just outside the external os where columnar epithelium is replaced by squamous; the site of HPV-related dysplasia and the target of the **Pap smear**.

Cervical mucus changes with hormones: watery and abundant at ovulation (helps sperm), viscous in the luteal phase (hinders them), and a thick **mucus plug** in pregnancy. Before birth, **effacement** dissolves collagen so the cervix softens and dilates.

Vagina

Stratified squamous nonkeratinized epithelium **150–200 µm** thick and rich in **glycogen**, with **no glands** (lubrication comes from cervical and Bartholin glands) and a lamina propria with elastic fibers and venous plexuses. **Lactobacilli** metabolize glycogen to lactic acid, keeping the pH low and protecting against pathogens.

External Genitalia

The **vulva**: the **labia majora** (skin with hair and sebaceous glands, containing the homologue of the scrotum), **labia minora** (thin skin without hair, with sebaceous glands), the **clitoris** (erectile tissue homologous to the penis), and the **vestibule** containing the urethral and vaginal openings and the openings of the **greater vestibular (Bartholin) glands**, which provide lubrication.

Mammary Glands

A modified **apocrine sweat gland** whose functional unit is the **terminal duct lobular unit (TDLU)**.

- **Non-lactating** — mostly dense connective tissue and fat with small ducts and lobules.
- **Pregnancy** — lobules proliferate under estrogen and progesterone.
- **Lactation** — alveoli distend with secretion; **protein is released by exocytosis** and **lipid droplets by apocrine secretion**, so milk fat is wrapped in apical plasma membrane. Prolactin drives synthesis, oxytocin the ejection reflex.
- **Duct system** — intralobular → interlobular → **lactiferous ducts** → **lactiferous sinuses** beneath the areola → nipple.

Hormonal control: **estrogen** for ductal growth, **progesterone** for lobuloalveolar development, **prolactin** for milk synthesis, **oxytocin** for milk ejection.

CLINICAL CORRELATION

Polycystic ovary syndrome — multiple arrested antral follicles, anovulation, hyperandrogenism. **Endometriosis** — functional endometrial tissue outside the uterus, bleeding cyclically. **Leiomyoma (fibroid)** — a benign smooth-muscle tumor of the myometrium. **Endometrial carcinoma** — unopposed estrogen, with atypical hyperplasia. **Cervical carcinoma** — HPV in the transformation zone. **Ectopic pregnancy** — implantation outside the uterine cavity, usually in the ampulla. **Breast carcinoma** — arises from ductal or lobular epithelium; the **terminal duct lobular unit** is the functional unit to remember.

UNDER THE MICROSCOPE

Place the ovary in the cycle by the **dominant structure**: many small primordial follicles = inactive/early; a large antral follicle = late follicular; a corpus luteum with folded, pale, luteinized cells = secretory phase; a hyaline scar = post-ovulatory. In the uterus, look at the **glands**, not the stroma.

MEMORY HOOK

"Proliferative = Pencil-thin glands; Secretory = Saw-tooth and swollen."

COMMON ERROR — Some summaries reverse the two lutein cell populations, giving estrogen to the granulosa-lutein cells. The correct assignment, and the one that follows from the two-cell mechanism, is **progesterone from the granulosa-lutein cells and estrogens from the theca-lutein cells**.

EXAM TIP
The corpus luteum secretes **progesterone**, and it is maintained by **LH** (and, in pregnancy, by **hCG**). Do not say estrogen maintains the corpus luteum.

Chapter 22 — summary

- Ovary has three regions: cortex with follicles, medulla with vessels, and a surface of **germinal epithelium** over a tunica albuginea.
- Follicle sequence: primordial → primary → secondary (antral) → mature (Graafian) → ovulation.
- **Zona pellucida** = glycoprotein coat around the oocyte; **corona radiata** = granulosa cells carried with it at ovulation.
- **Theca interna** makes androgens; **granulosa cells** aromatize them to estrogen under FSH.
- **Cumulus oophorus** = granulosa mound carrying the oocyte into the antrum.
- After ovulation: **corpus luteum** (progesterone) → **corpus albicans** (hyaline scar).
- Uterine tube: **infundibulum with fimbriae → ampulla → isthmus**; fertilization in the ampulla.
- Endometrium has a **functional layer** that sheds and a **basal layer** that regenerates it.
- Proliferative phase = estrogen, straight narrow glands, mitoses; secretory phase = progesterone, saw-toothed glands, edema.
- Menstrual phase = spiral artery constriction and functionalis shedding.
- Cervix: columnar endocervix, stratified squamous ectocervix, squamocolumnar junction = transformation zone.
- Vagina = stratified squamous nonkeratinized epithelium with glycogen, no glands; acidity comes from lactobacilli.
- Mammary gland: **TDLU** is the functional unit; estrogen drives ducts, progesterone drives lobuloalveolar growth, prolactin drives milk synthesis.

Active recall

1. **Which cells produce estrogen, and from what precursor?** — Granulosa cells aromatize androgens from theca interna cells.
2. **What is released at ovulation, and at what meiotic stage?** — The secondary oocyte, arrested in metaphase II, with the corona radiata.
3. **What prevents polyspermy?** — Cortical granule exocytosis modifies the zona pellucida into the perivitelline barrier.
4. **Which hormone maintains the corpus luteum early in pregnancy?** — hCG.
5. **How do proliferative and secretory endometrium differ?** — Proliferative: straight narrow glands, mitoses. Secretory: tortuous dilated glands with secretion and edematous stroma.

Exam-style question

A curettage specimen shows tortuous, saw-toothed glands with secretion and a coiled spiral artery. Which phase is this?

A. Menstrual
B. Proliferative
C. Secretory
D. Postmenopausal

Answer: C — The secretory phase is defined by tortuous secreting glands under progesterone.

Practical identification

In the ovary, stage the cycle by the dominant follicle or corpus luteum; identify the oocyte, zona pellucida, granulosa, theca interna and externa, and the two lutein cell populations. In the uterus, date the endometrium by gland shape and stromal edema, and identify spiral arteries. In the cervix, find the transformation zone; in the mammary gland, distinguish resting, proliferative and lactating lobules.

23 · The Eye & Ear: Special Sense Organs

In this chapter: Eyes: The Photoreceptor System · Ears: The Vestibuloauditory System

Learning objectives

By the end of this chapter you should be able to:

1. Name the three tunics of the eye and the five layers of the cornea.
2. Trace aqueous humor and explain glaucoma.
3. Describe the layers of the retina and the path of a light signal.
4. Compare the vestibular and cochlear sensory epithelia and their transduction mechanisms.

The core idea

Both organs convert a physical signal into a nerve impulse through a **specialized epithelium** — the retina for light, the organ of Corti for sound. Their histology is therefore about **transduction**, not merely structure.

Numbers to anchor the chapter: the adult eyeball is about **22 mm** in diameter; the **sclera** is 0.5 mm thick (1 mm posteriorly); the **cornea** is the anterior one-sixth and is **avascular**; the retina has **nine neural layers plus a pigmented layer**; the cochlea is about **35 mm** long with **2¾ turns**.

Eyes: The Photoreceptor System

Three tunics:

Tunic	Components	Function
Fibrous	**Cornea** and **sclera**	Protection and support; the cornea also refracts
Vascular (uvea)	**Choroid, ciliary body, iris**	Nutrition, accommodation, pupil control
Nervous	**Retina**	Phototransduction

The cornea has five layers, and the order is a standard question:

1. **Corneal epithelium** — stratified squamous, nonkeratinized, **5–6 cells thick**, about **10%** of corneal thickness; its stem cells sit in the **limbus**, and it has one of the richest sensory innervations in the body.
2. **Bowman's membrane** — a thick (**8–10 µm**) basement membrane.
3. **Stroma** — about **90%** of the thickness, some **60 layers** of type I collagen at right angles to each other, with flattened **keratocytes** between them; **avascular**.
4. **Descemet's membrane** — the thick basement membrane of the endothelium.
5. **Corneal endothelium** — simple squamous, the most metabolically active cells of the cornea; their **Na^+/K^+ ATPase** pumps control stromal hydration and therefore transparency.

The cornea is **transparent because it is avascular and its collagen is regularly arranged**. When it becomes edematous or scarred, it loses transparency — the anatomical explanation for corneal blindness.

- **Choroid** — pigmented, vascular; two layers: the inner **choroidocapillary lamina** (nourishes the outer retina) and **Bruch's membrane**, an elastin sheet between two collagen layers, next to the retinal pigment epithelium.
- **Ciliary body** — **ciliary muscle** (fibers in three complementary orientations; accommodation) and numerous **ciliary processes** covered by a **bilaminar epithelium**: a pigmented layer next to the stroma and a nonpigmented superficial layer. The nonpigmented cells pump aqueous humor and also secrete the **fibrillin** microfibrils of the **ciliary zonule** that suspends the lens.

- **Iris** — a stroma with melanocytes and, posteriorly, two pigmented epithelial layers. The anterior surface has **no epithelium**, only fibroblasts and melanocytes. The **dilator pupillae** is a layer of **myoepithelial cells** (sympathetic); the **sphincter pupillae** is a circular smooth-muscle bundle near the pupil (parasympathetic).

Aqueous humor is secreted by the ciliary processes into the **posterior chamber**, flows through the **pupil** into the **anterior chamber**, and drains at the **iridocorneal angle** through the **trabecular meshwork** into the **scleral venous sinus (canal of Schlemm)**, then into scleral veins. It has the ions of plasma but almost no protein. Blockage raises intraocular pressure — **glaucoma**; a narrow angle that closes when the pupil dilates gives **angle-closure glaucoma**.

The lens: capsule (10–20 µm, type IV collagen and proteoglycans) → **subcapsular epithelium** (simple cuboidal, anterior surface only) → **lens fibers** (7–10 mm long, 2 × 8 µm in cross-section, packed with **crystallins**, their organelles and nuclei removed by autophagy). New fibers are added at the equator throughout life, so the lens grows and its nucleus becomes denser. It is **avascular** and suspended by the zonule.

Accommodation: at rest the ciliary muscle is relaxed, the zonule is taut, and the lens is **flat** for distant vision. To focus close, the ciliary muscle **contracts**, the ciliary body moves forward, tension on the zonule falls, and the lens becomes **rounder**. Loss of lens elasticity in the fourth decade is **presbyopia**; loss of transparency is **cataract**.

The retina — the neural retina has **nine layers**, and with the pigment epithelium there are ten. From the vitreous inward-outward (following the path of light): **inner limiting membrane → nerve fiber layer → ganglion cell layer → inner plexiform layer → inner nuclear layer → outer plexiform layer → outer nuclear layer → outer limiting layer → rod and cone layer → pigmented layer.**

Layer	Contents
Retinal pigment epithelium (RPE)	Cuboidal, pigmented; absorbs stray light, forms the **blood–retina barrier** with tight junctions, **phagocytoses shed rod discs**, regenerates **11-cis-retinal**, and secretes trophic factors
Photoreceptor layer	**Rods** (rhodopsin, dim light, ~92 million) and **cones** (iodopsin, color, ~4.6 million). Rod discs are **membrane stacks**; cone outer segments are **conical**.
Outer nuclear layer	Photoreceptor nuclei
Inner nuclear layer	Bipolar, horizontal, amacrine, and Müller cell nuclei
Ganglion cell layer	Ganglion cells whose axons form the optic nerve

Rods versus cones at the level of the disc — a classic distinction: in **cones** the membranous discs of the outer segment are **continuous with the plasma membrane**; in **rods** they are **separate**, pinched-off discs. A single rod contains about **one billion rhodopsin** molecules, and each day it loses and replaces about **90 discs** in a cycle that takes **10 days**.

Phototransduction in one paragraph. In darkness, cGMP keeps **cGMP-gated Na^+ channels** open, so the photoreceptor is depolarized and steadily releases **glutamate**. A photon isomerizes **11-cis-retinal** to **all-trans-retinal** within a picosecond; the opsin changes shape and activates **transducin**, a heterotrimeric G protein whose α subunit stimulates **phosphodiesterase** to hydrolyze cGMP. The channels close, the cell **hyperpolarizes**, and transmitter release falls — which **depolarizes** the bipolar cells and sends signals to the ganglion cells. The free all-trans-retinal is carried to the RPE, re-isomerized to 11-cis, and returned; this **bleaching** cycle takes a minute or more and underlies slow dark adaptation.

The retina is **inverted**: light passes through the ganglion cell and inner layers before reaching the photoreceptors, which sit against the pigment epithelium. The **fovea centralis**, at the center of the **macula lutea**, contains **no rods** and a very high density of **cones**; the overlying layers are displaced sideways so light reaches the cones with little scattering. That is why acuity is highest there, but only in good light. The **optic disc** is the **blind spot**: ganglion-cell axons converge there to form the optic

nerve, and photoreceptors are absent.

Müller cells are the principal glia — modified astrocytes whose cell bodies lie in the inner nuclear layer and whose processes span the whole retina. They provide trophic support, buffer ions and water, regulate blood flow, form the **inner limiting membrane**, and join photoreceptors by zonulae adherentes to form the **outer limiting layer**.

Ears: The Vestibuloauditory System

External ear — the **auricle** is a plate of **elastic cartilage** covered by tightly adherent skin; the **external acoustic meatus** is lined by skin with hairs, sebaceous glands, and modified apocrine **ceruminous glands**. Their product, **cerumen**, contains proteins, saturated fatty acids, and shed keratinocytes and is antimicrobial. Elastic cartilage supports the outer third of the meatus; the temporal bone encloses the inner part. The **tympanic membrane** is fibroelastic connective tissue covered by epidermis outside and simple cuboidal mucosa inside.

Middle ear — an air-filled **tympanic cavity** lined by simple cuboidal epithelium, containing the **ossicles** (malleus, incus, stapes) linked by synovial joints, with two small muscles (**tensor tympani** on the malleus, **stapedius** on the stapes) that damp loud sounds. The **auditory (eustachian) tube** connects to the pharynx and equalizes pressure; the **oval window** receives the stapes and the **round window** relieves the pressure wave.

Inner ear — bony labyrinth containing the membranous labyrinth:

- **Cochlea** — a **35-mm** canal with **2¾ turns** around a bony core, the **modiolus**. It has three compartments: **scala vestibuli** (perilymph) above, **scala media / cochlear duct** (endolymph) in the middle, and **scala tympani** (perilymph) below. Vestibuli and tympani communicate at the apex through the **helicotrema**. The **vestibular (Reissner) membrane** separates vestibuli from media with tight junctions that keep the two fluids apart; the **basilar membrane** separates media from tympani.
- The **organ of Corti** sits on the basilar membrane: **about 12,000 outer hair cells in three rows (five near the apex)** with V-shaped stereocilia bundles, and a **single row of inner hair cells** with a linear bundle. The inner row is far more heavily innervated. **Phalangeal cells** support the hair cells and, with them, form the **apical plate**; **pillar cells** enclose the stiff **inner tunnel**. The tallest stereocilia of the outer cells are embedded in the acellular **tectorial membrane** (collagens II, V, and IX, proteoglycans, and tectorin).
- The **stria vascularis** in the lateral wall secretes endolymph, pumping K^+ into the lumen.
- **Vestibule — utricle** and **saccule**, each with a **macula** containing hair cells and otoliths (gravity and linear acceleration).
- **Semicircular canals — cristae ampullares** with hair cells in a cupula (rotational acceleration).

Endolymph fills the membranous labyrinth: **K^+ about 150 mM, Na^+ about 16 mM** — like intracellular fluid, and the source of the driving force for transduction. **Perilymph** fills the bony labyrinth and resembles cerebrospinal fluid. Perilymph drains through the perilymphatic duct to the subarachnoid space; endolymph drains through the endolymphatic duct to dural venous sinuses.

Transduction is the same in the macula, crista, and organ of Corti: bending the stereocilia toward the tallest opens **mechanically gated K^+ channels**, K^+ enters, the cell depolarizes, voltage-gated Ca^{2+} channels open at the base, and transmitter release rises. Bending the other way closes the channels and silences the cell.

CLINICAL CORRELATION

Glaucoma — raised intraocular pressure from impaired drainage through the trabecular meshwork, damaging the optic nerve. **Cataract** — lens opacity from crystallin denaturation. **Age-related macular degeneration** — degeneration of the RPE and photoreceptors at the macula. **Retinal detachment** — separation of the neural retina from the RPE, which starves the photoreceptors. **Diabetic retinopathy** — microvascular damage with hemorrhage and neovascularization. **Otosclerosis** — ossification of the stapes footplate, producing conductive deafness. **Presbycusis** — age-related loss of hair cells and neurons. **Ménière disease** — endolymphatic hydrops with vertigo, tinnitus, and fluctuating deafness.

Eyelids and tear film — the inner surface of the eyelid and the anterior sclera are covered by **conjunctiva**, a **stratified columnar epithelium with goblet cells** that is continuous with the corneal epithelium. **Lacrimal glands** produce the tear film, which drains through the lacrimal apparatus into the nasal cavity.

UNDER THE MICROSCOPE

The **cornea** is identified by five layers and a regular, avascular stroma. The **retina** by its ten layers with a pigmented layer at the outer border and ganglion cells at the inner border. The **organ of Corti** by the tunnel of Corti between inner and outer hair cells on the basilar membrane, with the tectorial membrane above.

EXAM TIP

Rods detect **dim light**, cones detect **color and detail**; the **fovea has only cones**, which is why acuity is highest there and color vision is lost first in the periphery.

Chapter 23 — summary

- The eyeball is about 22 mm; the sclera is 0.5 mm thick (1 mm posteriorly).
- Cornea: epithelium, Bowman membrane, stroma, Descemet membrane, endothelium — avascular and transparent.
- Aqueous humor: ciliary processes → posterior chamber → pupil → anterior chamber → trabecular meshwork → canal of Schlemm.
- The lens is avascular, suspended by the fibrillin zonule; it becomes rounder when the ciliary muscle contracts.
- The retina has nine neural layers plus the pigmented epithelium.
- Rods (rhodopsin, dim light, ~92 million) and cones (iodopsin, color, ~4.6 million); the fovea contains cones only.
- Phototransduction: retinal isomerization → transducin → phosphodiesterase → cGMP falls → channels close → hyperpolarization.
- The RPE absorbs light, forms the blood–retina barrier, phagocytoses discs, and regenerates 11-cis-retinal.
- Endolymph is high-K^+; perilymph resembles CSF.
- Organ of Corti: about 12,000 outer hair cells in 3–5 rows and a single row of inner hair cells.
- Maculae sense gravity and linear acceleration; cristae sense rotation; the cochlea hears.

Active recall

1. **Name the five layers of the cornea, outside in.** — Epithelium, Bowman membrane, stroma, Descemet membrane, endothelium.
2. **Where is aqueous humor produced and where does it drain?** — Ciliary processes; drains through the trabecular meshwork into the canal of Schlemm.
3. **Which retinal cells form the optic nerve?** — Ganglion cells.
4. **What is the fovea, and why is acuity highest there?** — A cone-only pit at the macula; it has the highest photoreceptor density and minimal overlying tissue.

5. **Endolymph versus perilymph?** — Endolymph (high K^+) fills the membranous labyrinth; perilymph (high Na^+) fills the bony labyrinth.

Exam-style question

A patient loses vision from increased intraocular pressure. Where is the obstruction?

A. Ciliary processes
B. Pupil
C. Trabecular meshwork and canal of Schlemm
D. Central retinal artery

Answer: C — Glaucoma results when aqueous humor cannot drain through the trabecular meshwork into the canal of Schlemm.

Practical identification

In the eye, identify the five corneal layers, the sclera, choroid, ciliary body with its bilaminar epithelium, iris, lens and the ten retinal layers. Locate the fovea and optic disc and explain the differences in photoreceptor density. In the ear, identify the three cochlear compartments, the organ of Corti with inner and outer hair cells, the tectorial and basilar membranes, and the cristae and maculae.

PART VI — RAPID REVIEW AND REFERENCE

Chapter 24, the master tables, and the reference appendices

24 · The Last-24-Hours Rapid Review

HIGH-YIELD PEARL
This chapter is designed to be read in one sitting, twice. Read it once slowly, then again out loud. Everything in it is either a rule that generates other facts or a fact worth being able to state from memory.

The subject as a concept map

If you remember nothing else, remember the shape of the subject. Each level is generated by the level above it, and each arrow is a question the subject naturally poses.

- **Molecules** — proteins, lipids, carbohydrates, nucleic acids
- **Organelles** — membrane, cytosol, nucleus (Chapters 2–3)
- **Cells** — each specialized by which organelle it expands
- **Epithelium** — sits on a basal lamina, avascular, joined by junctions (Chapter 4)
- **Surface epithelium** — number of layers and shape of the top cell
- **Glandular epithelium** — merocrine, apocrine, holocrine
- **Connective tissue** — defined by its matrix (Chapter 5)
- **Loose / dense / reticular / elastic / adipose** — amount and arrangement of fibers
- **Cartilage and bone** — matrix stiffened by proteoglycan or mineral (Chapters 7–8)
- **Muscle** — converts ATP into movement (Chapter 10)
- **Skeletal** — voluntary, peripheral nuclei, triads at the A–I junction
- **Cardiac** — branched, central nucleus, intercalated discs, diads at the Z line
- **Smooth** — no striations, calmodulin, dense bodies
- **Nervous tissue** — converts stimulus into impulse (Chapter 9)
- **Neuron** — soma, dendrites, axon, synapse
- **Glia** — oligodendrocyte, Schwann, astrocyte, microglia, ependyma
- **Organs** — one tissue arrangement plus a supporting stroma
- **Tubes** — concentric layers (gut, vessels, ducts, ureter)
- **Solid organs** — capsule, cortex, medulla, hilum (kidney, ovary, lymph node)
- **Glands** — ducts or no ducts (exocrine versus endocrine)
- **Systems** — organs that share a transport medium
- **Circulatory** — heart, vessels, blood, lymph (Chapters 11–13)
- **Immune** — thymus, lymph node, spleen, MALT (Chapter 14)
- **Digestive** — gut tube plus liver, pancreas, salivary glands (Chapters 15–16)
- **Respiratory** — conducting airway plus alveoli (Chapter 17)
- **Integumentary** — skin and appendages (Chapter 18)
- **Urinary** — filtration, reabsorption, storage (Chapter 19)
- **Endocrine** — pituitary, thyroid, adrenal, islets, parathyroid, pineal (Chapter 20)
- **Reproductive** — testis and ovary, then ducts and glands (Chapters 21–22)
- **Special senses** — eye and ear (Chapter 23)

HIGH-YIELD PEARL
Read the map downward when you revise a chapter, and upward when you face an unknown slide. Downward gives you the function; upward gives you the name.

The 60-second organ identification algorithm

Step 1 — Is there a lumen?

- **No lumen, solid cords or follicles** → thyroid (follicles), adrenal (cords), liver (cords + central veins), parathyroid (cords), pituitary (cell sheets), lymphoid organ (nodules).
- **Yes, a lumen** → go to Step 2.

Step 2 — What is the epithelium?

- **Stratified squamous** → esophagus, oral cavity, vagina, anal canal, epidermis (no lumen).
- **Simple columnar with crypts** → stomach, intestine, gallbladder, uterus.
- **Pseudostratified ciliated** → respiratory tract. **Pseudostratified with stereocilia** → epididymis, vas deferens.
- **Transitional** → ureter, bladder, renal pelvis.
- **Simple cuboidal/columnar tubules** → kidney, salivary ducts, epididymis.

Step 3 — What surrounds it?

- **Cartilage rings** → trachea. **Cartilage plates** → bronchus. **No cartilage** → bronchiole.
- **Three muscle layers** → stomach. **Taeniae coli** → colon. **Thick muscle wall** → vas deferens, ureter.
- **Serosa** → intraperitoneal; **adventitia** → retroperitoneal.

The ten master tables

1. Epithelium by location

Epithelium	Where
Simple squamous	Alveoli, endothelium, mesothelium, Bowman capsule
Simple cuboidal	Kidney tubules, thyroid follicles, small ducts, ovary surface
Simple columnar	Stomach, intestine, gallbladder, uterine tube
Pseudostratified ciliated	Trachea, bronchi, nasal cavity
Pseudostratified with stereocilia	Epididymis, vas deferens
Stratified squamous keratinized	Epidermis
Stratified squamous non-keratinized	Oral cavity, esophagus, vagina, cornea, anal canal
Transitional	Renal pelvis, ureter, bladder, proximal urethra
Epithelioid	Adrenal cortex, islets, liver

2. Collagen types

Type	Where	Disease
I	Bone, skin, tendon, cornea	Osteogenesis imperfecta
II	Cartilage, vitreous	Chondrodysplasia
III	Reticulin: lymphoid organs, marrow, liver	Ehlers–Danlos vascular
IV	Basement membrane	Alport syndrome
V	Cornea, bone, placenta	Ehlers–Danlos

3. Muscle

Feature	Skeletal	Cardiac	Smooth
Striations	Yes	Yes	**No**
Nuclei	Many, **peripheral**	1–2, **central**	1, **central**

Feature	Skeletal	Cardiac	Smooth
Junctions	—	**Intercalated discs**	Gap junctions
T-tubules	Triads at A–I	Diads at Z	Caveolae
Ca^{2+} sensor	Troponin C	Troponin C	**Calmodulin**
Regeneration	Limited (satellite)	**None**	Hypertrophy/hyperplasia

4. Cartilage

Feature	Hyaline	Elastic	Fibrocartilage
Collagen	II	II + elastic	**I + II**
Perichondrium	Yes	Yes	**No**
Where	Trachea, ribs, joints, nose	Epiglottis, pinna	Disc, symphysis, menisci
Cells	Isogenous groups	Numerous	**Rows**

5. Bone cells

Cell	Origin	Job
Osteoprogenitor	Mesenchyme	Precursor
Osteoblast	Mesenchyme	**Make osteoid**, RANKL
Osteocyte	Osteoblast	Maintain, sense load
Osteoclast	**Monocyte**	**Resorb**, ruffled border

6. Blood cells

Cell	%	Nucleus	Signature
Neutrophil	50–70	3–5 lobes	Pale granules
Eosinophil	1–4	**Bilobed**	**Orange-red** granules
Basophil	0.5–1	S-shaped, hidden	Dark blue granules
Lymphocyte	20–40	Round, dense	Scant cytoplasm
Monocyte	2–8	**Kidney-shaped**	Largest cell

7. Lymphoid organs

Organ	Follicles	Afferent lymph	Unique
Thymus	**No**	No	**Hassall corpuscles**
Lymph node	Yes	**Yes**	Cortex/paracortex/medulla
Spleen	Yes	**No**	White + red pulp
Tonsil	Yes	No	Crypts, partial capsule

8. Gut segments

Segment	Diagnostic feature
Esophagus	Stratified squamous, skeletal muscle above
Stomach	**Three muscle layers**, parietal + chief cells, no goblet
Duodenum	**Brunner glands (submucosa)**

Segment	Diagnostic feature
Jejunum	Tallest villi; no Brunner glands; Peyer's patches belong to the ileum
Ileum	**Peyer's patches**
Colon	**No villi**, crypts, many goblet, taeniae
Appendix	Lymphoid tissue all round, blind tube

9. Endocrine

Gland	Pattern	Hormone
Thyroid	**Follicles with colloid**	T3, T4; C cells → calcitonin
Parathyroid	Cords of chief cells	PTH
Adrenal cortex G/F/R	Cords	Aldosterone / cortisol / androgens
Adrenal medulla	Chromaffin	Epinephrine
Pituitary anterior	Acidophils + basophils	GH, PRL / ACTH, TSH, FSH, LH
Pituitary posterior	Fibrillary, Herring bodies	Oxytocin, ADH
Islets	Clusters	Insulin, glucagon, somatostatin

10. Airway

Level	Cartilage	Glands	Muscle
Trachea	**Rings**	Yes	Trachealis
Bronchus	**Plates**	Yes	Complete ring
Bronchiole	**None**	**None**	Prominent
Respiratory bronchiole	None	None	+ alveoli in wall

One hundred facts in one hundred lines

1. Basophilic = binds hematoxylin = nucleic acid.
2. Eosinophilic = binds eosin = protein.
3. PAS stains glycogen and basement membrane.
4. Silver stains reticulin (type III collagen).
5. Light microscope resolution = 0.2 µm.
6. Frozen section = rapid diagnosis + lipid preservation.
7. Plasma membrane thickness = 7–10 nm.
8. Na^+/K^+-ATPase pumps 3 Na^+ out, 2 K^+ in.
9. Clathrin mediates receptor-mediated endocytosis.
10. Mitochondria are maternally inherited.
11. Steroid-secreting cells have **tubular** cristae.
12. Nissl bodies = rough ER + polysomes.
13. Golgi **modifies and sorts**; RER **synthesizes**.
14. Lysosomes are acidic (pH ≈ 5).
15. Lipofuscin = residual body pigment.
16. Peroxisomes contain catalase.
17. Proteasomes degrade ubiquitin-tagged protein.
18. Kinesin = anterograde; dynein = retrograde.

19. Cilia = 9 + 2; basal body = 9 + 0.
20. Keratin = epithelium; vimentin = mesenchyme; desmin = muscle; GFAP = astrocyte.
21. Lamin A mutation = progeria.
22. Nucleolus makes rRNA; NORs on 13, 14, 15, 21, 22.
23. Barr body = inactivated X.
24. p53 = guardian of the genome.
25. Meiosis = four haploid cells; crossing over in pachytene.
26. Apoptosis = no inflammation.
27. Labile tissue = epidermis, gut, marrow; permanent = neurons, cardiac muscle.
28. Epithelium is avascular.
29. Basal lamina = lamina lucida + lamina densa; the basement membrane adds the reticular lamina (type III collagen).
30. Type IV collagen is in the basal lamina.
31. Desmosomes link to keratin; hemidesmosomes to laminin via integrin.
32. E-cadherin is calcium-dependent.
33. Gap junctions = connexins.
34. Microvilli core = actin.
35. Stereocilia are long microvilli (epididymis).
36. Endothelium and mesothelium are simple squamous.
37. Pseudostratified epithelium is truly one layer.
38. Transitional epithelium has umbrella cells and uroplakins.
39. Goblet cell = unicellular mucous gland.
40. Sebaceous gland = holocrine.
41. Mammary gland = modified apocrine gland; lipid by apocrine secretion, protein by merocrine secretion.
42. Pancreas = merocrine.
43. Myoepithelial cells compress acini.
44. Collagen synthesis needs **vitamin C** and **copper**.
45. Scurvy = defective hydroxylation.
46. Marfan = fibrillin.
47. Ehlers–Danlos = collagen processing.
48. Plasma cell nucleus = clock-face, eccentric.
49. Mast cell = histamine + heparin.
50. Brown fat = multilocular + UCP1 + central nucleus.
51. Leptin reduces appetite; adiponectin increases insulin sensitivity.
52. Hyaline cartilage = type II collagen.
53. Fibrocartilage = **no perichondrium**.
54. Interstitial growth = from within lacunae.
55. Osteoclast = monocyte lineage, TRAP-positive.
56. Osteoblast = RANKL; OPG is the decoy.
57. Osteon = concentric lamellae + Haversian canal.
58. Epiphyseal zones: reserve → proliferate → hypertrophy → calcify → ossify.
59. Intramembranous ossification = flat bones.
60. PTH raises calcium; calcitonin lowers it.
61. Oligodendrocyte = CNS, many axons; Schwann = PNS, one segment.

62. Microglia are mesodermal.
63. Wallerian degeneration occurs distal to injury.
64. Regeneration ≈ 1 mm/day.
65. A band never shortens.
66. Skeletal T-tubules form triads at the A–I junction.
67. Cardiac T-tubules form diads at the Z line.
68. Smooth muscle uses calmodulin, not troponin.
69. Intercalated discs = fascia adherens + desmosome + gap junction.
70. Purkinje fibers conduct; they are not neurons.
71. Elastic artery = elastic lamellae in media.
72. Muscular artery = prominent internal elastic lamina.
73. Sinusoidal capillaries = liver, spleen, marrow.
74. Blood–brain barrier = continuous capillaries + tight junctions.
75. Lymphatic capillaries are blind-ended.
76. Erythrocyte lifespan = 120 days.
77. Neutrophil = 3–5 lobes, acute inflammation.
78. Eosinophil = parasites and allergy.
79. Basophil = histamine, IgE.
80. Monocyte = largest leukocyte.
81. Platelets = 2–4 µm, from megakaryocytes.
82. EPO from kidney; TPO from liver.
83. M:E ratio = 2:1 to 4:1.
84. CD4 ↔ MHC II; CD8 ↔ MHC I.
85. Thymus has no follicles; spleen has no afferent lymphatics.
86. Hassall corpuscles = thymus.
87. PALS = T-cell zone of spleen.
88. Germinal center = B-cell proliferation.
89. Enteric plexuses: Meissner (submucosa), Auerbach (myenteric).
90. Parietal cell = HCl + intrinsic factor.
91. Chief cell = pepsinogen.
92. Paneth cell = lysozyme and defensins.
93. Brunner glands = duodenum.
94. Peyer's patches = ileum.
95. Portal triad = artery, portal vein, bile duct.
96. Bile flows toward the triad; blood flows away.
97. Type I pneumocyte = gas exchange; type II = surfactant.
98. Club (formerly Clara) cells = bronchioles; they secrete surfactant-like material and detoxify inhaled compounds.
99. Podocytes form the filtration slit diaphragm (nephrin).
100. Macula densa = distal tubule; JG cells = afferent arteriole.

Twenty-five distinctions worth knowing

Do not confuse	With	The difference
Basophilic	Eosinophilic	DNA/RNA versus protein
Mucous	Mucosa	Secretion versus lining layer
Endothelium	Mesothelium	Vessels versus body cavities
Pseudostratified	Stratified	One layer versus many
Hyaline cartilage	Fibrocartilage	Type II + perichondrium versus I + II + none
Osteoblast	Osteoclast	Builds versus resorbs; mesenchymal versus monocyte
Astrocyte	Oligodendrocyte	Support/BBB versus myelination
Schwann cell	Oligodendrocyte	One segment versus many axons
Neutrophil	Eosinophil	3–5 lobes versus bilobed
Monocyte	Lymphocyte	Kidney-shaped versus round
Proximal tubule	Distal tubule	Brush border, eosinophilic versus clean, pale
Type I pneumocyte	Type II	Thin for gas versus cuboidal for surfactant
Parietal cell	Chief cell	HCl versus pepsinogen
Brunner gland	Peyer's patch	Duodenum submucosa versus ileum lymphoid
Zona glomerulosa	Zona fasciculata	Aldosterone versus cortisol
Adrenal cortex	Adrenal medulla	Steroid versus catecholamine
Anterior pituitary	Posterior pituitary	Glandular versus neural
Sertoli cell	Leydig cell	Barrier/nurture versus testosterone
Granulosa cell	Theca cell	Estrogen (aromatase) versus androgen
Proliferative	Secretory endometrium	Straight glands versus tortuous secreting glands
Follicular	Luteal phase	Estrogen-dominant versus progesterone-dominant
Rod	Cone	Dim light versus color and acuity
Endolymph	Perilymph	High K^+ versus high Na^+
Apoptosis	Necrosis	No inflammation versus inflammation
Labile	Permanent tissue	Continuous turnover versus no division

Eight common errors

1. Saying the osteoclast is mesenchymal. It is **hematopoietic**.
2. Saying the A band shortens. It **does not**.
3. Saying the thymus has follicles. It **does not**.
4. Saying the spleen has afferent lymphatics. It **does not**.
5. Saying the adrenal medulla makes steroids. It makes **catecholamines**.
6. Saying FSH acts on Leydig cells. It acts on **Sertoli** cells; LH acts on Leydig.
7. Saying rods are at the fovea. The fovea has **cones only**.
8. Saying the stomach has goblet cells. It has **surface mucous cells**, not goblet cells.

Appendix A · Stains and What They Prove

Stain	Target	Color	Typical use
Hematoxylin	Nucleic acids	Blue-purple	Nuclei, rough ER
Eosin	Proteins	Pink-red	Cytoplasm, collagen, RBC
PAS	Glycogen, mucin, glycoproteins, basement membrane	Magenta	Goblet cells, liver glycogen, fungal walls
Masson trichrome	Collagen versus muscle	Blue/green versus red	Liver fibrosis, kidney biopsy
Silver (reticulin)	Type III collagen, nerve fibers	Black	Lymph node, spleen, marrow
Orcein / resorcin–fuchsin	Elastin	Dark brown-black	Aorta, lung, skin
Sudan black / oil red O	Lipid	Black/red	Steatosis, adipocytes (frozen)
Osmium tetroxide	Lipid, myelin	Black	Fat, myelin
Cresyl violet (Nissl)	Rough ER	Violet	Neurons
Feulgen	DNA	Magenta	DNA quantification
Prussian blue	Iron (hemosiderin)	Blue	Iron overload, old hemorrhage
Congo red	Amyloid	Apple-green birefringence	Amyloidosis
Gram	Bacterial wall	Purple (positive) / pink (negative)	Infection
Ziehl–Neelsen	Acid-fast bacilli	Red	Tuberculosis
Immunohistochemistry	Specific antigen	Brown (DAB) or fluorescent	Tumor typing, cell identification

The stains in detail

The table above is the quick-recall version; this one adds the chemistry behind each stain, so that you can predict what a stain will show even when you meet an unfamiliar one.

Stain	Chemistry	What it stains	Why it is used
Hematoxylin & eosin (H&E)	Hematoxylin is a **basic** dye; eosin is an **acidic** counterstain	Nuclei and rough ER blue-purple; collagen and most cytoplasmic proteins, especially mitochondria, pink-red	The default survey stain; the source of the basophilic/eosinophilic vocabulary
Pararosaniline–toluidine blue (PT)	Basic dye combination	Chromatin purple; cytoplasm and collagen lighter violet	Penetrates **plastic (resin) sections**, giving finer detail than H&E
Mallory trichrome	Sequential stain combination	Nuclei purple; cytoplasm, keratin, and erythrocytes red-orange; **collagen blue**	Demonstrates cells and small vessels of connective tissue. Masson and Gomori trichromes are equivalent, with collagen blue-green or green
Picrosirius–hematoxylin (PSH)	Sirius red in picric acid	Collagen red; cytoplasm pink-violet; nuclei purple	Collagen becomes **birefringent** under polarized light — a specific collagen stain

Stain	Chemistry	What it stains	Why it is used
Periodic acid–Schiff (PAS)	Periodic acid oxidizes vicinal diols; Schiff reagent detects them	Carbohydrate-rich structures **magenta**	Mucin granules, glycogen, glycocalyx, basement membranes
Wright–Giemsa	Two related Romanowsky combinations	Nuclei purple; erythrocytes pink-orange; leukocyte granules differentially	Blood and bone-marrow smears
Silver and gold impregnation	Metal salts reduce on filaments	Nerve filaments and **reticulin (type III collagen)** dark brown or black	Largely replaced by immunohistochemistry
Elastin stains — Weigert resorcin fuchsin, aldehyde fuchsin, orcein	Dye binds elastin	Elastic fibers brown or purple	Distinguishing elastic from collagen (aorta, lung, skin)
Lipid stains — oil red O, Sudan black, osmium tetroxide	Lipophilic dye partitions into lipid	Lipid droplets and myelin	Requires frozen sections or special processing, because solvents used in paraffin embedding extract lipid
Other aniline dyes — azures, cresyl violet, brilliant cresyl blue, luxol fast blue, light green	Basic dyes	Various cellular and extracellular components	Permanence and color contrast in paraffin sections; many began as textile dyes

UNDER THE MICROSCOPE

Read the stain label before you interpret the section. "PT" tells you the section is plastic-embedded and shows fine detail; "Mallory" tells you collagen will be blue and cytoplasm red; "PSH" tells you collagen is red and can be examined under polarized light. The stain is part of the evidence.

Appendix B · Vocabulary You Should Know

Term	Meaning
Parenchyma	The functional cells of an organ
Stroma	The supportive connective tissue framework
Capsule	Dense connective tissue enclosing an organ
Hilum	The entry/exit point for vessels, ducts, and nerves
Cortex / medulla	Outer versus inner region
Trabecula	A partition or supporting beam of connective tissue
Sinusoid	A wide, discontinuous capillary
Acinus	A berry-like cluster of secretory cells
Tubuloacinar	A gland with both tubular and acinar secretory units
Metaplasia	Change from one mature cell type to another
Dysplasia	Disordered growth with atypical cells
Neoplasia	New, autonomous growth
Hypertrophy	Increase in cell size
Hyperplasia	Increase in cell number
Atrophy	Decrease in size or number
Apoptosis	Programmed cell death
Necrosis	Pathological cell death with inflammation
Autolysis	Self-digestion after death
Fixation	Chemical preservation of tissue
Artifact	A structure produced by processing, not present in life

Appendix C · Sources and Further Reading

This book is a synthesis of standard histology, and its facts belong to the common scientific literature rather than to any single work. The first work below is the one against which the text was principally checked; the others are standard references offered for deeper reading and were not used as verification sources. Editions of the further-reading works are not listed: use the current edition of each. The principal reference is cited with its edition, because every check was made against that edition.

- Mescher AL. Junqueira's Basic Histology: Text and Atlas. 17th ed. New York: McGraw Hill; 2024.
- Ross MH, Pawlina W. Histology: A Text and Atlas. Wolters Kluwer.
- Young B, O'Dowd G, Woodford P. Wheater's Functional Histology. Elsevier.
- Standring S, editor. Gray's Anatomy. Elsevier.
- Alberts B, et al. Molecular Biology of the Cell. W. W. Norton.
- Kumar V, Abbas AK, Aster JC. Robbins and Cotran Pathologic Basis of Disease. Elsevier.

How the content was checked. Numerical values, structural descriptions and named cell types were compared with the principal reference; where standard accounts give different ranges or counts, the text gives the range or says which value is being used. Mechanisms and clinical correlations were reasoned through for consistency with standard physiology and pathology rather than checked against a second text. No claim is made that every sentence was individually verified, and no reference is cited for a statement it does not contain.

Limits. This book contains no histological images. The descriptions tell you what to look for, but recognition is learned at a microscope, and a description cannot certify what is on your particular slide. Clinical notes are for orientation and are not medical advice.

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PART VII — PRACTICE EXAMINATION

One hundred examination questions with explained answers

100 Examination Questions

EXAM TIP
Answer all 100 before turning to the key. Mark every guess with a dot; guesses, not errors, are what you must revise. A score of 80 or more suggests that the material is secure; below 65, return to the tables in Part VI. Neither figure predicts an examination result.

Cells and techniques (1–12)

1. A tissue section is treated with a basic dye. Which structure will stain most intensely?
A. Nuclei and rough endoplasmic reticulum
B. Erythrocytes
C. Elastic fibers
D. Collagen bundles

2. Which stain would you choose to demonstrate a basement membrane?
A. Eosin
B. Orcein
C. PAS
D. Sudan black

3. A surgeon requires an immediate diagnosis during an operation. The correct specimen preparation is:
A. Resin embedding
B. Whole-mount preparation
C. Routine paraffin section
D. Frozen section

4. The resolving power of the light microscope is approximately:
A. 2 µm
B. 0.2 µm
C. 20 µm
D. 0.2 nm

5. Which technique localizes a specific protein in a tissue section using an antibody?
A. Autoradiography
B. In situ hybridization
C. Immunohistochemistry
D. Polarizing microscopy

6. A cell specialized for steroid hormone synthesis would show abundant:
A. Rough ER and Golgi
B. Smooth ER and lipid droplets
C. Lysosomes and residual bodies
D. Free ribosomes and proteasomes

7. Which organelle contains catalase and performs β-oxidation of very long-chain fatty acids?
A. Mitochondrion
B. Lysosome
C. Peroxisome
D. Proteasome

8. Retrograde axonal transport of viruses toward the neuronal soma is mediated by:
A. Myosin
B. Dynein and kinesin equally
C. Kinesin

D. Dynein

9. Which intermediate filament is characteristic of astrocytes?

A. Vimentin
B. GFAP
C. Neurofilament
D. Desmin

10. A cell with a 9 + 0 arrangement of microtubules is:

A. A basal body
B. A sperm flagellum
C. A microvillus
D. A motile cilium

11. A mutation in lamin A produces:

A. Alport syndrome
B. Progeria
C. Duchenne dystrophy
D. Marfan syndrome

12. The nucleolar organizer regions are located on which chromosomes?

A. 16, 17, 18, 19, 20
B. X and Y
C. 1, 2, 3, 4, 5
D. 13, 14, 15, 21, 22

Epithelium, connective tissue, cartilage and bone (13–32)

13. Which junctional complex is principally responsible for sealing the intercellular space?

A. Hemidesmosome
B. Desmosome
C. Gap junction
D. Tight junction

14. Pemphigus vulgaris results from autoantibodies against:

A. Connexin
B. Laminin
C. Desmoglein
D. Integrin

15. Which epithelium lines the normal ureter?

A. Simple columnar
B. Pseudostratified ciliated
C. Stratified squamous
D. Transitional

16. Stereocilia are found in the:

A. Epididymis
B. Fallopian tube
C. Intestinal villi
D. Trachea

17. The basement membrane's lamina reticularis is composed mainly of:

A. Type I collagen
B. Type II collagen
C. Type III collagen
D. Type IV collagen

18. Which collagen type is defective in osteogenesis imperfecta?

A. Type II
B. Type III
C. Type IV
D. Type I

19. Hydroxylation of proline and lysine during collagen synthesis requires:

A. Vitamin C
B. Vitamin D
C. Copper
D. Vitamin A

20. A cell with an eccentric nucleus showing a clock-face chromatin pattern and a pale perinuclear halo is a:

A. Macrophage
B. Eosinophil
C. Mast cell
D. Plasma cell

21. The main function of UCP1 in brown adipose tissue is:

A. ATP synthesis
B. Uncoupling of oxidative phosphorylation to produce heat
C. Fatty acid storage
D. Leptin secretion

22. Adiponectin, secreted by adipocytes, acts to:

A. Increase insulin sensitivity
B. Increase blood pressure
C. Stimulate lipolysis in brown fat
D. Reduce appetite

23. Which cartilage type lacks a perichondrium?

A. Elastic
B. Fibrocartilage
C. Tracheal hyaline
D. Hyaline

24. The intervertebral disc's annulus fibrosus is composed of:

A. Dense regular connective tissue
B. Hyaline cartilage
C. Elastic cartilage
D. Fibrocartilage

25. Which cell is responsible for bone resorption?

A. Osteocyte
B. Osteoclast
C. Osteoprogenitor cell
D. Osteoblast

26. Osteoclasts are derived from:

A. Monocyte–macrophage lineage
B. Osteoblasts
C. Neural crest
D. Mesenchymal cells

27. In endochondral ossification, the zone in which chondrocytes enlarge and the matrix calcifies is the:

A. Ossification zone
B. Reserve zone
C. Proliferative zone
D. Hypertrophic zone

28. Which hormone lowers serum calcium?
A. Calcitriol
B. Calcitonin
C. Growth hormone
D. PTH

29. The primary ossification center of a long bone appears in the:
A. Diaphysis
B. Metaphysis
C. Periosteum
D. Epiphysis

30. A multinucleated cell lying in a scalloped resorption pit is:
A. An osteocyte
B. An osteoclast
C. A macrophage
D. An osteoblast

31. Dense regular connective tissue is characterized by:
A. Parallel collagen bundles
B. Abundant elastic fibers
C. A reticular network
D. Randomly oriented collagen bundles

32. Which tissue is the site of hematopoiesis in the adult?
A. Liver
B. Spleen
C. Red bone marrow
D. Yellow bone marrow

Nerve and muscle (33–44)

33. Nissl bodies are composed of:
A. Rough ER and polysomes
B. Neurofilaments
C. Synaptic vesicles
D. Smooth ER and Golgi

34. Which glial cell myelinates several axons in the central nervous system?
A. Schwann cell
B. Microglia
C. Astrocyte
D. Oligodendrocyte

35. Microglia are derived from:
A. Neural tube
B. Mesoderm (monocyte lineage)
C. Endoderm
D. Neural crest

36. The blood–brain barrier is formed principally by:
A. Pericytes alone
B. Astrocyte end-feet alone

C. Tight junctions between capillary endothelial cells
D. Ependymal cells

37. In a peripheral nerve, the connective tissue sheath surrounding a single axon is the:

A. Perineurium
B. Endoneurium
C. Mesoneurium
D. Epineurium

38. During muscle contraction, which band does not change in length?

A. I band
B. H band
C. A band
D. None of these

39. Calcium binds which molecule in skeletal muscle to initiate contraction?

A. Myosin light chain
B. Actinin
C. Calmodulin
D. Troponin C

40. Intercalated discs contain all of the following except:

A. Fascia adherens
B. Desmosomes
C. Gap junctions
D. Tight junctions

41. Which feature distinguishes cardiac from skeletal muscle?

A. Mitochondria
B. Striations
C. Central nucleus
D. Myofibrils

42. Smooth muscle contraction differs from striated muscle because it uses:

A. Tropomyosin only
B. No calcium
C. Troponin C
D. Calmodulin and myosin light-chain kinase

43. T-tubules and terminal cisternae form triads at the:

A. H band
B. Z line
C. A–I junction
D. M line

44. A muscle fiber that is rich in mitochondria and myoglobin and resistant to fatigue is:

A. All equally
B. Type I
C. Type IIa
D. Type IIx

Blood, hemopoiesis and immunity (45–60)

45. The most abundant leukocyte in normal peripheral blood is the:

A. Neutrophil
B. Monocyte
C. Eosinophil

D. Lymphocyte

46. A bilobed nucleus with bright orange-red granules identifies the:

A. Eosinophil
B. Monocyte
C. Basophil
D. Neutrophil

47. The largest leukocyte, with a kidney-shaped nucleus, is the:

A. Neutrophil
B. Basophil
C. Lymphocyte
D. Monocyte

48. Platelets are derived from:

A. Erythroblasts
B. Megakaryocytes
C. Monocytes
D. Lymphocytes

49. Erythropoietin is produced mainly by the:

A. Kidney
B. Spleen
C. Bone marrow
D. Liver

50. The normal myeloid-to-erythroid ratio in bone marrow is:

A. 10:1
B. 1:1
C. 2:1 to 4:1
D. 1:2

51. Which immunoglobulin is the main secretory antibody at mucosal surfaces?

A. IgA
B. IgE
C. IgG
D. IgM

52. CD4-positive T cells recognize antigen presented by:

A. Antibody alone
B. MHC class I
C. MHC class II
D. MHC class III

53. Hassall corpuscles are found in the:

A. Thymus
B. Tonsil
C. Spleen
D. Lymph node

54. Which organs lack afferent lymphatic vessels?

A. Lymph node
B. Thymus and spleen
C. Tonsil
D. Appendix

55. The periarteriolar lymphoid sheath of the spleen contains mainly:

A. Plasma cells
B. B cells
C. T cells
D. Macrophages

56. A germinal center is a site of:
A. Erythropoiesis
B. Platelet production
C. T-cell maturation
D. B-cell proliferation

57. The red pulp of the spleen functions mainly to:
A. Secrete antibodies
B. Produce T cells
C. Filter blood and remove aged erythrocytes
D. Store lymphocytes

58. The lifespan of a normal erythrocyte is approximately:
A. 120 days
B. 1 year
C. 7 days
D. 30 days

59. During the second trimester, hemopoiesis occurs mainly in the:
A. Liver and spleen
B. Bone marrow
C. Thymus
D. Yolk sac

60. Which cell is the principal antigen-presenting cell of the epidermis?
A. Keratinocyte
B. Melanocyte
C. Langerhans cell
D. Merkel cell

Digestive system (61–72)

61. Which layer of the gut wall contains the myenteric (Auerbach) plexus?
A. Between the circular and longitudinal muscle layers
B. Serosa
C. Mucosa
D. Submucosa

62. Parietal cells of the gastric glands secrete:
A. HCl and intrinsic factor
B. Mucus
C. Gastrin
D. Pepsinogen

63. Brunner glands are found in the:
A. Duodenum
B. Jejunum
C. Ileum
D. Colon

64. Peyer's patches are characteristic of the:
A. Jejunum
B. Ileum

C. Stomach
D. Duodenum

65. Paneth cells are located at the base of intestinal crypts and secrete:

A. Lysozyme and defensins
B. Gastrin
C. Intrinsic factor
D. Mucus

66. Which feature is absent in the colon but present in the small intestine?

A. Villi
B. Enteroendocrine cells
C. Goblet cells
D. Crypts

67. The portal triad contains all of the following except:

A. Hepatic artery
B. Portal vein
C. Bile duct
D. Central vein

68. Kupffer cells are:

A. Fat-storing cells
B. Endothelial cells
C. Hepatocytes
D. Sinusoidal macrophages

69. Bile flows in which direction relative to the portal triad?

A. Away from the triad
B. Toward the triad
C. Parallel to blood flow
D. Into the central vein

70. The exocrine pancreas is identified by:

A. Basophilic acini and zymogen granules
B. Mucous acini only
C. Follicles of colloid
D. Islets and striated ducts

71. Which cells secrete insulin?

A. Delta cells
B. PP cells
C. Alpha cells
D. Beta cells

72. The esophagus is lined by:

A. Transitional epithelium
B. Pseudostratified ciliated epithelium
C. Simple columnar epithelium
D. Stratified squamous non-keratinized epithelium

Respiratory system and skin (73–82)

73. Type II pneumocytes are characterized by:

A. Cilia and goblet granules
B. Phagocytic vacuoles
C. A squamous shape and no organelles

D. Lamellar bodies and surfactant production

74. Clara (club) cells are found in the:

A. Bronchioles
B. Pleura
C. Alveoli
D. Trachea

75. Cartilage in the bronchial tree changes in which sequence?

A. Rings → none → plates
B. None → plates → rings
C. Rings → plates → none
D. Plates → rings → none

76. The blood–air barrier does not include:

A. Basement membrane
B. Type II pneumocyte
C. Surfactant
D. Type I pneumocyte

77. The layer of the epidermis present only in thick skin is the:

A. Lucidum
B. Basale
C. Spinosum
D. Granulosum

78. Melanocytes are derived from:

A. Endoderm
B. Ectoderm
C. Neural crest
D. Mesoderm

79. Sebaceous glands secrete by which mechanism?

A. Apocrine
B. Holocrine
C. Endocrine
D. Merocrine

80. Pacinian corpuscles are located in the:

A. Basal epidermis
B. Dermal papillae
C. Deep dermis and hypodermis
D. Stratum corneum

81. Which cell type is most numerous in the epidermis?

A. Langerhans cell
B. Keratinocyte
C. Merkel cell
D. Melanocyte

82. Which receptor is responsible for fine tactile discrimination in the fingertips?

A. Pacinian corpuscle
B. Meissner corpuscle
C. Ruffini ending
D. Free nerve ending

Urinary and endocrine (83–92)

83. The filtration slit diaphragm between podocyte foot processes contains:

A. Desmoglein
B. Connexin
C. Laminin only
D. Nephrin

84. The macula densa is located in the:

A. Proximal tubule
B. Thin limb
C. Distal tubule
D. Collecting duct

85. Renin is secreted by:

A. Mesangial cells
B. Macula densa cells
C. Juxtaglomerular cells
D. Podocytes

86. Which tubule segment is most eosinophilic because of abundant mitochondria?

A. Collecting duct
B. Proximal convoluted tubule
C. Thin descending limb
D. Distal convoluted tubule

87. The epithelium of the ureter is:

A. Pseudostratified
B. Simple columnar
C. Transitional
D. Stratified squamous

88. Thyroid follicles are lined by:

A. Stratified cuboidal epithelium
B. Pseudostratified epithelium
C. Transitional epithelium
D. Simple cuboidal epithelium

89. Calcitonin is secreted by:

A. Follicular cells
B. Parafollicular C cells
C. Chief cells
D. Oxyphil cells

90. The zona fasciculata of the adrenal cortex secretes:

A. Epinephrine
B. Aldosterone
C. Cortisol
D. Androgens

91. Which is derived from neural crest?

A. Adrenal cortex
B. Adrenal medulla
C. Thyroid follicle
D. Parathyroid chief cells

92. Herring bodies are found in the:

A. Pineal gland
B. Thyroid

C. Anterior pituitary
D. Posterior pituitary

Reproductive system, eye and ear (93–100)

93. The blood–testis barrier is formed by tight junctions between:
A. Peritubular myoid cells
B. Spermatogonia
C. Sertoli cells
D. Leydig cells

94. LH acts primarily on:
A. Spermatogonia
B. Epididymal cells
C. Sertoli cells
D. Leydig cells

95. The usual site of fertilization is the:
A. Uterine cavity
B. Ampulla of the uterine tube
C. Isthmus
D. Cervix

96. Straight, narrow endometrial glands with numerous mitoses indicate which phase?
A. Menstrual
B. Proliferative
C. Secretory
D. Gestational

97. The corpus luteum secretes mainly:
A. FSH
B. Estrogen
C. Progesterone
D. hCG

98. The fovea centralis contains:
A. Both equally
B. Neither
C. Rods only
D. Cones only

99. Aqueous humor drains into the:
A. Canal of Schlemm
B. Ciliary body
C. Choroid
D. Vitreous body

100. The sensory cells of the organ of Corti are:
A. Spiral ganglion cells
B. Tectorial cells
C. Hair cells
D. Supporting cells

Answer Key with Explanations

1. A — Basic dyes such as hematoxylin bind acidic structures, and nucleic acids (nucleus, rough ER) are the most acidic components of the cell. **Trap:** D — collagen is basic and binds eosin, not hematoxylin. B — hemoglobin is basic. C — elastin is not particularly acidic.

2. C — PAS stains the glycoprotein-rich basement membrane magenta; eosin stains it only weakly and orcein stains elastin. **Trap:** A — eosin stains protein, not carbohydrate. B — orcein stains elastin. D — Sudan black stains lipid.

3. D — A frozen section gives a diagnosis within minutes, at the cost of some morphological distortion. **Trap:** C takes a day or more. A and B are research or teaching preparations, not urgent diagnostics.

4. B — The light microscope resolves about 0.2 µm; electron microscopy is required for anything smaller. **Trap:** D (0.2 nm) is the order of a TEM's point resolution, not the light microscope's. A and C are far too coarse.

5. C — Immunohistochemistry uses a specific antibody, visualized by an enzyme reaction or a fluorophore. **Trap:** A detects newly synthesized molecules. B detects nucleic acid sequence. D detects birefringence.

6. B — Steroid synthesis requires smooth ER and lipid (cholesterol) stores; such cells appear pale and foamy. **Trap:** A is the profile of a protein-secreting cell. C is a phagocyte. D is a stem or rapidly dividing cell.

7. C — Peroxisomes contain catalase and carry out β-oxidation of very long-chain fatty acids. **Trap:** B is the lysosome (acid hydrolases). D degrades ubiquitinated proteins. A is the main ATP source.

8. D — Dynein moves cargo toward the minus end, i.e. toward the cell body (retrograde). **Trap:** C moves cargo the other way (anterograde). A is a muscle motor. B is wrong because transport is directional.

9. B — GFAP is the characteristic intermediate filament of astrocytes. **Trap:** D is muscle. A is connective tissue and some neural cells. C is the neuronal cytoskeleton.

10. A — The basal body is 9 + 0; the motile cilium and flagellum are 9 + 2. **Trap:** D and B are 9 + 2. C has no microtubules at all.

11. B — Lamin A mutations destabilize the nuclear envelope and cause Hutchinson–Gilford progeria. **Trap:** C is dystrophin. D is fibrillin. A is type IV collagen.

12. D — The acrocentric chromosomes 13, 14, 15, 21, and 22 carry the nucleolar organizer regions. **Trap:** C and A are metacentric or submetacentric autosomes. B carries few rRNA genes.

13. D — The tight junction (zonula occludens) seals the intercellular space and creates polarity. **Trap:** B anchors cells to each other but does not seal. C communicates. A anchors epithelium to basal lamina.

14. C — Anti-desmoglein antibodies disrupt desmosomes and cause intraepithelial blistering. **Trap:** D links integrins to matrix (epidermolysis bullosa). A is a gap junction protein. B is a basal lamina protein.

15. D — The ureter is lined by transitional epithelium, which allows distension. **Trap:** C is the vagina and esophagus. A is the gallbladder. B is the trachea.

16. A — Long, stiff microvilli (stereocilia) are found in the epididymis and inner ear. **Trap:** D has motile cilia. B has motile cilia. C has microvilli of normal length.

17. C — The lamina reticularis contains type III (reticular) collagen, produced by fibroblasts. **Trap:** A is dense connective tissue. B is cartilage. D is the lamina densa.

18. D — Osteogenesis imperfecta results from defective type I collagen. **Trap:** A causes chondrodysplasias. B causes Ehlers–Danlos type IV. C causes Alport syndrome.

19. A — Vitamin C is required for prolyl and lysyl hydroxylase; deficiency causes scurvy. **Trap:** D is needed for vision. B is calcium homeostasis. C is a cofactor for lysyl oxidase.

20. D — The plasma cell has an eccentric clock-face nucleus and a pale perinuclear Golgi zone. **Trap:** C has granules and a central nucleus. A has a bean-shaped nucleus and vacuoles. B has a bilobed nucleus.

21. B — UCP1 uncouples the proton gradient so that energy is released as heat. **Trap:** A is the normal function that UCP1 bypasses. C is unilocular white fat. D is a white-adipocyte hormone.

22. A — Adiponectin increases insulin sensitivity; it is reduced in obesity. **Trap:** D is leptin. B and C are not adiponectin actions.

23. B — Fibrocartilage has no perichondrium, which contributes to its poor healing. **Trap:** D and A have perichondrium. C is hyaline and is covered by perichondrium.

24. D — The annulus fibrosus is fibrocartilage, with type I and type II collagen. **Trap:** B is the articular surface. C is the epiglottis. A is tendon.

25. B — The osteoclast resorbs bone using a ruffled border and acid secretion. **Trap:** D forms bone. A maintains it. C is the precursor.

26. A — Osteoclasts arise from the monocyte–macrophage lineage, not from mesenchyme. **Trap:** D gives osteoblasts. B are the bone-forming cells themselves. C gives melanocytes and Schwann cells.

27. D — In the hypertrophic zone chondrocytes enlarge and the matrix begins to calcify. **Trap:** B is small, sparse cells. C is active division. A is where osteoid is deposited.

28. B — Calcitonin from thyroid C cells inhibits osteoclasts and lowers serum calcium. **Trap:** D raises calcium. A raises calcium by increasing absorption. C has no direct effect.

29. A — The primary center appears in the diaphysis; secondary centers appear in the epiphyses. **Trap:** D is a secondary center. B is the growth plate region. C is not an ossification center.

30. B — A multinucleated cell in a Howship lacuna is an osteoclast. **Trap:** D is cuboidal and lines the surface. A sits in a lacuna without a pit. C does not make Howship lacunae.

31. A — Dense regular connective tissue has parallel collagen bundles, as in tendon. **Trap:** D is dense irregular. B describes elastic tissue. C is reticular tissue.

32. C — Red bone marrow is the adult hematopoietic site. **Trap:** A is the fetal site. B is a lymphoid filter. D is a fat store, not hemopoietic.

33. A — Nissl bodies are stacked rough ER and polysomes. **Trap:** D describes steroid-producing cells. B is the axonal cytoskeleton. C are synaptic, not somatic.

34. D — One oligodendrocyte myelinates segments of several CNS axons. **Trap:** C is a supporting astrocyte. A myelinates in the PNS. B is phagocytic.

35. B — Microglia are mesodermal, derived from monocytes — the exception among glia. **Trap:** D gives melanocytes and Schwann cells. A gives neurons and macroglia. C gives epithelia.

36. C — Tight junctions between endothelial cells are the anatomical basis; astrocyte feet and pericytes reinforce it. **Trap:** B contributes a barrier but is not the seal. D lines ventricles. A support the capillary.

37. B — Endoneurium surrounds individual nerve fibers; perineurium surrounds fascicles. **Trap:** D surrounds the whole nerve. A surrounds a fascicle. C does not exist as a standard layer.

38. C — The A band is the length of the thick filaments and does not change during contraction. **Trap:** A and B shorten. D is wrong because the A band is fixed.

39. D — Calcium binds troponin C, which moves tropomyosin and exposes the myosin-binding site. **Trap:** C is the smooth-muscle calcium sensor. A is the smooth-muscle target. B is a structural protein.

40. D — Intercalated discs contain fascia adherens, desmosomes, and gap junctions. **Trap:** A anchors actin, B resists shear, and C couples the cells electrically; a tight junction is not part of the disc.

41. C — Cardiac muscle has one or two central nuclei; skeletal muscle nuclei are peripheral and multiple. Striations, myofibrils, and mitochondria are shared, so they cannot distinguish the two. **Trap:** B is shared. D and A are shared.

42. D — Smooth muscle uses calmodulin and myosin light-chain kinase; it has no troponin. **Trap:** C refers to striated muscle. A is part of the thin filament but not the calcium sensor. B is wrong.

43. C — In mammalian skeletal muscle, triads lie at the A–I junction; in cardiac muscle, diads lie at the Z line. **Trap:** B is the site of cardiac diads. D is the middle of the sarcomere. A is the H zone, which lies inside the A band.

44. B — Type I fibers are slow, oxidative, and fatigue-resistant. **Trap:** C is fast, oxidative-glycolytic and fatigable. D is fast glycolytic. A is wrong.

45. A — Neutrophils constitute 50–70% of circulating leukocytes. **Trap:** D is second in adults. B and C are minorities.

46. A — The eosinophil has a bilobed nucleus and bright orange-red granules. **Trap:** C has basophilic granules. D has a multilobed nucleus and pale granules. B is agranular and larger.

47. D — The monocyte is the largest leukocyte and has a kidney-shaped nucleus. **Trap:** C has a round dark nucleus and scant cytoplasm. A has a segmented nucleus. B is smaller with coarse granules.

48. B — Platelets are fragments of megakaryocyte cytoplasm. **Trap:** A is an erythroid precursor. C and D do not make platelets.

49. A — Peritubular interstitial cells of the kidney produce erythropoietin. **Trap:** D produces it in fetal life. B and C are not the main source.

50. C — The normal M:E ratio is 2:1 to 4:1, as Chapter 13 states. **Trap:** A (10:1) suggests myeloid hyperplasia; B and D are not normal adult values.

51. A — IgA is the main secretory immunoglobulin; it exists as a dimer with a secretory component. **Trap:** C crosses the placenta and dominates serum. D is the first antibody response. B mediates allergy.

52. C — $CD4^+$ helper T cells recognize antigen on MHC class II. **Trap:** B is read by CD8 cells. D is not a classical antigen-presenting molecule. A is wrong.

53. A — Hassall corpuscles are concentric epithelial whorls unique to the thymic medulla. **Trap:** C has no corpuscles. D has follicles and sinuses. B has crypts and lymphoid tissue.

54. B — Neither the thymus nor the spleen has afferent lymphatics. **Trap:** A has afferent lymphatics. C and D are MALT, not encapsulated organs.

55. C — The PALS is the T-cell zone of the spleen; follicles are the B-cell zone. **Trap:** B occupies follicles. D and A are present but not the defining population.

56. D — Germinal centers are sites of B-cell proliferation and affinity maturation. **Trap:** C occurs in the thymus. A occurs in marrow. B occurs in marrow.

57. C — The red pulp filters blood and removes aged or damaged erythrocytes. **Trap:** B is thymic. D and A describe white pulp.

58. A — Erythrocytes circulate for about 120 days. **Trap:** C is a platelet. D and B are wrong.

59. A — In the second trimester, the liver and spleen are the main hemopoietic organs. **Trap:** D is the earliest site. B and C are later or lymphoid.

60. C — Langerhans cells are the epidermal antigen-presenting cells; they originate in bone marrow. **Trap:** A is structural. B is pigment-producing. D is a mechanoreceptor.

61. A — Auerbach's plexus lies between the circular and longitudinal layers of the muscularis externa. **Trap:** D contains the submucosal (Meissner) plexus. C is the mucosa; B is the outer covering.

62. A — Parietal cells secrete hydrochloric acid and intrinsic factor. **Trap:** D is secreted by chief cells. B by mucous cells. C by G cells.

63. A — Brunner glands lie in the submucosa of the duodenum. **Trap:** B, C, and D contain no Brunner glands.

64. B — Peyer's patches are characteristic of the ileum. **Trap:** Peyer's patches are not a feature of the duodenum or jejunum, and the stomach has no organized lymphoid nodules of this kind.

65. A — Paneth cells secrete lysozyme and defensins, contributing to innate immunity. **Trap:** D is goblet cells. B is G cells. C is parietal cells.

66. A — Villi are present in the small intestine and absent in the colon. **Trap:** C, D, and B are present in the colon.

67. D — The central vein is at the center of the lobule, not in the portal triad. **Trap:** A, B, and C form the triad. The central vein is at the lobule center.

68. D — Kupffer cells are the resident macrophages of the hepatic sinusoids. **Trap:** C are the parenchymal cells. A are Ito cells. B line the sinusoid.

69. B — Bile flows toward the portal triad while blood flows away from it toward the central vein. **Trap:** A is the direction of blood. C is wrong for bile. D is the direction of blood.

70. A — The exocrine pancreas has basophilic acini with eosinophilic zymogen granules. **Trap:** D describes salivary gland. B describes mucous glands. C describes thyroid.

71. D — Beta cells secrete insulin and are about twice as abundant as alpha cells. **Trap:** C is glucagon. A is somatostatin. B is pancreatic polypeptide.

72. D — The esophagus is lined by stratified squamous non-keratinized epithelium. **Trap:** C is the stomach and gut. A is the urinary tract. B is the respiratory tract.

73. D — Type II pneumocytes contain lamellar bodies and secrete surfactant. **Trap:** C is the type I cell. A describes airway epithelium. B describes a macrophage.

74. A — Clara (club) cells line bronchioles. **Trap:** The alveoli are lined by type I and type II pneumocytes, the trachea by respiratory epithelium with goblet cells, and the pleura by mesothelium.

75. C — Cartilage forms rings in the trachea, plates in bronchi, and is absent in bronchioles. **Trap:** B and D reverse the sequence. A is not the pattern.

76. B — Type II pneumocytes do not form part of the thin blood–air barrier. **Trap:** C, D, and A are all part of the barrier.

77. A — The stratum lucidum is present only in thick skin of palms and soles. **Trap:** B, C, and D are present in thin skin too.

78. C — Melanocytes derive from the neural crest. **Trap:** B is the origin of the neural tube. D gives the dermis. A gives gut epithelium.

79. B — Sebaceous glands are holocrine; the whole cell disintegrates into the secretion. **Trap:** D is eccrine sweat. A is mammary lipid. C is hormonal release.

80. C — Pacinian corpuscles lie deep in the dermis and hypodermis and sense vibration. **Trap:** B contains Meissner corpuscles; A is supplied by free nerve endings. D has no corpuscles.

81. B — Keratinocytes make up about 90% of epidermal cells. **Trap:** D, A, and C together are a small minority.

82. B — Meissner corpuscles in the dermal papillae mediate fine tactile discrimination. **Trap:** A senses deep pressure and vibration. C senses stretch. D senses pain and temperature.

83. D — Nephrin is the key protein of the slit diaphragm. **Trap:** A is a desmosomal protein. B is a gap junction protein. C is a basal lamina protein.

84. C — The macula densa is the thickened plaque of the distal straight tubule (thick ascending limb) where it touches the vascular pole. **Trap:** A, B, and D do not sense tubular fluid.

85. C — Juxtaglomerular cells of the afferent arteriole secrete renin. **Trap:** B senses NaCl. D are podocytes. A are mesangial cells.

86. B — The proximal tubule is intensely eosinophilic because of abundant mitochondria and brush border. **Trap:** C is thin. D is pale. A is pale.

87. C — The ureter is lined by transitional epithelium. **Trap:** B and A are respiratory or gut. D is the distal urethra and vagina.

88. D — Thyroid follicles are lined by a simple epithelium whose cells range from squamous (inactive) to low columnar (active), classically described as cuboidal. The key point is that the epithelium is always simple; squamous and columnar are activity states, not different cell types. **Trap:** C lines the urinary tract. A and B are not found in thyroid follicles.

89. B — Parafollicular C cells secrete calcitonin. **Trap:** A make thyroid hormone. C are parathyroid. D are parathyroid.

90. C — The zona fasciculata secretes cortisol under ACTH control. **Trap:** B is glomerulosa. D is reticularis. A is medulla.

91. B — The adrenal medulla is neural crest derived; the cortex is mesodermal. **Trap:** A is mesoderm. C is endoderm. D is pharyngeal pouch endoderm.

92. D — Herring bodies are neurosecretory dilatations in the posterior pituitary. **Trap:** C contains no Herring bodies. A and B are not neural.

93. C — Tight junctions between adjacent Sertoli cells form the blood–testis barrier. **Trap:** B are germ cells. D are interstitial. A surround the tubule.

94. D — LH stimulates Leydig cells; FSH acts on Sertoli cells. **Trap:** C responds to FSH. A and B are not the primary LH targets.

95. B — Fertilization normally occurs in the ampulla of the uterine tube. **Trap:** A is where the embryo implants. C is a narrow transport segment. D is the birth canal.

96. B — Straight, narrow glands with mitoses are typical of the proliferative phase. **Trap:** A shows shedding. C shows tortuous secreting glands. D shows decidual change.

97. C — The corpus luteum secretes mainly progesterone. **Trap:** B is a minor product. D comes from trophoblast. A comes from pituitary.

98. D — The fovea contains cones only, giving maximal acuity and color vision. **Trap:** C is the periphery. A is false. B is the optic disc, not the fovea.

99. A — Aqueous humor drains through the trabecular meshwork into the canal of Schlemm. **Trap:** D is behind the lens. B produces it. C is not a drainage route.

100. C — The inner and outer hair cells of the organ of Corti are the transducing receptors. **Trap:** D support the hair cells. A are afferent neurons. B is a membrane, not cells.

Scoring and What to Do Next

Score	Interpretation	Action
90–100	Excellent	Skim the rapid review once daily; do slide identification only
80–89	Exam-ready	Reread the twenty-five distinctions table; revisit missed chapters
65–79	Adequate	Redo the master tables in Part VI, then retake the questions
50–64	At risk	Read each chapter's "core idea" and tables, not the prose
Below 50	Start again from Part II	Build the ten master tables from memory before anything else

HIGH-YIELD PEARL

The final revision loop is not reading — it is **recall**. Close the text and write the ten master tables from memory. Whatever you cannot reproduce is exactly what you must study. Students who can reproduce the tables usually answer the recall questions far more quickly.